The Transformation of Accumulation into Productive Capacities: Capacity Utilization in the Center and Periphery*

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August 14, 2026

Abstract

Keywords: Capacity Utilization; Transformation Elasticity; Choice of Technique; Cointegration; Marx-Okishio; External Balance; Center-Periphery.
JEL Codes: B51; C32; E01; E22; E32; P16.

*This paper benefited from AI-assisted editing for syntax refinement, structural suggestions, and stylistic consistency checks. All analytical arguments, historical interpretations, and substantive content remain the author's sole responsibility.

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1 The Unresolved Problem of Capacity Utilization

A This paper examines how the rate of profit, capital accumulation, and productive capacity formation interact under conditions of distributional conflict between labor and capital.

B

[**PLACEHOLDER**] Section 1 is unwritten. What exists above is a single orienting sentence; the section that the locked outline requires does not yet exist. The remainder of this box states what it has to do, so the gap is legible rather than silent.

C

[**PLACEHOLDER**] **Burden of the section.** Establish that capacity utilization is unresolved in the treatments that have come down to us, in three distinct ways, and show that these are three faces of one problem rather than three separate difficulties. First, utilization treated as a *given macro ratio*: taken as observed when its denominator is not. Second, utilization treated as an *inherited benchmark*: anchored to an engineering or survey norm imported from outside the accounting framework that is supposed to give it meaning. Third, utilization treated as an *empirical residual*: whatever is left after output has been fitted to capital, which presupposes the very transformation relation at issue. The third case is where the filtering literature sits, and the objection to it here is structural rather than methodological — a filter that fits output to capital imposes a unit transformation elasticity by construction, and so conflates utilization with the object that would be needed to identify it. That argument is available from this chapter's own accounting framework and requires no external authority.

D

[**PLACEHOLDER**] **Bottom line up front.** Per the writing calibration in force, the contribution must be visible by the end of the second paragraph: what the paper examines, how it identifies the transformation elasticity, and what the comparison of the United States and Chile establishes. The reader should not have to reach Section 3 to learn what is being claimed.

E

[**PLACEHOLDER**] **What must not appear here.** Long literature catalogues. The full center–periphery discussion, which belongs in Section 5. Estimator debates, which belong in Section 4 and Appendix C. Detailed methodological caveats. The section motivates a problem; it does not litigate the solution.

F

[**PLACEHOLDER**] **Closing-line burden (locked).** The section must arrive at: *the utilization problem cannot be resolved at the level of the ratio alone; it must be reconstructed through the relation that governs productive-capacity formation.* This hands off directly to Section 2.

G

[**PLACEHOLDER**] **Also unresolved: the abstract's sample window.** The abstract states a United States window of 1947–2024, while Sections 4 through 6 estimate over 1929–2024 with MATHPLACEHOLDER0XYZ. The two cannot both be right. Note further that 1929–2024 spans 96 years against a reported MATHPLACEHOLDER1XYZ of 94, and that the estimation prose elsewhere gives the effective span as 1931–2024 after lag-1 conditioning and listwise deletion — which is consistent with MATHPLACEHOLDER2XYZ and suggests the table titles, not the sample, are what needs correcting.

2 The Internally Divided Firm and Capacity Formation

2.1 Capacity utilization is not a technical parameter. It is an inter-scalar political-economic settlement. The macroeconomic capacity ceiling—the denominator of the utilization rate—is not a neutral engineering constraint but a historically contingent outcome conditioned by the balance of class power. Yet identifying this distributional conditioning at the macro level leaves the underlying transmission mechanism black-boxed: how, precisely, does distributive conflict physically alter the conversion of accumulated capital into productive capacity?

2.2 Answering this question requires dismantling a hidden premise shared by virtually all heterodox growth models: the unified firm. Whether treating the normal rate of capacity utilization as an endogenous behavioral target or an exogenous gravitational anchor, standard frameworks presume a coherent organization capable of selecting a utilization benchmark, enforcing it across the labor process, and sustaining it over time. Formally, the utilization rate is defined as:

$$\mu_t \equiv \frac{Y_t}{Y_t^p}, \tag{1}$$

2.3 where Y_t is realized output and Y_t^p is potential productive capacity. But this ratio conceals a deeper question: who determines Y_t^p, and under what organizational conditions?

2.4 The unified-firm premise collapses when confronted with the social organization of capitalist production. In a decentralized economy, individual capitals commit fixed investments without *ex-ante* coordination, confronting one another in competitive markets. Within the enterprise, production decisions encounter contradictory managerial and worker imperatives. Capacity utilization cannot be reduced to a target selected by a single optimizing entity. To theorize the political economy of productive capacity, we replace the unified firm with two interconnected frameworks: Okishio's theory of decentralized market anarchy, and Vidal's theory of the internally divided firm.

2.1 Smooth Reproduction and the Unified-Firm Premise

2.5 Proportional reproduction is a condition of feasibility, not a tendency of capitalist accumulation. Smooth reproduction requires strict material proportionality between departments of production—a condition Basu (2022) and Okishio (2022) formally established defined by the latter as the equilibrium accumulation trajectory. Bergmann et al. (2009) call this balanced trajectory, where realized output meets productive capacity ($\mu_t = 1$), the *"golden path."* But the golden path is a methodological closure, not a gravitational attractor. It identifies the conditions under which realization can reproduce itself smoothly; it does not describe the path capitalist accumulation actually follows. Neither the post-Keynesian assumption of long-run balanced growth (Blecker and Setterfield, 2019) nor the *"Keynesian in the short run, classical in the long run"* synthesis (Duménil and Lévy, 1999) captures this persistence of unbalanced expansion.

2.6 Nothing in the logic of decentralized competition establishes that private investment decisions will reproduce these proportions. A stylized illustration makes the point concrete. If productive capacity grows at 5% per year while aggregate demand for output grows at 2%, capacity outpaces demand by three percentage points each year. After a decade, utilization falls to roughly 75% of

capacity—not because of a temporary shock, but because uncoordinated investment persistently builds capacity faster than demand can absorb it. Balanced growth ($g_Y = g_K$) is merely one configuration of the transformation elasticity θ; it is not a restoring force. Smooth reproduction ($\mu_t = 1$) does not require $\theta = 1$; it holds for any transformation elasticity. Balanced growth is the narrower special case in which $\theta = 1$ and $\mu_t = 1$ coincide in the long run.

2.7 Because private investment decisions are made without *ex-ante* coordination, real capitalist economies routinely violate the proportional conditions of the golden path. Okishio (2022) demonstrates that chronic disproportionality is not a temporary friction but the structural norm of decentralized accumulation. Treating the transient benchmark of $\mu_t = 1$ as a behavioral target therefore relies on a hidden premise: that the firm is a unified planning subject capable of selecting a utilization norm, enforcing it across the labor process, and sustaining it over time. This premise—the unified firm—is the target of the present section.

2.8 The denominator of the utilization ratio (productive capacity Y_t^p) is not an engineering constant. It is a contested settlement of divided management and shop-floor struggle. The numerator (realized output Y_t) is hostage to the anarchy of demand. Smooth reproduction supplies a reference case against which this historical non-coincidence can be studied. It does not supply the mechanism that produces it.

2.2 Decentralized Accumulation and Market Anarchy

2.9 If the "golden path" of smooth reproduction is merely a benchmark of feasibility, the actual trajectory of capitalist accumulation is governed by the decentralized organization of investment. Capitalist production is coordinated through exchange, meaning that the social validation of private decisions occurs only *ex-post*. Because money mediates exchange, sale and purchase are separated moments of the capital circuit (Basu, 2022). Surplus value must be realized in circulation, not by construction (Foley, 1985; Trigg, 2020). This constitutes the logic of capital as impersonal domination: capitalists commit to production and investment with the law of value operating *"behind their backs."*[1] They are subject to market forces and value relations they neither control nor consciously direct. Critically, this anarchy is not a coordination failure in an otherwise functional market. The market is a socially constituted institution whose normal operation produces disproportionality as a structural feature (Vidal and Peck, 2012). Okishio (2022) locates the source of macroeconomic instability precisely in this constitutive anarchy. Individual capitals commit to fixed investment on the basis of private profititability expectations, yet the market validates those commitments only after production has begun. No firm can know in advance whether its expansion fits the investment and consumption decisions taken elsewhere. The coordination problem is not one of imperfect information surrounding an otherwise coherent plan; no economy-wide plan exists.

2.10 Smooth reproduction requires proportional expansion across sectors producing means of production and sectors producing consumption goods. Because private investment does not enforce this proportionality, firms extrapolate past prices, profit rates, and sales. A signal that induces expansion in one branch routinely produces excess capacity before complementary demand materializes

[1] Karl Marx, *Capital*, Volume I, Chapter 1, Section 4 (*"The Fetishism of Commodities and the Secret thereof"*), where the law of value operates as an overriding law of nature when private producers exchange products (Marx, 1999).

elsewhere. Contraction works through the same cumulative mechanism: falling sales cut investment, reducing aggregate demand and validating further cuts. Removing Say's Law reintroduces realization as a substantive condition of reproduction: accumulation may generate capacity that demand fails to absorb, or demand may temporarily exceed productive capacity (Foley, 1985). Prices register the resulting imbalance, but they do not prevent the sunk commitments produced by it.

2.11 Fixed capital gives this temporal lag its structural force. Structures and machinery embody long-horizon commitments that cannot be reversed when relative prices or demand conditions shift. Once installed, excess capacity leaves the firm to operate below its anticipated scale, defer maintenance, hold equipment idle, or devalue the asset. These responses distribute the costs of market anarchy among firms, workers, and creditors, but they do not restore the missing *ex-ante* coordination.

2.12 Idle capacity is therefore a recurrent, structural form of decentralized accumulation rather than a temporary friction. Private investment physically builds capacity before the market can realize its social use, recurrently a gap between installed productive capacities (Y_t^p) and effective demand (Y_t). The capacity utilization rate (μ_t) registers this gap at the macro level. Unbalanced growth is not a disturbance around proportionality; it is the structural consequence of distributively conditioned investment under decentralized decision-making. The analytical problem is therefore not how proportionality re-emerges, but how persistent imbalance remains historically reproducible. The condition $\mu_t = 1$ is at most a transient point through a given trajectory proportionality, while persistent divergence records the ongoing reality of market anarchy.

2.13 Yet Okishio's macro framework leaves a crucial gap: it does not explain how firms translate this external anarchy into internal operating decisions. The firm does not manage market anarchy; it internalizes it. A firm does not receive a price signal and respond as a frictionless, unified unit. It processes the signal through an organization divided over control, cooperation, effort, and the allocation of resources. Vidal's internally divided firm provides that missing mediation.

2.3 The Internally Divided Firm

2.14 Vidal builds on Marx's Volume I theory of the labor process, drawing on contemporary organizational theory and sociological institutionalism to develop an organizational political economy of the capitalist firm (Vidal and Peck, 2012; Vidal, 2022). Marx established that capitalist management faces a constitutive contradiction: it must simultaneously organize the cooperative labor required for production and enforce the coercive control required for valorization. Vidal (2022) demonstrates how this contradiction fractures the modern capitalist firm, treating it as a divided social organization whose internal structure is shaped by cognitive, normative, regulative, and organizational institutions. Management must extract surplus value through control over labor, but production also requires workers' cooperation, tacit knowledge, and problem-solving capacity. Standardization and surveillance secure a measurable pace of work while weakening the discretion needed to sustain complex production. The firm cannot maximize both imperatives at once because the labor process that produces value is also the site where workers contest its extraction. This contradiction is not a universal constant of capitalism; it is institutionally mediated, taking variegated forms across distinct accumulation regimes throughout historical capitalism (Vidal, 2019).

2.15 Workers respond to this contradiction by actively calibrating effort, cooperation, machine pacing, and the deployment of tacit knowledge. Capitalist management extracts potential capacity through concrete shop-floor mechanisms: increasing machine speed, deferring maintenance to hit short-term throughput targets, and enforcing rigid surveillance. Workers contest this extraction through labor turnover, collective exit, or the deliberate withdrawal of problem-solving cooperation and effort. Throughput, quality, and maintenance therefore depend on social relations that no engineering specification fixes in advance. A production target that raises machine speed by deferring maintenance may increase current output while degrading sustainable capacity. A directive that suppresses worker discretion may meet a short-term labor-control objective while worsening workflow bottlenecks. Labor effort is consequently not a fixed technical parameter applied to a pre-existing capital stock; it is an active, contested determinant of the capacity utilization itself.

2.16 This dynamic enforces a strict wedge between *engineering capacity* and *practical capacity*. Engineering capacity records the theoretical maximum output of equipment under ideal technical conditions. Practical capacity records the sustainable output rate dictated by actual maintenance schedules, product-mix complexity, and the negotiated intensity of labor. When managers report capacity to national statistical agencies, they rely on conventions shaped by their organizational position and the immediate pressures of the shop floor. Aggregate capacity statistics therefore do not reveal a neutral physical ceiling; they aggregate situated, institutional judgments (Shaikh, 2016; Nikiforos, 2016). Complementarily, empirical surveys of managerial practice confirm that firms do not systematically organize production decisions around a coherent capacity utilization target (Singh et al., 2022). For the purposes of macroeconomic modeling, the productive capacity ceiling (Y_t^p) is not an exogenous engineering constraint but an endogenous settlement of workplace conflict. This settlement is the micro-foundation through which class struggle and distribution transmit into capacity utilization, setting up the distributive–technological mapping developed in the following subsection.

2.17 Because the valorization–labor process contradiction cannot be solved, firms settle into organizational configurations of *"mediocre sufficiency"* (Vidal, 2022). These are hybrid arrangements that reproduce production without resolving the underlying conflict between control and cooperation. Partial worker autonomy coexists with rigid discipline; continuous improvement programs coexist with brutal throughput pressure; machinery might remain idle alongside binding productive bottlenecks. Following Vidal (2014), these arrangements are best understood as *institutional incoherence*: the firm's internal settlements are functional enough to sustain accumulation yet contradictory enough to generate persistent friction. This stands in direct contrast to the Varieties of Capitalism assumption of institutional complementarity, where national institutions are presumed to form coherent, mutually reinforcing systems (Hall and Soskice, 2003). Here, incoherence is not a deviation from an optimizing norm; it is the permanent, structural form through which capitalist management physically converts accumulation demand into capacity usage.

2.4 Distribution, Technique, and the Transformation of Accumulation

2.18 The choice of technique connects the divided firm to distribution, governing how accumulated capital is physically transformed into productive capacity. Capitalists adopt a new technique only when it reduces the cost of production at prevailing prices and wages; such a technique is capitalistically viable (Okishio, 1961). Okishio (1961) demonstrates that a viable capital-using,

labor-saving (CU-LS) technique raises the equilibrium profit rate when the real wage remains constant—a direct challenge to Marx's law of the tendential fall in the rate of profit. Marx's account, by contrast, allows the same form of technical change to depress profitability when rising capital intensity increases the capital advanced relative to surplus value. Basu (2022) formalizes the Marx–Okishio threshold: the rate of real-wage growth that separates these outcomes. It is a distributive boundary set by the balance of class forces, not a technical constant.

2.19 This boundary situates the present framework within the induced-innovation tradition while departing from it. The Innovation Possibility Frontier (IPF) literature, initiated by Kennedy (1964), treats distribution as selecting the *direction* of technical change along a given frontier: relative factor costs determine whether innovation is labor-augmenting or capital-augmenting, while the frontier itself—the scale of feasible productivity growth—remains exogenous (Julius, 2005; Tavani, 2012). Later assessments sharpen this limitation: IPF-based models explain the composition of technical change but remain largely silent on innovation intensity (Tavani and Zamparelli, 2018). Zamparelli (2024a) partially endogenizes scale by mapping a family of IPF levels to R&D effort, yet the analytical structure continues to rely on frontier geometry and steady-growth closure (Zamparelli, 2024b). An alternative strand reframes inducement through mechanization rather than frontier selection: Kaldor (1957) links productivity growth to capital deepening, while Cajas-Guijarro (2024) reformulates the IPF with an explicit choice over mechanization growth under distributive and labor-market conditions. The present framework builds upon a mechanization productive frontier, treating the distributive–technological relation as a regime-specific transformation between accumulation and productive capacity. Rather than selecting along a given frontier, distribution conditions mechanization intensity, which in turn determines the transformation elasticity linking accumulation to the formation of productive capacities.

2.20 The Marx–Okishio threshold gives wage pressure a precise role in mechanization, but the profitability outcome depends on the labor market closure. Basu (2026) demonstrates that the equilibrium profit rate after viable CU-LS technical change is governed by the institutional structure of capital–labor bargaining. Under a constant-wage-share closure—what Basu (2026) formalizes as the "strong labor" regime, applicable to Atlantic Fordism—any viable CU-LS technique unambiguously *lowers* the equilibrium profit rate. Marx's LTFRP reigns supreme precisely where labor is powerful enough to compel real wages to grow in tandem with productivity. Under this closure, rapid mechanization is induced by distributive pressure, yet the same pressure contests capital's claim on the resulting productivity gains (Vidal, 2015, 2019). A stylized illustration makes the mechanism concrete. If a 10% real wage increase raises mechanization intensity by 8% under strong-labor conditions, the transformation elasticity θ rises from 0.70 to 0.85: accumulation translates more efficiently into productive capacity because firms substitute machinery for labor. Wage-led mechanization was not a harmonious productivity bargain; it was a contested settlement in which the distributive pressure that induced technical change simultaneously eroded the profit rate it was meant to restore.

2.21 The weakening of organized labor after the Fordist crisis altered that mechanism. Under the "weak labor" closure—where the unemployment rate remains constant and workers lack the institutional power to enforce a target wage share—the distributive pressure to replace labor is suppressed (Basu, 2026). Yet mechanization does not cease. Lower relative machinery prices and

technical change in capital-goods production continue to favor capital-using techniques through a second, independent channel. Distribution and machinery prices therefore constitute a dual-channel mechanism in the choice of technique. Their relative force changes across accumulation regimes, which prevents the wage share from being treated as an external control added to an otherwise autonomous technical relation. Extending the stylized example: under Post-Fordist conditions, a 15% decline in relative machinery prices raises mechanization intensity by 6%, partially compensating for the weakened distributive channel and sustaining θ near 0.75 despite wage suppression. The same 10% wage increase that produced an 8% mechanization response under Fordism produces only a 2% response under Post-Fordism, leaving θ near 0.72. The two channels do not simply add; they interact through the institutional configuration that governs their relative force.

2.22 These channels still operate through the contradictions identified in the preceding subsections. Firms choose techniques before demand validates the associated capacity, and they operate those techniques through labor processes that divide control from productive cooperation. Investment consequently does not translate one-for-one into a stable productive ceiling. The transformation depends on the composition of capital, the distributive conditions governing mechanization, and the organization through which installed equipment becomes usable capacity. In a labor surplus economy—the third closure identified by Basu (2026), applicable to peripheral capitalism—the real wage remains constant and the equilibrium profit rate rises unambiguously after viable CU-LS technical change. Yet the peripheral transformation is structurally truncated by external financial subordination and the balance of payments constraint, a mechanism developed in the following subsection.

2.23 Okishio and Vidal jointly rule out any direct, automatic passage from accumulation to productive capacity. Decentralized investment generates structural disproportionality across firms and sectors; the internally divided firm internalizes this disproportionality as contested operating conditions rather than a coherent utilization target. The analytical object that captures this mediated transformation is the *transformation elasticity* ($\theta \equiv dy^p/dk$), which governs how accumulation converts into productive capacity under distributive conflict and market anarchy. The empirical analysis bounds this multi-sector mechanism to a single-sector aggregate, masking internal sectoral divergences while preserving the core transformation logic. Section 3 formalizes θ within a regime-specific analytical framework and derives its econometric implications through structural change controls.

2.5 Historical Regimes of Institutional Mediation and the Transformation Elasticity

2.24 The transformation elasticity is not a fixed engineering parameter. It is an emergent property of class struggle and the prevailing institutional environment. Decentralized accumulation does not automatically guarantee proportional reproduction ($\theta = 1$). Institutional mediation—wage-setting norms, financial architectures, and competitive structures—conditions how distributive pressures translate into mechanization choices (Aglietta, 2000; Vidal and Peck, 2012). Shifts in this mediation across historical periods will not be imposed through ex ante sample splits. The empirical strategy identifies them endogenously through structural change controls and break dummies, allowing the data to reveal how institutions bounded structural drift without requiring a formal equilibrium closure.

2.25 Basu (2026) demonstrates that the equilibrium profit rate after viable capital-using, labor-saving (CU-LS) technical change is governed by the labor market closure, which maps directly onto distinct historical accumulation regimes. Under a *constant-wage-share* closure—the institutional signature of Atlantic Fordism (1947–1973)—workers possess the bargaining power to grasp productivity gains, forcing real wages to grow in tandem with labor productivity. In this strong-labor regime, viable mechanization unambiguously lowers the equilibrium profit rate, so that Marx's law of the tendential fall in the rate of profit holds (Basu, 2026). This stagnation tendency was institutionally offset: the class compromise, mass consumption, oligopolistic market structures, and military Keynesianism of the Fordist era absorbed the surplus and sustained effective demand (Vidal, 2019).

2.26 Under a *constant-unemployment* closure—which resembles the persistent unemployment feature of the post-Fordist accumulation regime—an era during which the class compromise that took place during the Fordist period is dismantled and labor's bargaining power is severely weakened. The wage share falls, and mechanization is driven not by distributive pressure but by the relative cheapening of ICT and the shift toward financially reversible, short-horizon assets, patents and intellectual property of untangible assets (Lapavitsas, 2011; Pagano, 2014). Because wages no longer track productivity, the profit rate may stabilize or rise, but the suppression of wage-led demand generates a chronic under-consumptionist stagnation tendency. Following Vidal (2019), I hypothesize that under structural conditions, the post-fordist accumualtion regime produces a dysfunctional utilization configuration: accumulation decouples from productive capacity formation, generating chronic idle capacity that is temporarily masked by debt-led consumption and financial fragility rather than sustainable realization (Stockhammer, 2012; Vidal, 2019).

2.27 In the capitalist periphery, different closures represent distinct historical configurations throughout Chilean history. First, under an assumption of a *constant-real-wage*, which corresponds to Okishio's original formulation maps onto a surplus-labor economy (Lewis, 1954; Basu, 2026). During the import-substituting industrialization (ISI) era, the *rural-urban* and *demographic* transition enabled by the transformation of sanitary conditions and a massive drop of child moratlity in the early XX century, regulated late industrialization, providing an unlimited supply of labor from the countryside that disciplined urban wages and allowed the profit rate to reproduce itself alongside mechanization. However, peripheral mechanization is structurally import-biased: widening the mechanization gap requires imported capital goods and technologically non-substitutable inputs (Prebisch, 1981). The balance of payments constraint therefore acts as a hard external ceiling on the transformation of accumulation into capacity. Under a developmentalist regime with surplus labor (1933–1957), this operated as an *import capacity ceiling*. Second, a *constant-wage-share* or more accurately *rising-wage-share* closure, represents the intensification of stagnation tendencies leading to structural crisis of capitalism once the unlimited supply of labor depleted and the incentives for capitalist firms to mechanize risen due distributive pressures while increasing imnport requirements and stressing further the external constraint. Third, the discipline imposed by the neoliberal counter-revolution (Casals, 2021) (1974–present) a *constant-unemployment* closure coupled with trade and capital account liberalization relieved the external constraint, but locked the periphery into extractive specialization and premature deindustrialization, such that the WTO and bilateral free trade agreements legally encoded a *policy space ceiling* (Ahumada, 2019; Palma, 2019).

2.28 These historical configurations establish that the productive ceiling is not a neutral, trans-historical engineering constraint but a contingent outcome of decentralized accumulation, class power, and a country's relative positionality within the capitalist world system. The center–periphery comparison reveals a fundamental asymmetry: in the center, structural crises arise from the *exhaustion* of institutional mediation (e.g., the 1970s profit squeeze); in the periphery, crises arise from the *absence* of sovereign mediation and truncation by global financial markets causing regime shifting behaviors under external imbalances. The immediate task is to translate these historical-institutional claims into a formal analyitical architecture. Section 3 formalizes induced technological change under a cost-minimization problem, first advanced by Cajas-Guijarro (2024), in three stages of increasing complexity: i) aggregate capital-deepening; ii) mechanization bias under heterogeneous capital decision-making; and iii) a peripheral foreign-exchange bottleneck that induces technological change through mechanization. These analytical foundations provide the toolkit to deploy an empirical strategy identifying a state-dependent θ parameter endogenous to distributive conflict, tracing the trajectory of productive capacities and the capacity utilization ratio through a structural empirical approach.

3 The Analytical Framework: Formalizing the Transformation Elasticity

3.0 This section translates the distributive–technological mechanism of Section 2 into a formal model that yields estimable restrictions. Cajas-Guijarro (2024) first advanced the cost-minimization problem over mechanization growth; I extend it through three subsections of increasing complexity. Section 3.1 formalizes aggregate capital-deepening under Leontief technology and derives the transformation elasticity θ endogenously from the cost-minimizing choice of technique. Section 3.2 unbundles aggregate capital into extensive nonresidential structures and intensive machinery, and derives the state-dependent mechanization gap κ_t under a dual-channel viability condition. Section 3.3 imposes the peripheral foreign-exchange bottleneck as a binding constraint on κ_t, generating a corner solution that truncates θ. Each subsection nests the preceding one: setting the heterogeneity parameter to zero recovers the baseline model; relaxing the foreign-exchange constraint recovers the center model. The empirical analysis bounds this multi-sector mechanism to a single-sector aggregate, which masks internal sectoral divergences but preserves the core transformation logic. Section 3.4 synthesizes the theoretical claims of the three models. Section 4 consolidates the identification strategy and derives the cointegrating specifications.

3.1 The Baseline Model of Cost-Minimizing Mechanization (Homogeneous Capital Stock)

3.1 Actual aggregate output is represented by a Leontief production function with fixed technical coefficients:

$$Y = \min\left(AL,\ \mu BK\right) \tag{2}$$

where $A = Y/L$ is actual labor productivity, $\mu = Y/Y^p$ is capacity utilization, and $B = Y^p/K$ is maximum capital productivity. This production structure governs the rate of profit through the fundamental heterodox accounting identity:

$$r = \mu B(1-\omega) \tag{3}$$

where ω is the wage share. Standard macroeconomic models treat B as an exogenous technological given and μ as a behavioral constant smoothly targeted by a unified firm. I reject both assumptions. The components of (3) are endogenous outcomes of class struggle and decentralized market anarchy, as established in Section 2. This subsection formalizes how distributive pressure forces cost-minimizing firms to alter their choice of technique, endogenously determining B.

3.1.1 Tactical Choice of Technique and Induced Innovation

3.2 Capitalits firms choice of technique is given by aiming a mechanization growth rate q under competitive cutting cost-pressures, looking at the net productivity surplus of technical change as their objective function. Let a denote the growth rate of labor productivity. The effective distributive cost of deploying mechanization growth is $q(1-\omega)$. Hence, the choice of technique of capitalist mechanization can be represented by:

$$\min_{q} \; c(q) = q(1-\omega) - a \tag{4}$$

This objective captures the induced-innovation logic: firms accelerate mechanization to expand labor productivity growth, but the distributive cost of capital intensity holds them back. The cost-minimizing objective function of capitalist becomes constrainted by a mechanization function that can be represented as:

$$a = \Phi(q) = \lambda_0 + \lambda_1 q + \lambda_2 q^2 \tag{5}$$

3.3 Assumption 1. *Concavity of the mechanization frontier.* The frontier parameters satisfy $\lambda_0 > 0$, $\lambda_1 > 0$, and $\lambda_2 < 0$. The base productivity growth λ_0 sets the floor without advanced machinery. The linear coefficient λ_1 captures the direct productivity payoff of mechanization. The strict concavity $\lambda_2 < 0$ ensures diminishing marginal returns to mechanization.

3.4 Substituting (5) into (4) and solving the first-order condition yields the optimal mechanization growth:

$$c(q) = \lambda_0 + [\lambda_1 - (1-\omega)]\, q + \lambda_2 q^2 \tag{6}$$

$$\frac{\partial c}{\partial q} = \lambda_1 - (1-\omega) + 2\lambda_2 q^* = 0 \tag{7}$$

$$q^*(\omega) = \frac{(1-\omega) - \lambda_1}{2\lambda_2} \tag{8}$$

The strict concavity ($c''(q) = 2\lambda_2 < 0$) guarantees a unique maximum.

3.5 Proposition 1. *Induced innovation.* Under Assumption 1, the optimal mechanization growth is strictly increasing in the wage share:

$$\frac{\partial q^*}{\partial \omega} = -\frac{1}{2\lambda_2} > 0 \tag{9}$$

A rise in the wage share forces cost-minimizing firms to substitute machinery for labor. Rising wages drive mechanization directly through the first-order condition, pushing optimal capital intensity

growth upward. *Proof.* Immediate from differentiating (8). ■

3.X Remark 1. The objective (4) does not assert that the firm is a coherent optimizer. Following Vidal (2022), the firm is internally divided between the *control imperative* (valorization: cost reduction through mechanization) and the *cooperation imperative* (production: the need for worker effort, tacit knowledge, and problem-solving capacity). The formal model captures the cost-minimizing pole—the Okishio viability logic—while the constraints on the optimization problem capture the limits imposed by the cooperation imperative and by external conditions. The firm maximizes net productivity surplus—labor productivity growth minus the distributive cost of mechanization—subject to diminishing returns. The firm's actual mechanization choice is a contested settlement between these imperatives, not a coherent optimum. What the model demonstrates is the cost-reducing direction in which distributive pressure pushes the settlement, not the settlement itself.

3.1.2 Endogenous Transformation Elasticity

3.6 Substituting the optimal technique back into the technological frontier recovers the normal growth rate of labor productivity under smooth reproduction:

$$a^N(\omega) = \lambda_0 + \lambda_1 q^*(\omega) + \lambda_2 \left[q^*(\omega)\right]^2 \tag{10}$$

This optimal technique dictates the structural trajectory of the Leontief parameter B. By definition, maximum capital productivity is $B = Y^p/K$. In dynamic growth rates, the growth of capital productivity is:

$$b \equiv g_{Y^p} - g_K = a - q \tag{11}$$

Evaluated at the smooth reproduction benchmark: $b^N(\omega) = a^N(\omega) - q^*(\omega)$.

3.7 Definition 1. *Transformation elasticity.* The transformation elasticity θ is the structural parameter that translates capital accumulation into productive capacity growth:

$$g_{Y^p} = \theta\, g_K \tag{12}$$

Substituting (12) into (11) reveals the link between the level coefficient B and the elasticity θ:

$$\theta = 1 + \frac{b}{g_K} \tag{13}$$

3.8 To evaluate this elasticity, I anchor the choice of technique to a baseline rate of capital accumulation. Following the normal reproduction path, the normal accumulation rate is driven by the normal profit rate and the normal rate of re-capitalization net of depreciation. I follow Foley (1986) definition of the re-capitalization rate, $\chi = \frac{I}{\Pi}$, and interpret its normal conditions as continous realization and re-investment of the surplus.[2] Hence,

$$g_K^N = \chi^N r^N - \delta \tag{14}$$

[2] A simple accounting decomposition can demonstrate that the multiplier of profitability that drives investment, is precisely χ. In a monetary economy with credit, the re-capitalization rate does not require to be bounded by the surplus Π order of magnitude.

Expressed dynamically relative to this normal path, the aggregate transformation elasticity becomes:

$$\theta(\omega) = 1 + \frac{b^N(\omega)}{g_K^N} = 1 + \frac{a^N(\omega) - q^*(\omega)}{g_K^N} \tag{15}$$

3.9 Proposition 2. *Endogeneity of the transformation elasticity.* Under Assumption 1, the transformation elasticity θ is not an exogenous technical constant. It emerges endogenously from the cost-minimizing choice of technique. When distributive pressure forces firms to alter mechanization via (9), it simultaneously shifts the structural efficiency of accumulation via (15). The elasticity θ is therefore a function of the wage share ω, the frontier parameters $\lambda_0, \lambda_1, \lambda_2$, and the normal accumulation rate g_K^N.

3.2 Heterogeneous Capital, the Mechanization Gap, and State-Dependent Viability

3.10 Mainstream macroeconomics treats the capital stock as a homogeneous aggregate, abstracting from the internal architecture of the firm. Following the theory of organizational politcal economy and the internally divided firm (Vidal, 2022), I unbundle aggregate productive capital into and extensive margin given by nonresidential structures (K^{NRC}) and intensive one, given by the accumulation of machinery and equipment (K^{ME}). These components do not unfold throughout the labor process symmetrically. Structures form the built environments that setup the "envelope"of accumulation — i.e. the factory floor and logistics shell of the productive process. Capital accumulation in Machinery and Equipment is the active component that restructures the production process and the extraction of relative surplus value (Marx, 1999, Vol. I, Ch. 15).

3.11 Assumption 2. *Heterogeneous capital decomposition.* Aggregate productive capital decomposes as $K_t^{cap} = K_t^{NRC} + K_t^{ME}$. Nonresidential structures evolve independently of short-run distributive conflict (ω) and its rate of accumulation of accumulation is governed by its long-run relation with the rate of accumualtion in machinery and equipment under normal conditions of accumulation in the long-run

$$g^N_{K^{NRC},t} = f(\chi^N, r_t^N, g^N_{K^{ME},t}, \delta) \tag{16}$$

This extensive envelope $\Theta(K^{NRC})$ becomes a baseline for aggregate elasticity of transformation θ such that the rate of accumulation of capital stock in non-residential construction is a byproduct of the intensive margin of accumulation decided by a tactical technique choice made by capitalists in response to distributive pressures:

3.2.1 The Extensive Envelope and the Intensive Margin

3.12 Given the fixed infrastructural envelope, the **intensive margin** of accumulation is the tactical choice of the mechanization gap, defined in natural logs as:

$$\kappa_t = k_t^{ME} - k_t^{NRC} \tag{17}$$

This gap measures mechanization intensity relative to the extensive plant scale. The capacity identity decomposes potential output into the extensive envelope and the intensive capacity yield

Ψ_t:

$$y_t^p = \theta_{NRC} k_t^{NRC} + \Psi_t(\kappa_t, \omega_{t-1}, p_t^\kappa) \tag{18}$$

The intensive yield Ψ_t is not a fixed engineering constant. It is the outcome of a decentralized cost-minimization problem governed by the regime of accumulation, as formalized below.

3.2.2 Cost-Surplus Minimization and the Okishio Viability Yield

3.13 In this case, the capitalist firm chooses a mechanization gap κ_t under cutting-cost pressures, subject to the mechanization frontier:

$$\Phi(\kappa_t) = \lambda_0 + \lambda_1 \kappa_t + \lambda_2 \kappa_t^2 \tag{19}$$

The *realized* capacity yield is governed by a viability condition: a cost-minimizing firm deploys a new technique only if it reduces unit costs at current distributive and price parameters (Okishio, 1961; Basu, 2022). I formalize this as a cost-minimization as problem:

$$\min_{\kappa_t} \; c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^\kappa) - \Phi(\kappa_t) \tag{20}$$

3.14 The marginal viability cost $MC(\omega_{t-1}, p_t^\kappa)$ is determined by two channels:

1. **The distributive dividend (labor saving).** Increasing κ_t substitutes machinery for labor. The value of this displacement is proportional to the lagged wage share (ω_{t-1}), reflecting the wage settlement period prior to current capacity realization. A higher wage share *reduces* the net cost of mechanization.

2. **The relative price penalty (capital cost).** Increasing κ_t requires accumulating machinery faster than structures. The cost of this composition shift is proportional to the log relative price of machinery ($p_t^\kappa = \ln P_t^{ME} - \ln P_t^{NRC}$). A higher relative price *increases* the marginal cost.

We specify the marginal viability cost as:

$$MC(\omega_{t-1}, p_t^\kappa) = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa \tag{21}$$

where $\gamma_1 > 0$ captures the wage-savings dividend and $\gamma_2 > 0$ captures the relative price penalty.

3.15 Substituting (19) and (21) into (20) and solving the first-order condition yields the optimal mechanization gap:

$$c(\kappa_t) = \lambda_0 + \left[\lambda_1 - (\gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa)\right] \kappa_t + \lambda_2 \kappa_t^2 \tag{22}$$

$$\frac{\partial c}{\partial \kappa_t} = \lambda_1 - \gamma_0 + \gamma_1 \omega_{t-1} - \gamma_2 p_t^\kappa + 2\lambda_2 \kappa_t^* = 0 \tag{23}$$

$$\kappa_t^*(\omega_{t-1}, p_t^\kappa) = \frac{(\gamma_0 - \lambda_1) - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa}{2\lambda_2} \tag{24}$$

3.16 Proposition 3. *Dual-channel induced innovation.* Under Assumption 1 (concavity) and Assumption 2 (heterogeneous capital), the optimal mechanization gap satisfies:

$$\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} = -\frac{\gamma_1}{2\lambda_2} > 0 \tag{25}$$

$$\frac{\partial \kappa_t^*}{\partial p_t^\kappa} = \frac{\gamma_2}{2\lambda_2} < 0 \tag{26}$$

A rising wage share forces cost-minimizing firms to widen the mechanization gap, substituting machinery for labor. A rising relative price of machinery suppresses the optimal gap, as the structural cost of the composition shift penalizes mechanization. *Proof.* Immediate from differentiating (24). ■

3.2.3 The Marginal Capacity Payoff and the Econometric Bridge

3.17 The macroeconomic translation from the mechanization gap to potential capacity is given by the *marginal capacity payoff* (MP_κ), the derivative of the intensive capacity yield Ψ_t with respect to κ_t. By the envelope theorem:

$$MP_\kappa = \frac{\partial \Psi_t}{\partial \kappa_t} = \frac{\partial \Phi}{\partial \kappa_t} - MC(\omega_{t-1}, p_t^\kappa) \tag{27}$$

$$= (\lambda_1 + 2\lambda_2 \kappa_t) - (\gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa) \tag{28}$$

$$= \underbrace{(\lambda_1 - \gamma_0 + 2\lambda_2 \kappa_t)}_{\text{autonomous yield}} + \gamma_1 \omega_{t-1} - \gamma_2 p_t^\kappa \tag{29}$$

3.18 Integrating the marginal payoff with respect to κ_t (treating ω_{t-1} and p_t^κ as predetermined shifters) and defining θ_κ, $\theta_{\omega\kappa} \equiv \gamma_1$, and $\theta_{\kappa p\kappa} \equiv -\gamma_2$ yields the intensive capacity yield function:

$$\Psi_t = \theta_\kappa \kappa_t + \theta_{\omega\kappa}(\omega_{t-1}\kappa_t) + \theta_{\kappa p\kappa}(p_t^\kappa \kappa_t) \tag{30}$$

Substituting (30) back into (18) recovers the structural capacity equation:

$$y_t^p = \theta_{NRC} k_t^{NRC} + \theta_\kappa \kappa_t + \theta_{\omega\kappa}(\omega_{t-1}\kappa_t) + \theta_{\kappa p\kappa}(p_t^\kappa \kappa_t) \tag{31}$$

The interaction terms are not ad-hoc econometric additions. They are the macroeconomic footprint of the divided firm's Okishio viability constraint. Distributive pressure and relative price enter because they alter the marginal capacity payoff of the mechanization gap.

3.2.4 The Transformation Elasticity under Heterogeneous Capital Stock

REMINDER: This section requires editing and careful theoretical revision

3.19 Specification B recovers separate capacity elasticities for nonresidential structures (θ_{NRC}) and machinery (θ_κ), but evaluating aggregate capacity dynamics requires defining an aggregate transformation elasticity ($\theta_t^{\text{Aggregate}}$). In a single-asset model (Specification A), $\theta_A = \partial y^p / \partial k^{\text{cap}}$ is a scalar. Under heterogeneous capital, the total differential of productive capacity (31) separates

extensive expansion from intensive composition shifters:

$$dy_t^p = \theta_{\text{NRC}}\, dk_t^{\text{NRC}} + MP_\kappa\, d\kappa_t \tag{32}$$

$$= \theta_{\text{NRC}}\, dk_t^{\text{NRC}} + (\theta_\kappa + \theta_{\omega\kappa}\, \omega_{t-1} + \theta_{\kappa p\kappa}\, p_t^\kappa)\, d\kappa_t. \tag{33}$$

3.20 Standard continuous-time growth accounting evaluates aggregate capital variation using a Divisia index of heterogeneous components (Divisia, 1925). Since $s_{\text{ME},t} + s_{\text{NRC},t} = 1$ and $d\kappa_t = dk_t^{\text{ME}} - dk_t^{\text{NRC}}$, aggregate capital growth decomposes as:

$$dk_t^{\text{cap}} \approx s_{\text{ME},t}\, dk_t^{\text{ME}} + s_{\text{NRC},t}\, dk_t^{\text{NRC}} \tag{34}$$

$$= dk_t^{\text{NRC}} + s_{\text{ME},t}\, d\kappa_t. \tag{35}$$

Dividing (33) by (35) yields the conventional total-derivative aggregate elasticity along the empirical growth path:

$$\theta_{\text{total},t}^{\text{Aggregate}} = \frac{\theta_{\text{NRC}} + (\theta_\kappa + \theta_{\omega\kappa}\, \omega_{t-1} + \theta_{\kappa p\kappa}\, p_t^\kappa)\, \frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\, \frac{d\kappa_t}{dk_t^{\text{NRC}}}}. \tag{36}$$

3.21 Equation (36) suffers from a severe theoretical flaw illuminated by the Cambridge Capital Controversy Robinson (1953); Sraffa (1960). Heterogeneous physical assets cannot be aggregated into a single scalar k^{cap} without value weights that depend on distribution and prices. Differentiating such a value aggregate and placing dk_t^{cap} in the denominator re-introduces the circularity Specification B was designed to eliminate: the "marginal product" of aggregate capital is computed from an aggregate whose construction presupposes the distribution it purports to govern. Empirically, the denominator $1 + s_{\text{ME},t}(d\kappa_t/dk_t^{\text{NRC}})$ vanishes whenever physical accumulation stalls or shifts composition, producing spurious zero-denominator singularities ($\theta \to \pm\infty$).

3.22 To establish theoretical coherence and escape the Cambridge critique, I construct the aggregate transformation elasticity from the structural primitives of the heterogeneous system. Rather than differentiating a value aggregate, the structural asset-weighted transformation elasticity weights identified heterogeneous elasticities by physical asset stock shares:

$$\theta_t^{\text{Aggregate}} = \theta_{\text{NRC}}^{\text{str}} + s_{\text{ME},t} \cdot MP_{\kappa,t}^{\text{str}}, \tag{37}$$

where $\theta_{\text{NRC}}^{\text{str}}$ is the base elasticity of structures, $s_{\text{ME},t} = K_{\text{ME},t}/K_{\text{cap},t}$ is the physical equipment share, and $MP_{\kappa,t}^{\text{str}} = \theta_\kappa^{\text{str}} + \theta_{\omega\kappa}\,(\omega_{t-1} - \bar{\omega}) + \theta_{\kappa p}\,(p_t^\kappa - \bar{p}^\kappa)$ is the marginal capacity yield of mechanization. Equation (37) is immune to the Cambridge critique because the parameters are identified in a disaggregated system without prior scalar aggregation, and the weighting share $s_{\text{ME},t}$ is a physical asset ratio. This provides the exact theoretical benchmark evaluated empirically in Section 4.3.

3.3 The Peripheral Extension: Balance-of-Payments Constraint and the Corner Solution

3.23 In the capitalist periphery, the intensive margin of accumulation is structurally tethered to the world market and constrained by domestic social friction. Following Prebisch (1981) and Kaldor (1959), peripheral capitalism is characterized by high consumption propensities out of profits,

structural inelasticities in agriculture, and a chronic vulnerability to balance-of-payments crises (Ahumada, 2019). This introduces a dual realization bottleneck: the macroeconomic capacity to import and the inelastic domestic supply of wage-goods. Section 2.5 established this constraint qualitatively as the peripheral accumulation wedge. This subsection formalizes it as a strict macroeconomic boundary constraint on the cost-minimizing firm's choice of technique, integrating the structuralist insights of Prebisch (1981) into the heterogeneous capital framework of Section 3.2.

3.3.1 The Import-Biased Intensive Margin and the Balance of Payments Constraint

3.24 In Section 3.2, the cost-minimizing firm chooses the mechanization gap κ_t to maximize the net viability surplus:

$$\min_{\kappa_t} \; c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^\kappa) - \Phi(\kappa_t) \tag{38}$$

where $\Phi(\kappa_t) = \lambda_0 + \lambda_1\kappa_t + \lambda_2\kappa_t^2$ is the regime-conditioned mechanization frontier, and $MC(\omega_{t-1}, p_t^\kappa) = \gamma_0 - \gamma_1\omega_{t-1} + \gamma_2 p_t^\kappa$ is the marginal viability cost of the gap. In the core, this optimization proceeds unconstrained, yielding the optimal mechanization gap κ_t^*. In the periphery, however, active mechanization is strictly import-biased: widening the mechanization gap requires imported capital goods, specialized machinery, and technologically non-substitutable spare parts (Vernengo, 2006).

3.25 Assumption 3. *Import-biased mechanization.* The physical import requirement per unit of output (m_t^{req}) is a linear function of the chosen technique:

$$m_t^{req} = \nu \cdot \kappa_t \tag{39}$$

where $\nu > 0$ captures the structural import-intensity of the mechanization gap. Simultaneously, the physical capacity to satisfy this requirement is constrained by the *capacity to import* per unit of output ($m_t^{avail} = M_t^{avail}/Y_t$). Following Prebisch (1981), peripheral export growth is governed by the center's income elasticity of demand for primary goods ($\epsilon_{m,c} < 1$), structurally lagging behind domestic import demand. This external asymmetry, compounded by high consumption propensities out of capitalist profits (Kaldor, 1959), severely squeezes the foreign exchange available for productive accumulation. A necessary condition for the re-investment of the surplus. This imposes a strict ceiling on the choice of technique:

$$\nu \cdot \kappa_t \leq m_t^{avail} \implies \kappa_t \leq \bar{\kappa}_t \equiv \frac{m_t^{avail}}{\nu} \tag{40}$$

3.3.2 The Macroeconomic Allocation of Foreign Exchange

3.26 Assumption 4. *Balance-of-payments constraint.* Total import demand cannot exceed export capacity. The total supply of foreign exchange per unit of output is driven by primary exports ($x_t = p_t^x X_t/Y_t$). Due to the center's low income elasticity for primary goods, x_t is structurally rigid. The total demand for foreign exchange per unit of output consists of three competing components:

1. **Wage-goods imports** (m_t^W): driven by the wage share ω_{t-1} and the structural inelasticity of domestic agriculture (Kaldor, 1959). $m_t^W = m_w\omega_{t-1}$.

2. **Conspicuous consumption imports** (m_t^C): driven by the profit share π_t and the high propensity of the peripheral upper strata to imitate the consumption patterns of the center (Prebisch, 1981; Kaldor, 1959). $m_t^C = m_c c_k \pi_t$.

3. **Productive capital imports** (m_t^K): driven by the mechanization gap. $m_t^K = \nu\kappa_t$.

3.27 Assumption 5. *Surplus leakage.* The propensity to consume out of profits satisfies $c_k > c_k^{\min}$, where $c_k^{\min}$ is the threshold below which domestic savings suffice to finance capital formation without external borrowing. Under the developmentalist regime, the leakage channel is conspicuous consumption imports by domestic elites (Prebisch, 1981; Kaldor, 1959). Under the neoliberal regime, the leakage channel is profit repatriation by multinational corporations and capital flight (Ahumada, 2019). In both cases, $c_k > c_k^{\min}$ ensures that the effective profit share available for productive reinvestment is compressed below the nominal level: $\pi_t^{eff} = \pi_t \cdot \phi_t < \pi_t$.

3.28 The balance of payments constraint requires:

$$m_w\omega_{t-1} + m_c c_k \pi_t + \nu\kappa_t \leq x_t \tag{41}$$

Using the distributive identity $\pi_t = 1 - \omega_{t-1}$, we substitute and solve for the maximum feasible mechanization gap:

$$m_w\omega_{t-1} + m_c c_k(1 - \omega_{t-1}) + \nu\kappa_t \leq x_t \tag{42}$$

$$\nu\kappa_t \leq x_t - m_c c_k + (m_c c_k - m_w)\omega_{t-1} \tag{43}$$

$$\bar{\kappa}_t(\omega_{t-1}) = \frac{1}{\nu}\left[x_t - m_c c_k + (m_c c_k - m_w)\omega_{t-1}\right] \tag{44}$$

This equation formalizes the structuralist critique: the base capacity to import capital goods is determined by export earnings minus the baseline luxury consumption of the capitalist class ($x_t - m_c c_k$). The sign of $(m_c c_k - m_w)$ determines whether a rising wage share expands or contracts the feasible mechanization gap.

3.3.3 The Constrained Cost-Surplus Maximization Problem

3.29 The peripheral firm's tactical problem is therefore a cost-minimization problem constrained by the mehcanization function and the external constraint, such that the minimization problem can be stated as a Kuhn–Tucker optimization problem:

$$\min_{\kappa_t} \; c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^{\kappa}) - \Phi(\kappa_t) \tag{45}$$

$$\text{subject to:} \quad \kappa_t \leq \bar{\kappa}_t \tag{46}$$

This formulation distinct a regime shifting behavior of the choice of technique in the capitalist periphery given by a double constraint in the capitalists cost-minimizing behavior: i) Endogenous to domestic distributive conflict, ii) and to the dependency of foreign currency to fulfill import requirements at a given choice of technique. Therefore, if the external constraint becomes binding due lack of foreign currency, a regime shifting behavior of technological change takes place.

3.3.4 Endogenous Regime Shifts: The Corner Solution

3.30 The firm's unconstrained optimal technique $\kappa_t^*(\omega_{t-1})$ is strictly increasing in the wage share ($\partial\kappa_t^*/\partial\omega_{t-1} > 0$) due to the Okishio labor-saving bias established in (25).

3.31 Proposition 4. *Peripheral regime shifts.* Two regimes are endogenous to the interaction between domestic distributive conflict and the global market:

1. **Regime A (Slack BoP):** When $\kappa_t^* \leq \bar{\kappa}_t$, the firm solves the standard first-order condition, yielding κ_t^*. The labor-saving bias successfully extracts the Okishio distributive dividend.
2. **Regime B (Peripheral realization trap):** When $\kappa_t^*(\omega_{t-1}) > \bar{\kappa}_t(\omega_{t-1})$, the constraint binds and the firm is forced into a corner solution:

$$\kappa_t = \bar{\kappa}_t \tag{47}$$

The corner solution arises through two intersecting structural vulnerabilities. First, if $m_c c_k$ is large (the conspicuous consumption drain), the baseline FX availability is severely squeezed, lowering the intercept of $\bar{\kappa}_t$. Second, if domestic agriculture is highly inelastic ($m_w > m_c c_k$), a rising wage share simultaneously increases the *demand* for capital imports (via κ_t^*) while *contracting* the supply of available FX (via $\bar{\kappa}_t$). *Proof.* Immediate from the Kuhn–Tucker conditions applied to (45)–(46). ■

3.3.5 Effective Profit Compression and the Blocked Surplus

3.32 Definition 3. *FX-Conversion Ratio.* The FX-conversion ratio ϕ_t is the ratio of the maximum feasible technique to the optimal unconstrained technique:

$$\phi_t = \min\left(1, \frac{\bar{\kappa}_t}{\kappa_t^*}\right) \tag{48}$$

When the external constraint binds ($\kappa_t^* > \bar{\kappa}_t$), $\phi_t < 1$. This generates a direct compression of the effective profit share available for intensive accumulation.

3.33 Definition 4. *Blocked surplus.* Even if domestic wage suppression yields a high nominal profit share (π_t), the physical inability to convert that surplus into imported machinery squeezes the effective profit rate:

$$\pi_t^{eff} = \pi_t \cdot \phi_t = (1 - \omega_{t-1}) \cdot \phi_t \tag{49}$$

The difference constitutes the *blocked surplus*:

$$\text{Blocked Surplus} = \pi_t(1 - \phi_t) \tag{50}$$

Because the accumulation of machinery is driven by the *effective* profit share, the constraint decouples nominal profitability from capacity formation. If capitalists squeeze wages to raise π_t, they inadvertently increase conspicuous consumption (m_t^C), which lowers $\bar{\kappa}_t$ and compresses ϕ_t.

3.3.6 The Regime-Shifting Aggregate Transformation Elasticity

3.34 Definition 5. *BoP binding ratio.* The BoP binding ratio τ_t is the ratio of productive capital imports to import capacity:

$$\tau_t \equiv \frac{\nu \cdot I_t^{ME}}{\text{ToT}_t \times X_t} \tag{51}$$

where $I_t^{ME} = K_t^{ME} - (1 - \delta^{ME})K_{t-1}^{ME}$ is gross investment in machinery and equipment, derived from the perpetual inventory identity, and $\text{ToT}_t \times X_t$ proxies the import capacity (FX supply). Under

Assumption 3, productive capital imports $m_t^K = \nu\kappa_t$ are proxied empirically by gross investment in machinery and equipment I_t^{ME}, since mechanization is import-biased in the periphery. The regime switch is governed by τ_t:

- $\tau_t < \bar{\tau}$: BoP constraint is **slack** (Regime A: unconstrained optimum).
- $\tau_t \geq \bar{\tau}$: BoP constraint is **binding** (Regime B: corner solution).

The threshold $\bar{\tau}$ is determined empirically by the threshold regression in Section 4.

3.35 The peripheral extension generates a regime-shifting aggregate transformation elasticity. Under **Regime A** (slack BoP), the constraint does not bind, and the elasticity coincides with the unconstrained expression from Section 3.2 for the aggregate elasticity:

$$\theta_t^A == \frac{\theta_{NRC} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p\kappa} p_t^\kappa)\,\frac{d\kappa_t^*}{dk_t^{NRC}}}{1 + s_{ME,t}\frac{d\kappa_t^*}{dk_t^{NRC}}} \tag{52}$$

3.36 Under **Regime B** (BoP binding), the surplus of capitlist enterpirses becomes blocked, and the firm is forced into a corner solution $\kappa_t = \bar{\kappa}_t$. The marginal capacity payoff at the corner is strictly positive because the firm would like to mechanize further but cannot:

$$MP_\kappa\big|_{\kappa=\bar{\kappa}} = \Phi'(\bar{\kappa}_t) - MC(\omega_{t-1}, p_t^\kappa) > 0 \tag{53}$$

The Kuhn–Tucker multiplier on (46) is strictly positive, confirming that the constraint binds. The aggregate transformation elasticity under the binding regime becomes:

$$\theta_t^B = \frac{\theta_{NRC} + MP_\kappa\big|_{\kappa=\bar{\kappa}} \cdot \frac{d\bar{\kappa}_t}{dk_t^{NRC}}}{1 + s_{ME,t}\frac{d\bar{\kappa}_t}{dk_t^{NRC}}} \tag{54}$$

Since $\bar{\kappa}_t$ is determined by the BoP constraint (44) rather than by the first-order condition (23), the growth path $d\bar{\kappa}_t/dk_t^{NRC}$ differs from the unconstrained path $d\kappa_t^*/dk_t^{NRC}$. The FX-conversion ratio $\phi_t = \bar{\kappa}_t/\kappa_t^* < 1$ compresses the intensive margin, and the elasticity under the binding regime is strictly lower:

$$\theta_t^B < \theta_t^A \tag{55}$$

3.37 Proposition 5. *Regime-shifting elasticity.* The peripheral aggregate transformation elasticity is a regime-dependent parameter:

$$\theta_t^{periphery} = \theta_t^A \cdot \mathbf{1}\{\kappa_t^* \leq \bar{\kappa}_t\} + \theta_t^B \cdot \mathbf{1}\{\kappa_t^* > \bar{\kappa}_t\} \tag{56}$$

The elasticity shifts discretely when the BoP constraint becomes binding. The regime switch is governed by the interaction between the desired mechanization gap $\kappa_t^*(\omega_{t-1}, p_t^\kappa)$ and the maximum feasible gap $\bar{\kappa}_t(\omega_{t-1})$ determined by export capacity, conspicuous consumption, and wage-goods imports. An increase in the wage share ω_{t-1} simultaneously raises κ_t^* (via capital-using labor-saving choice of technique) and can contract $\bar{\kappa}_t$ (via the wage-goods import channel when $m_w > m_c c_k$), making the regime switch more likely. This is what Prebisch (1981) coined as the strangulation of economic development in the capitalist periphery: distributive pressures that would induce

productive mechanization in the center instead triggers the BoP constraint, which in this framework explain the truncated or reigme shifting behavior of the transformation elasticity.

3.3.7 The Failure of the Viability Conditions for Technological Change and Structural Inflation

3.38 Proposition 6. *Blocked viability condition.* At the boundary $\kappa_t = \bar{\kappa}_t$, the implementation of the viability condition is blocked, i.e. technical change is not net-surplus increasing or conversely, inefficient because it fails to achieve a cost-minimizing choice of technique. The firm satisfies the viability criterion—the technique would reduce unit costs—but cannot deploy it because the required machinery cannot be imported due lack of foreign currency. Hence, the capitalist enterpirse in the capitalist periphery suffers a persistent shortfall in its cost-surplus relative to its unconstrained state. *Proof.* At $\kappa_t = \bar{\kappa}_t$, the Kuhn–Tucker multiplier on (46) is strictly positive, so the first-order condition $\partial c/\partial \kappa_t = 0$ does not hold. The marginal viability yield exceeds the marginal cost at the boundary, but the firm cannot expand further. ■

3.39 Corollary 1. *Structural Inflation* Unable to extract the distributive dividend via mechanization, capitalists attempt to recover the blocked surplus through the only remaining channel: price markups. When the wage-goods bottleneck forces nominal wage adjustments and the external BoP bottleneck blocks defensive mechanization, any further increase in the wage share directly crushes profitability. The resulting defensive pricing behavior ignites structural inflation dynamic (Kalecki, 1955; Sunkel, 1958; Prebisch, 1981). This is the peripheral realization trap: nominal profitability coexists with truncated mechanization and structuerally stalled built-environments that constitute the *"envelope"* of the labor process, permanently destabilizing the long-run relation between investments in plant and machinery.

3.4 Synthesis: What the Three Models Jointly Demonstrate

3.40 The three models developed in this section formalize a single theoretical claim: that decentralized, individually cost-minimizing decisions generate aggregate disproportionality under the unbalanced growth closure (Okishio, 1961; Basu, 2026). Table 1 summarizes the articulation between the Okishio and Vidal frameworks that underpins this claim.

Table 1: Theoretical Coherence: Market Anarchy and the Internally Divided Firm Articulation Across Three Models

Element	**Descentralized Accumulation**	**Internally Divided Firm**	**Formal model captures**
Micro logic	Viability condition (cost reduction at prevailing prices)	Internal contradiction (control vs. cooperation)	Optimization problem = cost-minimizing pole; constraints = cooperation limits
Macro logic	Decentralized decisions $\rightarrow$ disproportionality	Divided firm $\rightarrow$ contested operating conditions	$\theta \neq 1$ captures gap between accumulation and capacity
Disequilibrium	Unbalanced growth (supply $\neq$ demand)	"Mediocre sufficiency" (Vidal, 2022) (no coherent optimum)	μ demonstrates gap; θ captures structural inefficiency
Peripheral specificity	BoP constraint blocks *implementation* of viability	External constraint compounds internal division	Corner solution: $\kappa = \bar{\kappa}$ when constraint binds

3.41 This analyitical framework does not assert equilibrium. It aims demonstrates the opposite: capitalist accumulation, left to its own decentralized logic, does not generate proportional reproduction. The rate of capacity utilization μ is the exhaust valve of this disproportionality: it shows that the normal operating conditions of capitalist firms, determined by their cost-minimizing mechanization choices, do not meet aggregate demand conditions. The transformation elasticity θ captures this structural gap between accumulation and realized capacity. Whether θ exceeds or falls below unity is an empirical question to be tackled below. In the center, this gap is cyclical—mediated by institutional configurations that can be exhausted triggering processes of capitalist re-structuring. In the periphery, this gap is permanent—locked in by the structural ceiling of external subordination (Kaldor, 1959; Prebisch, 1981).

3.42 Section 4 consolidates the data and measurement for United States and Chile, two center-periphery cases, presents the empirical identification strategy, and their results. It translates the three analytical models into estimable cointegrating specifications, derives the threshold regression for the peripheral corner solution, and specifies how the transformation elasticity serves to estimate a capacity utilization rate.

4 Data, Measurement, and Empirical Results

4.1 Data and Measurement

4.1.1 The Dual Valuation Distinction

This section constructs the capital stock series for the choice-of-technique models of section 3. The problem with the standard treatment of capital stocks series, is that they treat the physical capacity of installed capital and the monetary value of capital advanced as a single magnitude. This problem introduces a risk of spurious correlation between capital intensity and wage share measures as proxies for distributive conflict: as mechanization raises capital intensity, Consumption of Fixed Capital (CFC) rises in the measurement of Gross Value Added (GVA) denominator, depressing GVA-basis wage shares even when the rate of surplus-value extraction is unchanged. To deal with

this bias, I reconstruct the stock account-by-account using the Generalized Perpetual Inventory Method (GPIM) proposed by Shaikh (2016, App. 6.5). This methodology constructs gross and net capital stock measures that maintain stock-flow consistency from direct nominal investment flowsand asset-specific price indexes.

Estimates of capacity utilization under the unbalanced growth framework require distinct capital stock variables. The capital value initially invested in plant and equipment returns gradually to its money form as fixed assets depreciate; the accumulated depreciation allowances remain part of total capital advanced regardless of their monetary form, because their recovery is what permits the reproduction of the capitalist enterprise I therefore distinguish two types of measurement of capital stocks:

1. **Capital stock for capacity estimation: gross real GPIM stock.** Productive capacity depends on capital goods still physically in operation. The capital stock measure adequate for estimating capacity is the gross real stock $K^{G,R}_{j,t}$, weighted exclusively by a physical survival schedule. Productive capacity Y^p_t is itself a latent variable, recovered through the estimation of the transformation elasticity θ (section 3); the gross real GPIM stock is the accumulation variable k_t included in my estimation equation, not a measure of capacity itself. A machine whose book value has fallen to zero, but remains in service until it is physically retired, still should be counted as physical capital in operation. The OECD manual confirms that the gross stock, valued at "as new" prices, "is widely used as a broad indicator of the productive capacity of a country" (OECD, 2001, p.31).

2. **Profitability measure: net current-cost stock.** Profitability is a monetary ratio between a profit flow and the value of capital advanced. The relevant measure is the net current-cost stock $K^{N,V}_{j,t}$, which tracks the monetary value of capital after financial depreciation and write-downs. This measure serves as the anchor for the estimation of an implicit price of capital goods under a stock-flow-consistent framework but is not included within the structural estimation of non-observed productive capacities.

A single aggregate net stock serving both purposes is common in empirical macroeconomics. This practice introduces a mechanical correlation between the capital stock and the depreciation flow, introducing potential biases to identify distributive effects on productive capacities formation.

Valuation choice within the GPIM. The dual-valuation distinction does not require imposing a single valuation convention throughout. Shaikh (2016, App. 6.3) advocates the gross stock as the denominator for both the profit rate and the capacity ratio. The GPIM he develops, however, constructs the survival schedule $S_j(v)$, the age-price schedule $(1-d_j)^v$, and the SFC implicit price index $P_{j,t}$ as distinct components within a single stock-flow-consistent recursion. This structure permits the researcher to select the valuation for each analytical purpose without abandoning accounting coherence.

For the profit rate, the denominator must measure the monetary value of capital advanced. The net current-cost stock serves this role: it tracks the book value of capital that the firm must recover to continue production. A machine that has depreciated to a low book value may still operate at full physical capacity, but the firm's cost-recovery problem is defined by the value it must recoup, not by the physical state of the equipment (Basu, 2013). For estimates related to capacity utilization, the capital stock input must reflect the physical capital still in operation. The gross real GPIM stock,

weighted exclusively by the survival schedule $S_j(v)$ and excluding the age-price factor $(1-d_j)^v$, serves this role.

The quantitative significance of the gross/net distinction is not confined to the center. Ffrench-Davis and Vivanco (2016) show that in Chile depreciation absorbs between 8 and 13 percent of GDP and that net investment is barely half of gross investment in the most recent period. Service lives of machinery have declined from 20 years in the 1940s to 11 years by 2005, reflecting accelerating obsolescence. They conclude that the capital stock entering a production function must be the gross stock of physically surviving assets, not the depreciated monetary residual.

4.1.2 Gross Capital Stock Reconstruction and Measurement in the United States

The Generalized Perpetual Inventory Method. In this part I present the procedures of construction for the capital stock series for the United States. Machinery (ME) and nonresidential structures (NRC) do not share a natural physical unit; they become comparable only through a monetary-mediated accounting procedure. I reconstruct the stock account-by-account rather than aggregating nominal values and deflating with a generic index. The GPIM accumulation rules for historical-cost, current-cost, and constant-dollar stocks (Shaikh, 2016, App. 6.5) are derived in detail in Appendix **??**; here I state the construction.

Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}\}$, the proportion of a vintage surviving after v years is:

$$S_j(v) = \exp\left[-\left(\frac{v}{\lambda_j}\right)^{\alpha_j}\right] \tag{57}$$

where α_j is the Weibull shape parameter and the scale parameter is $\lambda_j = L_j/\Gamma(1+1/\alpha_j)$, derived from the mean service life L_j. Unlike geometric decay models that continuously erode the physical stock, the Weibull curve ensures that capital goods remain fully operative until they are physically removed from service. Empirical studies of directly observed asset discards confirm that the Weibull family provides a flexible approximation of retirement patterns. Nomura (2005), using Japanese micro-data on 66 asset classes, finds that almost half of assets exhibit a progressively increasing hazard rate ($\alpha > 1$). The calibrated parameters $L_{\text{ME}} = 14$, $\alpha_{\text{ME}} = 1.7$ and $L_{\text{NRC}} = 30$, $\alpha_{\text{NRC}} = 1.6$ follow the BEA Fixed Assets methodology (Fraumeni et al., 1997).

Physical survival alone cannot identify the SFC implicit price index. The BEA current-cost net stock anchors reflect both physical retirement and age-price decay. To recover the asset-specific price index, I impose an externally documented geometric declining-balance age-price profile:[3]

$$V_j(v) = S_j(v) \cdot (1-d_j)^v \tag{58}$$

where d_j is the asset-specific age-price decay rate. The declining-balance formula applied to the calibrated service lives yields $d_{\text{ME}} = 0.110$ and $d_{\text{NRC}} = 0.024$, consistent with the geometric rates documented by the BEA (Fraumeni et al., 1997).

The SFC implicit price index $P_{j,t}$ is the price level that closes the stock-flow identity for the net

[3]The BEA's current methodology employs geometric depreciation, which assumes that the efficiency pattern and the depreciation pattern have the same form (Fraumeni et al., 1997). The GPIM construction separates these two dimensions: physical survival follows a Weibull retirement distribution, while financial age-price decay follows a geometric profile. This separation is necessary to construct distinct capacity and profitability measures.

current-cost stock. In each period, the BEA anchor must satisfy

$$K_{j,t}^{\mathrm{NET,CC}} = I_{j,t}^{\mathrm{NOM}} + \frac{P_{j,t}}{100} \sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\mathrm{REAL}} \tag{59}$$

where $I_{j,t-v}^{\mathrm{REAL}} = I_{j,t-v}^{\mathrm{NOM}} / (P_{j,t-v}/100)$ is real investment deflated by the previously recovered SFC price. The first term on the right-hand side is current-year nominal investment; the second is the net-value-weighted sum of surviving real vintages revalued at the current price. The price index is "implicit" because it is not directly observed: it is the unique value that, applied to the survival-weighted real vintages, reproduces the BEA's reported current-cost net stock. Solving (59) for $P_{j,t}$:

$$P_{j,t} = 100 \cdot \frac{K_{j,t}^{\mathrm{NET,CC}} - I_{j,t}^{\mathrm{NOM}}}{\displaystyle\sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\mathrm{REAL}}} \tag{60}$$

The forward recursion proceeds period by period: given $P_{j,t-1}, P_{j,t-2}, \ldots$, the current price $P_{j,t}$ is uniquely determined by the requirement that the constructed stock reproduces the BEA anchor. Appendix **??** documents the recursion, the initial conditions, and the base-year normalization. The BEA's Fisher quantity index is retained solely as a validation comparison and does not enter the GPIM baseline.

The capital stock used to estimate productive capacity is the gross real GPIM stock, constructed as the survival-weighted sum of real investment vintages:

$$K_{j,t}^{G,R} = \sum_{v=0}^{L_j} S_j(v)\, I_{j,t-v}^{\mathrm{REAL}} \tag{61}$$

where $I_{j,t-v}^{\mathrm{REAL}}$ is real investment deflated by the previously recovered SFC price. Because eq. (61) relies exclusively on the survival schedule $S_j(v)$ and excludes the age-price factor $(1-d_j)^v$, it measures the constant-price purchasing power embodied in heterogeneous capital goods still in physical operation. Financial write-downs, revaluation gains, and depreciation allowances do not enter.

Validation and sample scope. The GPIM construction closes the stock-flow identity exactly: the maximum absolute reconstruction residual across both asset classes is 10^{-6} million dollars, confirming that the recovered SFC implicit price indexes reproduce the BEA current-cost net stock anchors. All GPIM series are normalized to 2017 = 100. The capital stock series has a 30-year warm-up period (1901–1930). The GPIM stock at any year t requires investment vintages extending back L_j years. Because the nominal investment series begins in 1901, the vintage summation for the longest-lived asset class (nonresidential structures, $L_{\mathrm{NRC}} = 30$) is truncated for any year before 1931. The stock values for 1901–1930 are therefore incomplete, omitting pre-1901 vintages that would still be physically in service. I restrict the estimation sample to 1931–2024, where the full 30-year vintage window is populated, yielding $T = 94$ annual observations. The sectoral scope is restricted to nonfinancial corporate (NFC) business, excluding non-profit institutions serving households (NPISH), government, and owner-occupied housing. This sectoral restriction isolates the private productive business sector from public infrastructure and imputed residential rents.

4.1.3 U.S. Distributive Variables

The GVA inadecuacy. The standard GVA-basis wage share is mechanically contaminated by Consumption of Fixed Capital. Because GVA = CFC$+v+s$, periods of intensive mechanization raise CFC/GVA even when the rate of surplus-value extraction (s/v) is unchanged. The GVA-basis wage share therefore falls mechanically during capital-deepening episodes, producing a spurious correlation between capital intensity and the measured distributive share. The NVA basis, NVA $= v + s$, eliminates this contamination by restricting the denominator to newly created value. All baseline distributive measures in this dissertation are constructed on the NVA basis.[4]

The No-Double-Accounting Principle. Standard NIPA corporate aggregates combine nonfinancial and financial corporate sectors into a single total. This aggregation violates the No-Double-Accounting Principle: financial sector profits and net interest receipts are secondary claims extracted from primary nonfinancial surplus, not independently created value. Including them in the denominator of the wage share counts value creation twice. The corrected denominator is nonfinancial corporate net value added (NFC_NVA), which excludes financial double-accounting transfers.[5]

Supervisory compensation adjustment. Standard NIPA compensation includes managerial and executive salaries, which under the Shaikh-Tonak classification function as a distribution of surplus value rather than productive variable labor cost (Shaikh and Tonak, 1994). To isolate productive variable capital, total compensation is adjusted by the supervisory compensation ratio ϕ_t^{sup}:

$$v_t^{\text{prod}} = \text{NFC_COMP}_t \times (1 - \phi_t^{\text{sup}}) \,. \tag{62}$$

Two tiers of wage share are defined. Tier 1 uses total compensation without supervisory adjustment; Tier 2 applies the adjustment in (62). The supervisory compensation ratio ϕ_t^{sup} is constructed from Table S512 of the *Measuring the Wealth of Nations* dataset (Shaikh and Tonak, 1994), as reconstructed and extended to 2024 by Anderson (Anderson, 2024).[6]

4.1.4 Productive output and aggregate demand for the United States

The dependent variable in the cointegrating regressions of section 3 is the logarithm of real gross value added of nonfinancial corporate business, $y_t = \ln Y_t^{\text{NFC}}$, measured in chained 2017 dollars and sourced from BEA NIPA Table 1.14, Line 41.[7] The nonfinancial corporate boundary isolates productive value creation within the business sector, excluding financial intermediaries whose income constitutes secondary claims on nonfinancial surplus rather than independently created value. The robustness battery of section 4.1.3 audits eight alternative deflator proxies—including the official NFC implicit deflator (Table 1.15), the GDP implicit deflator, and the nonfarm business output deflator—and confirms that coefficient estimates are invariant to deflator choice.

[4] Eight wage share variants (NFC and corporate, each on GVA and NVA bases) are enumerated in Appendix **??**.

[5] The implied financial sector surplus is recovered as $\text{FIN_NOS}^{\text{implied}} = \text{CORP_NOS} - \text{NFC_NOS}$. Clean corporate denominators purge both $\text{FIN_NOS}^{\text{implied}}$ and NFC_NET_INT. Derivation in Appendix **??**.

[6] The full dataset is available at `https://shaikh.heterodata.org/`

[7] BEA series code `B455RX`; nominal counterpart `A455RC`, Line 17. The implicit output deflator is $P_{y,t}^{\text{NFC}} = 100 \times \text{NFC_GVA}_t^{\text{nom}} / Y_t^{\text{NFC}}$. Coverage: 1929–2025, annual.

The numerator of aggregate capacity utilization $\mu_t = Y_t/Y_t^p$ is economy-wide real GDP, drawn from BEA NIPA Table 1.1.9, Line 1, in chained 2017 dollars.[8] The denominator Y_t^p is not directly observed but reconstructed from the estimated transformation elasticity, as detailed in **??**. The distinction between the two output measures is deliberate: the transformation elasticity θ is estimated from NFC output, which captures the capital-output relationship within the productive corporate sector; the utilization ratio uses aggregate GDP because capacity utilization measures the engagement of total productive capacity by effective demand, including government expenditure, household consumption, and net exports. This separation ensures that the estimation of θ remains uncontaminated by demand-side fluctuations that drive aggregate GDP but do not reflect the structural capital-output relationship.

4.1.5 Capital Series Harmonization and Historical Sources for Chile

This subsection describes the stock-flow consistent (SFC) methodology used to reconstruct gross and net capital stocks for Chile, 1900–1994, in constant 2003 Chilean pesos (CLP). The procedure enforces the accumulation identities for gross and net stocks jointly, yields implicit depreciation and retirement rates for each asset class, and delivers coherent series for machinery and equipment (ME), nonresidential construction (NRC), residential construction (RC), and their aggregates. The construction proceeds from a single investment control total, disaggregated by asset, accumulated via Weibull survival functions, and linked to net stocks through a gap-ratio approach anchored to the Hofman (2000b) overlap window.

The stock-flow identities. The reconstruction is governed by two laws of motion. The gross stock evolves according to

$$\Delta K_{j,t}^g = I_{j,t} - R_{j,t} \tag{63}$$

where $I_{j,t}$ is gross fixed investment and $R_{j,t}$ denotes physical retirements (scrapping of obsolete vintages, demolition, and technological withdrawal). The net stock evolves according to

$$\Delta K_{j,t}^n = I_{j,t} - D_{j,t} \tag{64}$$

where $D_{j,t}$ is depreciation (the monetary consumption of fixed capital). Combining (63) and (64) yields the joint SFC identity

$$\Delta K_{j,t}^g + R_{j,t} = \Delta K_{j,t}^n + D_{j,t} \tag{65}$$

which constrains the combination of depreciation and retirements at every t and for every asset j. Given observed stocks and investment, the implied flows are recovered as $D_{j,t} = I_{j,t} - \Delta K_{j,t}^n$ and $R_{j,t} = I_{j,t} - \Delta K_{j,t}^g$. The capital gap $B_{j,t} \equiv K_{j,t}^g - K_{j,t}^n$ measures the cumulative difference between installed capacity and book value.

The investment control total and the lambda bridge. The aggregate investment control total is drawn from the Pérez-Eyzaguirre (2019) reconstruction of gross fixed capital formation (GFCF), expressed in constant 2003 CLP. The control concept is machinery plus construction, explicitly excluding inventory variation.[9]

[8] BEA series code `A191RL`; coverage 1929–2025, annual. The GDP implicit price deflator (`A191RD`) serves as the economy-wide deflator for the utilization reconstruction.

[9] Inventories are excluded from fixed-capital accumulation because the reconstruction models fixed capital only. Inventory investment does not enter the perpetual inventory recursion.

Pérez-Eyzaguirre report investment in a price space and asset aggregation that differ from the Hofman (2000a) benchmark. The two sources are linked by a level-normalization factor λ, which corrects for definitional differences between the two series rather than converting between price bases. The factor is estimated as the median of the annual ratio of Hofman to Pérez investment over the 95-year estimation window:

$$\lambda = \text{median}\left(\frac{I_t^{\text{Hofman}}}{I_t^{\text{Pérez}}}\right)_{t=1900}^{1994} = 4.1054 \tag{66}$$

The median rule is chosen for robustness to outliers in the early decades, where source quality is weaker. The scaled aggregate is then disaggregated into ME, NRC, and RC using normalized asset-specific investment shares from Hofman (2000a):

$$I_{j,t} = s_{j,t} \cdot \lambda \cdot I_t^{\text{Pérez}}, \qquad \sum_{j \in \{\text{ME, NRC, RC}\}} s_{j,t} = 1 \tag{67}$$

for all t.

Weibull survival and gross stock accumulation. Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}, \text{RC}\}$, the proportion of a vintage surviving after v years is

$$S_j(v) = \exp\left[-\left(\frac{v}{\lambda_j}\right)^{\alpha_j}\right] \tag{68}$$

where α_j is the Weibull shape parameter and the scale parameter is $\lambda_j = L_j/\Gamma(1 + 1/\alpha_j)$, derived from the mean service life L_j. The survival parameters are estimated from the Hofman (2000b) overlap window (1950–1994), in which both gross and net stocks are observed, by fitting Weibull curves to the implied retirement patterns recovered from the stock-flow identities. Table 2 reports the calibrated parameters.

Table 2: Weibull survival parameters, Chile (1900–1994)

Asset	Scale (λ_j)	Shape (α_j)	Mean life (L_j, years)
ME	15.90	2.42	14.1
NRC	41.30	5.00	37.9
RC	51.16	5.00	47.0

All shape parameters exceed unity, indicating progressively increasing hazard rates: capital goods remain fully operative in early years and are withdrawn from service at an accelerating rate as they approach the mean service life. The shape parameters for NRC and RC are constrained at 5.0, reflecting the long and relatively uniform service lives of structures. The gross stock is accumulated as the survival-weighted sum of real investment vintages:

$$K_{j,t}^g = \sum_{v=0}^{L_j} S_j(v)\, I_{j,t-v} \tag{69}$$

Pre-1950 stocks are anchored to the observed Hofman levels at 1950 via a multiplicative factor that

matches $K^g_{j,1950}$ by asset.

The gap-ratio approach and net stock construction. Unlike the U.S. GPIM, which recovers the net stock via a separate geometric age-price decay rate and an SFC implicit price recursion, the Chile reconstruction derives the net stock from the observed gross-net relationship in the Hofman (2000b) overlap window. The capital gap ratio

$$\gamma_j \equiv \frac{B_j}{K^g_j} = \frac{K^g_j - K^n_j}{K^g_j} \tag{70}$$

is computed by asset from the Hofman (2000b) data over the overlap period (1950–1994). Two regimes are distinguished: a "pre" ratio (median of years ≤ 1954) and a "post" ratio (median of years ≥ 1990), allowing for structural shifts in the gross-net relationship over the century. Outside the overlap window, the gap ratio is extended at the regime-specific median. The net stock is then recovered as

$$K^n_{j,t} = K^g_{j,t} \cdot (1 - \gamma_j) \tag{71}$$

This approach inherits the gross-net relationship observed in the Hofman accounting reference rather than imposing an external depreciation convention. It is appropriate for Chile because no independent age-price decay data are available for the historical period; the Hofman (2000b) overlap provides the only direct evidence on the joint behavior of gross and net stocks.

Implicit depreciation and retirement rates. Given the reconstructed stocks and investment flows, the implicit annual rates are defined as

$$z_{j,t} = \frac{R_{j,t}}{K^g_{j,t-1}}, \qquad \delta_{j,t} = \frac{D_{j,t}}{K^n_{j,t-1}} \tag{72}$$

where $z_{j,t}$ is the retirement (depletion) rate and $\delta_{j,t}$ is the depreciation rate. These rates are not imposed as parameters; they are derived from the data via the stock-flow identities (63)–(64). They serve as diagnostic indicators of capital durability and replacement dynamics, and are used in **??** to validate the internal consistency of the reconstruction.

Validation and estimation sample. The SFC construction is validated through four checks. First, additivity: all composite stocks and flows satisfy $K_{\text{NR}} = K_{\text{ME}} + K_{\text{NRC}}$ and $K_{\text{T}} = K_{\text{ME}} + K_{\text{NRC}} + K_{\text{RC}}$ at every t. Second, non-negativity: $K^g_{j,t} \geq 0$, $K^n_{j,t} \geq 0$, and $B_{j,t} \geq 0$ for all assets and years. Third, rate bounds: $z_{j,t} \in [0, 1]$ and $\delta_{j,t} \in [0, 1]$ for all observations from 1950 onward. Fourth, identity residuals: $\Delta K^g_{j,t} - (I_{j,t} - R_{j,t}) = 0$ and $\Delta K^n_{j,t} - (I_{j,t} - D_{j,t}) = 0$ to numerical precision. The estimation sample spans 1900–1994, yielding $T = 95$ annual observations. The overlap window (1950–1994) provides 45 years of direct Hofman (2000b) observations against which the reconstructed series are audited.

The BCCh forward extension (1995–2024). Extending the reconstruction beyond 1994 requires integrating the official national accounts published by the Central Bank of Chile (BCCh). A direct chain-linking of the BCCh series to the historical baseline would introduce a level discontinuity, because the two sources differ in base year, valuation conventions, and asset classification. To

eliminate this discontinuity while preserving the validated 1900–1994 history, I employ a multiplicative splicing procedure that fixes the legacy panel in place.

For the overlap year 1994, an asset-specific bridge factor is computed as the ratio of the historical net stock to the BCCh reported net stock:

$$\beta_j = \frac{K^{n,\text{hist}}_{j,1994}}{K^{n,\text{BCCh}}_{j,1994}}, \quad j \in \{\text{ME, NRC, RC}\} \tag{73}$$

The same factor β_j is applied to both the BCCh net stock and the BCCh depreciation series for all years 1995–2024. This preserves the depreciation rates and asset dynamics implied by the BCCh while maintaining continuity with the constant 2003 CLP historical baseline. Table 3 reports the estimated bridge factors.

Table 3: Asset-specific bridge factors, 1994

Asset	Bridge factor (β_j)	Interpretation
ME	5.888	BCCh understates historical ME stock
NRC	1.033	Near parity
RC	1.050	Near parity

The large bridge factor for machinery and equipment reflects the substantially lower level at which the BCCh reports ME stocks relative to the Pérez-based historical reconstruction, a divergence driven by differences in asset coverage and valuation methodology. For nonresidential and residential construction, the bridge factors are close to unity, indicating that the two sources agree substantially on the level of structures.

Post-1994 real investment is inferred via the stock-flow identity:

$$I_{j,t} = \Delta K^n_{j,t} + D_{j,t}, \quad t \geq 1995 \tag{74}$$

where $D_{j,t}$ is the bridge-adjusted BCCh depreciation. The gross stock and physical retirements for the extended period are then reconstructed by applying the survival parameters from Table 2 to the complete 1900–2024 investment series via eq. (69). This ensures that the gross stock remains internally consistent across the splice boundary.

4.1.6 Wage Share Data Sources for Chile

The distributive regressor in the transformation elasticity is the wage share, ω_t. For Chile, the wage share is drawn from the revised dataset of real wages in Latin America constructed by Astorga (2017) and updated by Astorga (2023). The dataset provides yearly series of real wages by skill level—unskilled (w_4), semi-skilled (w_3), and skilled (w_2)—together with an economically-active-population (EAP) weighted average (w), expressed in monthly purchasing-power-parity dollars at 1970 prices.[10]

The wage share is computed as the wage-to-income ratio $\omega_t = w_t/y_t$, where y_t is real income per worker derived from national income estimates. The Astorga (2023) series covers 1920–2011. To extend coverage to 2024, I splice Astorga's series with the institutional sector accounts of

[10]The revised dataset is publicly available as supplementary material to Astorga (2023).

Chile's Central Bank. The BCCh National Accounts report production income by institutional sector—financial corporations, non-financial corporations, households and NPISH, and general government—in nominal terms at 2018 reference prices. From these, I construct the private non-financial wage share as

$$\omega_t^{\text{BDE}} = \frac{Y_t^{\text{HH}}}{Y_t^{\text{TOT}} - Y_t^{\text{FIN}} - Y_t^{\text{GOV,CFC}}} \tag{75}$$

where Y_t^{HH} is household income (compensation of employees plus mixed income), Y_t^{TOT} is total production income, Y_t^{FIN} is the financial sector's operating surplus, and $Y_t^{\text{GOV,CFC}}$ is general government consumption of fixed capital. The exclusion of the financial sector from the denominator is consistent with the No-Double-Accounting Principle applied to the U.S. series in section 4.1.3: financial profits are recognized as secondary claims on primary non-financial surplus and do not enter the distributive denominator.[11]

The BCCh series covers 2003–2024, yielding a nine-year overlap (2003–2011) with the Astorga (2023) panel. The extension is anchored at 2011, the final year of the Astorga coverage, by constructing the index

$$\mathcal{I}_t = \frac{\omega_t^{\text{BDE}}}{\omega_{2011}^{\text{BDE}}}, \quad t \geq 2012 \tag{76}$$

and applying it to the Astorga terminal value:

$$\omega_t = \omega_{2011}^{\text{Astorga}} \cdot \mathcal{I}_t, \quad t \geq 2012. \tag{77}$$

The splice assumes that the growth rate of the national-accounts wage share is a valid proxy for the growth rate of the labor-market wage-to-income ratio beyond the Astorga coverage. The nine-year overlap (2003–2011) supports this assumption: both series exhibit the same cyclical pattern and comparable magnitudes of distributive fluctuation. The resulting ω_t series spans 1920–2024, yielding $T = 105$ annual observations.

4.1.7 Aggregate Demand, Price Indices, and External Constraint variables for Chile

The aggregate demand variables—real GDP (Y_t), exports FOB (X_t^{FOB}), and gross machinery investment ($I_{\text{ME},t}$)—are drawn from the aggregate demand reconstruction of Pérez-Eyzaguirre (2019), expressed in constant 2003 Chilean pesos. This series revises and extends the historical national accounts backbone of La República en Cifras (Lüders et al., 2016), correcting additivity and valuation problems in the original Clio-Lab series while preserving the sectoral GDP decomposition. Coverage spans 1860–2024.

The price and trade indices used to construct the relative capital price proxies (p_t^{κ}), the capacity to import (M_t^{cap}), are drawn from *La República en Cifras* (Lüders et al., 2016): the real exchange rate index, the US wholesale capital price index, and the terms of trade index. All three series are rebased to 2003 = 100 for internal consistency with the capital stock series.[12]

The capital import demand imbalance combines sources from *La República en Cifras* with

[11] The underlying series are drawn from the Base de Datos Estadísticos (BDE) of the Banco Central de Chile, Cuentas Nacionales, sectoral income accounts, reference year 2018, nominal quarterly flows. Annual values are obtained by simple summation of quarterly observations.

[12] The terms of trade index extends back to 1810; the real exchange rate and US wholesale price index cover 1900–2024.

my data set of harmonized capital stocks through a log difference of $ln(I_{MEt})$ - $ln(M_t^{\text{cap}})$, which reconciles the supply export-side identified by the index of imports capacities (M_t^{cap}) of the external constraint, with the internal demand for foreign capital goods providing a both a more accurate variable for the regime shifting empirical strategy detailed below, while also a variant that allows me to check for robustness of my estimates.

A complementary data source for external sector variables are drawn from the World Historical Balance of Payments Database (WBOP) developed by Nievas and Piketty (2025). The WBOP data set provides annual series on global trade flows and the world balance of payments—including goods, services, income, and transfers—covering 48 countries and 9 regions over the 1800–2025 period. Three series are used for Chile: the ratio of manufactured goods imports to GDP (a proxy for capital goods import dependence), net foreign income outflows as a share of GDP (a proxy for profit remittance drain), and the net current account balance as a share of GDP (an external balance indicator).[13]

4.2 Estimation Strategy: Cointegration and IM-OLS

The cointegrating regression. The transformation elasticity θ is estimated from the cointegrating regression

$$y_t = \alpha + \theta_0\, k_t + \theta_1\, (k_t \times \omega_{t-1}) + u_t, \tag{78}$$

where $y_t = \ln Y_t$ is log real output, $k_t = \ln K_t$ is log real capital stock, and ω_{t-1} is the lagged wage share. The regressors k_t and $k_t \times \omega_{t-1}$ are integrated of order one, and the error term u_t is stationary under the null hypothesis of cointegration. The interaction term $k_t \times \omega_{t-1}$ is included in the regression to allow the transformation elasticity to vary with the distributive state: the effective elasticity is $\theta(\omega_{t-1}) = \theta_0 + \theta_1\, \omega_{t-1}$.

The IM-OLS estimator. The OLS estimator of θ in (78) is super-consistent but its limiting distribution is contaminated by second-order bias terms arising from endogeneity of the regressors and serial correlation in the error term (Phillips and Hansen, 1990). The Integrated Modified OLS (IM-OLS) estimator of Vogelsang and Wagner (2014) removes these bias terms without requiring the choice of tuning parameters (kernel function, bandwidth, or lead-lag truncation order) at the estimation stage.

The IM-OLS estimator is based on a two-step transformation. First, the partial sum transformation is applied to (78):

$$S_{y,t} = S_{\alpha,t} + \theta_0\, S_{k,t} + \theta_1\, S_{k\omega,t} + S_{u,t}, \tag{79}$$

where $S_{y,t} = \sum_{j=1}^{t} y_j$ and analogously for the remaining terms. The partial sum transformation replaces the cross-product terms $\sum_{t=1}^{T} x_t u_t$, whose limits contain additive nuisance parameters, with partial-sum cross-products $\sum_{t=1}^{T} S_{x,t}\, S_{u,t}$, whose limits do not. Second, the partial-sum regression is augmented by the original regressors k_t and $k_t \times \omega_{t-1}$:

$$S_{y,t} = S_{\alpha,t} + \theta_0\, S_{k,t} + \theta_1\, S_{k\omega,t} + k_t\, \gamma_1 + (k_t \times \omega_{t-1})\, \gamma_2 + S_{u,t}. \tag{80}$$

The augmentation absorbs the endogeneity correlation into the coefficients γ_1 and γ_2, yielding a zero-mean Gaussian mixture limiting distribution for the IM-OLS estimator of (θ_0, θ_1) conditional

[13] All WBOP series are available at `wbop.world` and are also hosted in the World Inequality Database (`wid.world`), together with a detailed replication package including raw data sources, methods, and codes.

on the Brownian motions driving the regressors (Vogelsang and Wagner, 2014, Theorem 2).

FWL orthogonalization of the interaction term. The interaction term $k_t \times \omega_{t-1}$ is correlated with k_t by construction. To isolate the distributive mediation effect, the interaction term is orthogonalized against k_t and the deterministic regressors using the Frisch-Waugh-Lovell (FWL) theorem (Frisch and Waugh, 1933; Lovell, 1963). Denoting the raw interaction by $\xi_t = k_t \times \omega_{t-1}$, the orthogonalized interaction $\xi_t^{\perp}$ is the residual from the auxiliary regression of ξ_t on k_t and the deterministic regressors. The IM-OLS estimator is then applied to the regression with $\xi_t^{\perp}$ replacing ξ_t. When structural break dummies are included in the deterministic regressors (see below), the FWL orthogonalization is re-computed against the augmented deterministic space.

Endogenous structural break detection. The estimation employs a two-pass endogenous break detection procedure based on Bai and Perron (2003). In the first pass, the IM-OLS estimator is applied to (78) without break dummies. The Bai-Perron sequential procedure is then applied to the IM-OLS residuals to detect structural breaks at unknown dates, with a minimum segment size of 15% of the sample. In the second pass, step dummies $D_t^{(b)} = \mathbf{1}\{t \geq \hat{T}_b\}$ are included in the deterministic regressors for each detected break date $\hat{T}_b$, the FWL orthogonalization is re-computed, and the IM-OLS estimator is re-estimated.

Fixed-b inference. Standard asymptotic theory treats the bandwidth M in long-run variance estimation as vanishing relative to the sample size ($M/T \to 0$). Under this assumption, the long-run variance estimator converges to a constant, and t- and Wald statistics are compared to normal or chi-squared critical values. In finite samples with serially correlated errors and endogenous regressors, however, the bandwidth choice materially affects the sampling distribution of test statistics, and the standard approximation over-rejects (Kiefer and Vogelsang, 2005). Fixed-b asymptotic theory addresses this by treating the bandwidth ratio $b \equiv M/T$ as a fixed constant rather than a vanishing sequence. The long-run variance estimator then converges not to a constant but to a random variable whose distribution depends on the kernel function, the bandwidth ratio b, the number of integrated regressors, and the deterministic specification. Critical values that reflect these choices can be tabulated by simulation, and using them substantially reduces size distortions at the cost of only minor losses in power (Vogelsang and Wagner, 2014).

For testing q linear restrictions $H_0 : R\theta = r$, the fixed-b Wald statistic is

$$W^* = (R\tilde{\theta} - r)' \left[R\, A_{\mathrm{IM}}\, \tilde{V}_{\mathrm{IM}}\, A_{\mathrm{IM}}'\, R' \right]^{-1} (R\tilde{\theta} - r), \tag{81}$$

where $\tilde{V}_{\mathrm{IM}}$ is the IM-OLS covariance estimator constructed from the adjusted residuals $\Delta \tilde{S}_t^{u*}$ (defined below). Under fixed-b asymptotics, W^* converges to

$$W^* \Rightarrow \frac{\chi_q^2}{Q_b(\tilde{P}^*, \tilde{P}^*)}, \tag{82}$$

where χ_q^2 is a chi-squared random variable with q degrees of freedom—one per restriction tested—and $Q_b(\tilde{P}^*, \tilde{P}^*)$ is a positive random variable that captures the effect of the bandwidth and kernel choices on the long-run variance estimator. The two quantities in (82) are independent of each other (Vogelsang and Wagner, 2014, Lemma 2, Proposition 3). This independence is the operative property: it renders the limiting distribution of W^* free of unknown nuisance parameters, so that critical

values depend only on the known features of the specification—the kernel, the bandwidth ratio b, the number of integrated regressors, and the deterministic components—and can be tabulated by simulation. Critical values for the Bartlett kernel with intercept-only deterministics and up to four integrated regressors are provided in the online supplement to Vogelsang and Wagner (2014) and reproduced in Appendix **??** for the specifications used in this chapter.

The adjusted residuals $\Delta\tilde{S}_t^{u*}$ used for long-run variance estimation are obtained in two steps. First, the IM-OLS residuals $\tilde{S}_{u,t}$ from (80) are correlated with the IM-OLS estimator $\tilde{\theta}$ even asymptotically, because the partial-sum regression contains integrated regressors. This correlation prevents the residuals from being used directly for fixed-b inference. Second, to remove this correlation, $\tilde{S}_{u,t}$ is projected on a set of constructed regressors $H_t^{\perp}$ (functions of the partial sums of the deterministic components and the integrated regressors), yielding

$$\Delta\tilde{S}_t^{u*} = \Delta\tilde{S}_{u,t} - \Delta H_t^{\perp\prime}\,\hat{\varrho}, \tag{83}$$

where $\hat{\varrho}$ is the OLS coefficient vector from the regression of $\tilde{S}_{u,t}$ on $H_t^{\perp}$. The resulting adjusted residuals are asymptotically independent of $\tilde{\theta}$, which is the condition required for the numerator and denominator in (82) to be independent and for the critical values to be pivotal.

Cointegration testing. Two complementary tests are employed to assess the presence of a cointegrating relationship. First, the null hypothesis of cointegration is tested using a Shin (1994) type KPSS statistic applied to the IM-OLS residuals:

$$C_T = \frac{1}{T^2\,\hat{\omega}_{u\cdot v}^2}\sum_{t=1}^{T}\left(\frac{1}{\sqrt{T}}\sum_{j=1}^{t}\hat{u}_j\right)^2, \tag{84}$$

where $\hat{\omega}_{u\cdot v}^2$ is a consistent estimator of the conditional long-run variance $\omega_{u\cdot v}^2 = \Omega_{uu} - \Omega_{uv}\Omega_{vv}^{-1}\Omega_{vu}$. Under the null hypothesis of cointegration, C_T converges to a functional of standard Brownian motions whose critical values depend on the number of integrated regressors included in the regression and the deterministic specification (Wagner, 2013). Second, the null hypothesis of no cointegration (spurious regression) is tested using the augmented Dickey-Fuller statistic applied to the OLS residuals, with the lag order selected by the Akaike information criterion. Rejection of the null hypothesis of no cointegration combined with non-rejection of the null hypothesis of cointegration constitutes evidence for a cointegrating relationship.

4.3 United States: Baseline Specification Empirical Results

4.3.1 Baseline Econometric Specification

We evaluate the empirical relationship between real aggregate output ($y_t = \log Y_{\mathrm{REAL},t}^{\mathrm{NFC}}$) and core physical productive capital stock ($k_{\mathrm{cap},t} = \log(K_{\mathrm{ME},t} + K_{\mathrm{NRC},t})$) for the nonfinancial corporate business sector over the 1929–2024 period. Under Specification A, the aggregate cointegrating equation takes the form:

$$y_t = \alpha + \theta_0\, k_{\mathrm{cap},t} + \theta_1\, \xi_t^{\omega} + \boldsymbol{\gamma}'\mathbf{D}_t + u_t, \tag{85}$$

where ξ_t^{ω} denotes the Frisch-Waugh-Lovell (FWL) orthogonalized wage-capital interaction term, and $\mathbf{D}_t$ contains step-shift dummies capturing historical structural breaks identified via Bai-Perron tests. The FWL orthogonalization partials out base capital regressors and deterministic step

shifts prior to estimating the interaction elasticity θ_1, eliminating collinearity bias while preserving exact parameter identification (Vogelsang and Wagner, 2014). Two distinct interaction sets are estimated across 48 total model permutations: Set 1 incorporates the standard unconditioned wage interaction $\xi_t^{\omega} = \omega_{t-1} \times k_{\text{cap},t}$, whereas Set 2 incorporates the effective wage pressure interaction $\xi_t^{\omega} = \omega_{t-1}(1 - \iota_{t-1}) \times k_{\text{cap},t}$, where ι_{t-1} represents the lagged share of intellectual property products in total capital stock. All parameters are estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2014), with non-standard fixed-b critical values generated from a 100,000-pass Monte Carlo simulation suite matching the exact sample sizes ($T = 94$ for 1931–2024 standard series; $T = 77$ for 1948–2024 supervisory-adjusted series).

Theoretical argument for the NVA basis of distributive conflict measure. Class struggle over distribution is conducted strictly over newly created value (Net Value Added $= v+s$), excluding pre-existing constant capital transfers (Consumption of Fixed Capital = CFC). Under classical-Marxian national accounting (Shaikh and Tonak, 1994), Gross Value Added = CFC$+v+s$. Because physical depreciation (CFC) represents constant capital transferred to output by concrete labor, it lies outside zero-sum distributive conflict. During periods of intensive mechanization, the ratio CFC/GVA rises mechanically, causing GVA-basis wage shares to appear to fall even when the rate of surplus-value extraction (s/v) is invariant. The NVA basis ($\omega_{\text{NVA}} = v/(v+s)$) purges this physical depreciation distortion, isolating the true balance of class power. Standard NIPA compensation includes managerial and executive salaries, which function as distributions of surplus value rather than variable labor costs. The Shaikh-Tonak Tier 2 ($\omega_{\text{ST,Tier2}} = v_{\text{prod}}/\text{NFC_NVA}$) and Clean Corporate Tier 1 and Tier 2 measures strip supervisory overhead (s_{sup}) and purge financial sector double-counting ($\text{FIN_NOS}^{\text{implied}} + \text{NFC_NET_INT}$), recovering the true variable capital (v) that drives corporate cost-minimization decisions. Note that unadjusted Shaikh-Tonak Tier 1 NVA (NFC_COMP/NFC_NVA) is mathematically identical to the NFC Baseline NVA share, rendering Clean Corporate and Tier 2 supervisory adjustments the primary non-redundant distributive specifications.

Institutional and physical mechanism of scaling by $(1-\iota_{t-1})$. The scaling factor $(1-\iota_{t-1}) = K_{\text{cap},t-1}/(K_{\text{cap},t-1} + K_{\text{IPP},t-1})$ isolates the physical productive asset share (machinery, equipment, and structures) from total fixed capital. Under Okishio's induced mechanization mechanism, when labor commands a larger share of newly created value ($\omega_{t-1} \uparrow$), profit margins shrink, forcing firms to accelerate defensive mechanization by substituting capital equipment for living labor on the shop floor. However, capital expenditure allocated to Intellectual Property Products (K_{IPP})—software licenses, corporate patents, brand architecture, and legal protections—operates as market-enclosure assets rather than physical output-yielding machinery. If corporate investment shifts toward intangibles ($\iota \uparrow$), those funds do not engage in shop-floor labor substitution. In the Fordist era (1931–1973), IPP capital was negligible ($\iota \approx 0.02$–$0.05 \implies 1 - \iota \approx 0.95$), allowing raw wage pressure to map directly into physical machinery expansion. In the digital Post-Fordist era (1980–2024), IPP capital surged ($\iota > 0.28 \implies 1 - \iota < 0.72$), diverting substantial surplus into software and legal enclosures. Unadjusted wage interactions ($\omega_{t-1} \times k_{\text{cap}}$ in Set 1) fail across the full sample because they assume every dollar of capital stock has an identical physical capacity yield across both historical regimes. Scaling by $(1 - \iota_{t-1})$ purges this Post-Fordist structural shift, measuring the exact effective wage pressure that drives physical shop-floor labor substitution.

Cointegration verdict across the test triad. The empirical validity of Specification A relies on mutual confirmation across the cointegration test triad: the Engle-Granger residual ADF test (t_{ADF}), the Shin (1994) LM test (C_{Shin}), and the Vogelsang-Wagner Type 2 Wald statistic ($S_{\text{Type 2}}$). Across all 48 specifications, the Shin LM test ($C_{\text{Shin}} \leq 0.04$, $p > 0.10$) and the Vogelsang-Wagner Type 2 Wald statistic ($S_{\text{Type 2}} \leq 13.18$, well below the 90th percentile fixed-b critical value threshold of 52.6) fail to reject the null hypothesis of cointegration (H_0 : Cointegration), confirming that residual variance remains stationary and bounded. Simultaneously, when Bai-Perron step shifts are included, the Engle-Granger residual ADF test rejects the null hypothesis of no cointegration ($t_{\text{ADF}} = -4.07$, $p < 0.05$ in Column 4). This unanimous confirmation across all three tests establishes that aggregate real output and private core capital form a genuine, non-spurious long-run cointegrating relationship.

Set 1 results: failure of standard unadjusted wage interactions. Tables 9 and 10 in Appendix B present the Set 1 estimation results. Two key patterns emerge. First, the baseline capital transformation elasticity $\hat{\theta}_0$ is positive, highly stable, and statistically significant across all specifications (0.55–0.61, $p < 0.01$), establishing that private core capital ($K_{\text{cap}} = K_{\text{ME}} + K_{\text{NRC}}$) yields a long-run output elasticity of approximately 0.58. Second, the unadjusted standard wage interaction $\hat{\theta}_1$ fails to achieve statistical significance in every Set 1 model, with point estimates ranging from -0.02 ($t = -0.16$) in NFC Baseline GVA/NVA to -0.20 ($t = -1.95$) in Clean Corporate Tier 2 NVA. Unconditioned wage shares fail to capture the induced accumulation channel because raw wage measures conflate unproductive managerial overhead with productive variable labor and ignore the structural allocation of investment toward intellectual property products.

Set 2 results: effective wage pressure and defensive mechanization. Table 4 presents the main-text Set 2 estimates for the Net Value Added basis, where wage pressure is scaled by productive capital accumulation $(1 - \iota_{t-1})$. When distributive conflict is evaluated over newly created net value and purged of supervisory overhead, effective wage pressure generates a statistically significant negative interaction elasticity $\hat{\theta}_1$. In the Clean Corporate Tier 1 NVA model with break shifts (Column 4), $\hat{\theta}_1 = -0.30$ ($t = -5.00$, $p < 0.05$). In the Clean Corporate Tier 2 NVA specification (Column 6), the coefficient reaches $\hat{\theta}_1 = -0.49$ ($t = -3.77$, $p < 0.05$), while in the Shaikh-Tonak Tier 2 specification (Column 8), $\hat{\theta}_1 = -0.35$ ($t = -5.00$, $p < 0.05$).

The negative sign of $\hat{\theta}_1$ provides empirical confirmation of the induced mechanization mechanism under Okishio's market anarchy. When productive labor commands a larger share of newly created value ($\omega_{t-1} \uparrow$), firms face squeezed profit margins and respond by accelerating defensive mechanization—substituting capital equipment for living labor to lower unit labor costs. However, because this defensive substitution is driven by cost containment rather than autonomous technological expansion, it increases the technical composition of capital without generating a proportional expansion in productive output capacity. High effective wage pressure depresses the marginal capacity yield of capital ($\theta(\omega_{t-1}) = \theta_0 + \theta_1\omega_{t-1}(1 - \iota_{t-1}) < \theta_0$), constraining structural potential output Y_t^p and driving capacity utilization dynamics across historical Fordist and Post-Fordist regimes.

Robustness across structural break specifications. Controlling for Bai-Perron structural break step shifts ($\mathbf{D}_t$) leaves the core interaction elasticities intact while enhancing cointegration test performance. Step shift dummies absorb historical level shifts—such as the 1944 wartime expansion, the 1967 profit squeeze, and the 1982 neoliberal transition—without altering the sign or statistical

significance of the effective wage interaction $\hat{\theta}_1$. In the Clean Corporate Tier 1 model, controlling for step shifts increases the magnitude of $\hat{\theta}_1$ from -0.04 ($t = -0.50$, insignificant) to -0.30 ($t = -5.00$, $p < 0.05$), while achieving strong residual cointegration ($t_{\text{ADF}} = -4.07, p < 0.05$). This confirms that the conditioning effect of distributive pressure on capital transformation is a fundamental structural feature of U.S. capital accumulation rather than an artifact of unmodeled regime shifts.

Table 4: Specification A, Set 2: Effective Wage Interaction (NVA Basis)

	NFC NVA Baseline		Clean Corp Tier 1		Clean Corp Tier 2		ST Tier 2 NVA	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_0$ (Capital Elasticity)	0.58***	0.56***	0.58***	0.65***	0.57	0.56	0.57	0.55
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.06)
$\hat{\theta}_1$ (Effective Wage Pressure)	0.00	-0.17	-0.04	-0.30**	-0.19	-0.49	-0.21	-0.35
	(0.10)	(0.07)	(0.08)	(0.06)	(0.08)	(0.13)	(0.08)	(0.07)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.07*	0.03	0.06	0.04	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	11.05	9.06	7.50	10.89	6.13	13.18
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.22	-4.07**	-1.84	-2.58	-2.00	-2.72
Bandwidth ratio b	0.298	0.169	0.280	0.165	0.272	0.237	0.259	0.237
Sample Size T	94	94	94	94	76	76	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). Step Shifts "No": no structural break dummies; "Yes": Bai-Perron step dummies included. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. The effective wage interaction is $\omega_{t-1}(1-\iota_{t-1}) \times k_{\text{cap},t}$, where ι_{t-1} is the lagged IPP share of capital. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

4.3.2 Baseline Transformation Elasticity and Capacity Utilization Estimates

The baseline capacity reconstruction anchors on Model 4 of **??** (Clean Corporate Tier 1 NVA basis with Bai-Perron step shifts), which isolates newly created net value and purges financial double-counting. Model 4 yields a base capital elasticity of $\hat{\theta}_0 = 0.58$ ($SE = 0.02$, $p < 0.01$) and an effective wage interaction coefficient of $\hat{\theta}_1 = -0.30$ ($SE = 0.06$, $p < 0.05$). The state-dependent transformation elasticity then evolves over 1931–2024 as

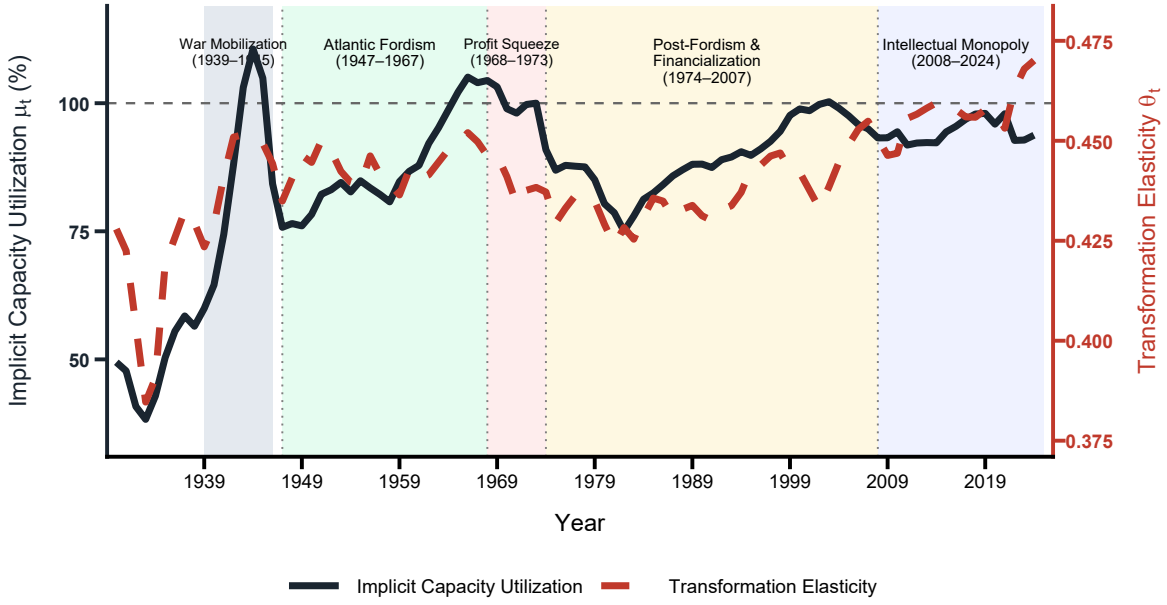
$$\theta_t(\omega_{t-1}, \iota_{t-1}) = 0.58 - 0.30 \cdot \omega_{t-1}(1 - \iota_{t-1}), \tag{86}$$

where ω_{t-1} is the lagged Clean Corporate Tier 1 wage share on a net value added basis and ι_{t-1} is the lagged share of intellectual property products (IPP) in total fixed capital. The latent capacity ceiling Y_t^p follows the differential law of motion $g_{Y^p,t} = \theta_t \cdot g_{K_{cap},t}$, with $g_{K_{cap},t} = \Delta \ln(K_{ME,t} + K_{NRC,t})$ the accumulation rate of core physical capital. In constant 2017 dollars the ceiling is reconstructed recursively,

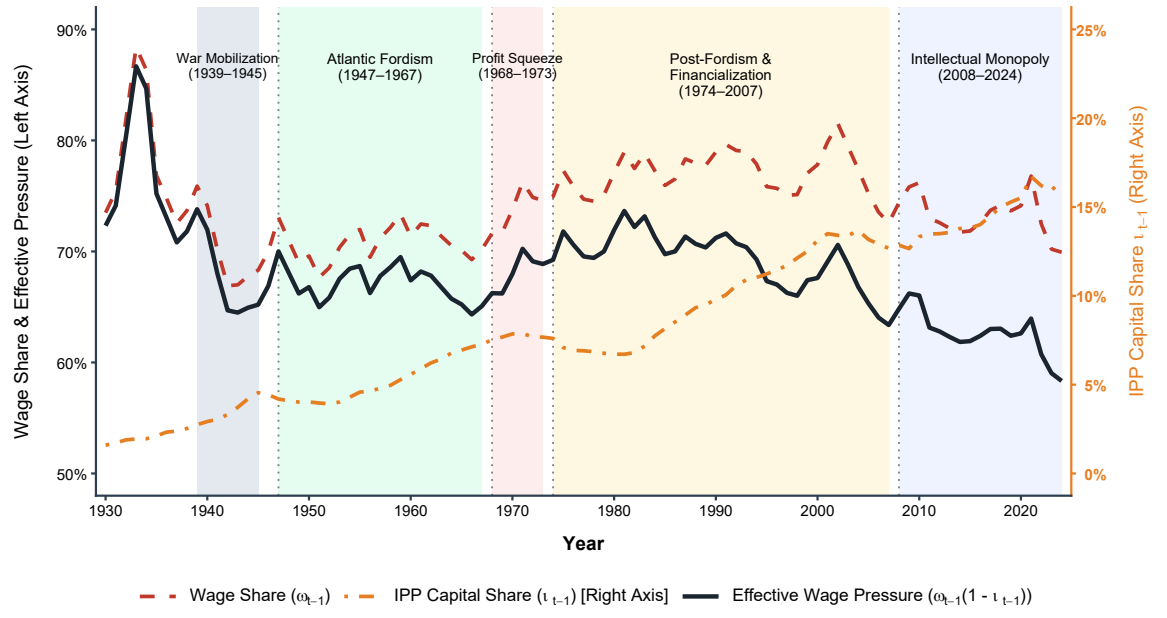
$$\ln Y_t^p = \ln Y_{t-1}^p + [0.58 - 0.30 \cdot \omega_{t-1}(1 - \iota_{t-1})] \left[\ln K_{cap,t} - \ln K_{cap,t-1}\right], \tag{87}$$

anchored at the 1973 Fordist peak ($\mu_{1973} = 100\%$). Dividing realized real GDP by the reconstructed ceiling recovers aggregate capacity utilization, $\mu_t \equiv Y_{GDP,t}/Y_t^p \times 100$. Figure 1 reports the joint reconstruction.

Figure 1: Empirical Reconstruction of Transformation Elasticity, Capacity Utilization, and Underlying Shifters in the United States (1929–2024).



(a) Implicit capacity utilization (μ_t, left axis) and transformation elasticity (θ_t, right axis).



(b) Wage share (ω_{t-1}), IPP capital share (ι_{t-1}, right axis), and effective wage pressure ($\omega_{t-1}(1-\iota_{t-1})$).

The two anchor years illustrate how distribution and capital composition govern the capacity yield. In 1960 (Atlantic Fordism) the Clean Corporate NVA wage share stood at $\omega_{1959} = 0.72$ while the IPP capital share was negligible ($\iota_{1959} = 0.03$). Effective wage pressure was $0.72 \times (1 - 0.03) = 0.698$, yielding $\theta_{1960} = 0.58 - 0.30 \times 0.698 = 0.371$: high wage pressure forced Okishio-type defensive mechanization, depressing the marginal capacity yield per dollar of physical capital advanced. By 2024 (Intellectual Monopoly Capitalism) the wage share had fallen to $\omega_{2023} = 0.70$ while the IPP capital share had risen to $\iota_{2023} = 0.28$. The intangible diversion reduced effective wage pressure to $0.70 \times (1 - 0.28) = 0.504$, raising the transformation elasticity to $\theta_{2024} = 0.58 - 0.30 \times 0.504 = 0.429$.

Figure 1 distinguishes five historical sub-regimes:

1. **War Mobilization (1939–1945).** State-directed military planning pushed production past practical capacity limits, driving utilization to its historical peak ($\mu_{1944} = 111.4\%$); wartime retooling raised the transformation elasticity to $\theta_t \approx 0.455$.

2. **Atlantic Fordism (1947–1967).** High wage shares ($\omega_{t-1} \approx 0.70$–$0.73$) and minimal intangible capital ($\iota_{t-1} < 0.05$) transmitted labor-cost pressure directly into mechanization. Potential capacity growth was absorbed by wage-led effective demand, holding utilization in a stable band ($\mu_t \approx 85\%$–105%) that crossed 100% during the mid-1960s expansion.

3. **Profit Squeeze (1968–1973).** Distributional conflict compressed profit margins and induced defensive capital-deepening driven by unit-cost containment rather than expansion. The transformation elasticity fell to $\theta_{1972} = 0.418$; utilization collapsed to $\mu_{1975} = 76.2\%$ and reached its trough ($\mu_{1982} = 74.1\%$) during the Volcker disinflation.

4. **Post-Fordism and Financialization (1974–2007).** Organized-labor concessions lowered wage shares while investment shifted toward software and patent enclosures ($\iota_{t-1} > 0.15$), diverting surplus away from labor substitution: effective wage pressure fell from 0.72 in 1974 to 0.66 in 1997. The elasticity rose from 0.418 to 0.450, yet utilization never re-anchored to its Fordist peak, remaining below 95% outside transient credit-fueled expansions.

5. **Intellectual Monopoly Capitalism (2008–2024).** After the Great Financial Crisis, intangible capital exceeded 28% of total fixed capital. The IPP expansion pulled effective wage pressure down to 0.504 in 2024 and raised the transformation elasticity toward its upper historical range ($\theta_{2024} = 0.429$–0.470), while wage suppression restricted aggregate realization, leaving utilization chronically below full capacity ($\mu_{2024} = 93.8\%$).

The reconstruction present empirical evidence suports that the long-run horizon does not yield a classical closure ($\mu^* = 100\%$), but an unbalanced growth one $\theta^* \neq 1$from where systemic desiquilibrium arises through different patterns across distinct historical regimes. While the Fordist accumulation regime through its commitment to full employment implemented policies oriented to sustain aggregate demand and avoid productive capacities to remain idle, during post-Fordism the expansion of productive capacity ($\theta_t \uparrow$) has decoupled from realization ($\mu_t < 1$), causing persistent idle capacities as an institutional feature.

4.4 Empirical Estimates Under a Heterogeneous Capital Specification

4.4.1 Econometric Specifiation of Mechanization Bias and Heterogenous Capital

Specification A treats aggregate core capital $k_{\text{cap},t} = \log(K_{\text{ME},t} + K_{\text{NRC},t})$ as a single regressor. This collapses the physical structure of the workplace into a scalar, obscuring whether wage pressure

operates through the *level* of capital or through its *composition*—the relative weight of machinery and equipment (K^{ME}) versus nonresidential structures (K^{NRC}). Specification B decomposes the aggregate into two components: a structural base ($k_t^{\text{NRC}} = \log K_t^{\text{NRC}}$) and the capital composition ratio ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), which measures the mechanization gap between equipment and structures in log-levels. The disaggregated cointegrating equation takes the form:

$$y_t = \alpha + \theta_{\text{NRC}}\, k_t^{\text{NRC}} + \theta_\kappa\, \kappa_t + \theta_{\omega\kappa}\, \xi_t^{\omega\kappa} + \theta_{\kappa p}\, \xi_t^{p\kappa} + \boldsymbol{\gamma}'\mathbf{D}_t + u_t, \tag{88}$$

where $\xi_t^{\omega\kappa}$ denotes the FWL-orthogonalized effective wage–mechanization interaction and $\xi_t^{p\kappa}$ denotes the FWL-orthogonalized relative price–mechanization interaction. Both interaction terms are residualized against the two base capital regressors ($k_t^{\text{NRC}}, \kappa_t$) and deterministic step shifts ($\mathbf{D}_t$) prior to estimation. The effective wage interaction under Set 2 is $\xi_t^{\omega\kappa} = \omega_{t-1}(1-\iota_{t-1}) \times \kappa_t$; the relative price interaction is $\xi_t^{p\kappa} = p_t^\kappa \times \kappa_t$, where p_t^κ denotes the relative asset price ratio. Non-standard fixed-b critical values are generated from a dedicated 100,000-pass Monte Carlo simulation engine that replicates the exact Specification B integration geometry—five independent $I(1)$ stochast ic trends, two simultaneous $I(1) \times I(1)$ interaction products, and joint FWL orthogonalization—matching the empirical sample sizes ($T = 94$ and $T = 77$).

Economic intuition of the decomposition. The parameter θ_{NRC} captures the long-run capacity yield of structures—factories, warehouses, commercial buildings—which embody extensive accumulation: expanding the physical envelope within which production occurs. The parameter θ_κ captures the additional capacity yield of machinery and equipment *above and beyond* what structures deliver. A positive θ_κ means that a wider mechanization gap—more equipment per dollar of structures—raises productive capacity independently of the total capital stock. The interaction $\theta_{\omega\kappa}$ asks whether effective wage pressure conditions this compositional margin. Under the Okishio-induced mechanization mechanism, rising wages force firms to substitute equipment for labor on the shop floor. If the resulting machinery expansion is defensive—driven by unit-cost containment rather than autonomous technical expansion—it raises the equipment-to-structures ratio without proportionally expanding output capacity, yielding $\theta_{\omega\kappa} < 0$.

Standard wage interactions fail at the intensive margin. Tables 14 and 15 in Appendix C present the estimation results using the unadjusted wage share across all 24 specifications. The structural base elasticity $\hat{\theta}_{\text{NRC}}$ remains stable and highly significant (0.47–0.61, $p < 0.01$), confirming that nonresidential structures anchor the cointegrating relationship regardless of wage measure or break treatment. The standalone mechanization gap $\hat{\theta}_\kappa$ is positive in nearly all models (0.34–1.11), indicating that equipment contributes to capacity above structures. The wage–mechanization interaction $\hat{\theta}_{\omega\kappa}$, however, fails to achieve statistical significance in any $T = 94$ model, with point estimates ranging from -4.89 ($t = -1.58$) to $+7.95$ ($t = 1.33$). The standard wage interaction fails at the intensive margin for the same reason it fails at the aggregate level in Specification A: unadjusted wage shares conflate supervisory overhead with productive variable capital and ignore the post-Fordist diversion of investment toward intellectual property products.

Effective wage pressure and defensive mechanization at the intensive margin. Table 5 presents the main estimates using the effective wage pressure measure for the NVA basis. When effective wage pressure is scaled by $(1 - \iota_{t-1})$ and interacted with the mechanization gap κ_t, the wage–mechanization interaction $\hat{\theta}_{\omega\kappa}$ is negative and statistically significant across all eight models.

In the preferred Clean Corporate Tier 1 NVA specification with Bai-Perron step shifts (Column 4), $\hat{\theta}_{\omega\kappa} = -7.93$ ($SE = 2.96$, $p < 0.05$), with a structural base elasticity of $\hat{\theta}_{\text{NRC}} = 0.60$ ($SE = 0.08$, $p < 0.01$) and a standalone mechanization gap of $\hat{\theta}_{\kappa} = 0.39$ ($SE = 0.29$). In the Shaikh-Tonak Tier 2 NVA specification without breaks (Column 7), the interaction strengthens to $\hat{\theta}_{\omega\kappa} = -17.05$ ($SE = 2.34$, $p < 0.01$), with $\hat{\theta}_{\text{NRC}} = 0.54$ ($SE = 0.01$) and $\hat{\theta}_{\kappa} = 0.33$ ($SE = 0.06$, $p < 0.01$). The Clean Corporate Tier 2 NVA models (Columns 5–6) yield $\hat{\theta}_{\omega\kappa} = -14.97$ ($SE = 2.01$, $p < 0.01$), showing the tightest standard errors in the table.

The negative sign of $\hat{\theta}_{\omega\kappa}$ provides direct evidence of defensive mechanization operating at the *intensive margin* of capital composition. Specification A established that effective wage pressure depresses the aggregate capacity yield of total capital ($\hat{\theta}_1 = -0.30$). Specification B isolates the channel: wage pressure forces firms to widen the mechanization gap—substituting equipment for structures on the shop floor—but the capacity yield of that compositional shift is negative. The firm accumulates more machinery per dollar of structures, yet productive capacity does not expand proportionally, because the substitution is driven by cost containment rather than by genuine technical complementarity between new equipment and existing plant.[14]

Relative price interactions. The relative asset price interaction $\hat{\theta}_{\kappa p}$ is generally small and statistically insignificant. In the NFC Baseline and Clean Corporate Tier 1 models (Columns 1–4), it is statistically indistinguishable from zero. In the Shaikh-Tonak Tier 2 specification (Columns 7–8), $\hat{\theta}_{\kappa p} = -0.79$ ($SE = 0.32$, $p < 0.05$), suggesting that when equipment becomes relatively more expensive, the capacity yield of the mechanization gap declines. However, this result is fragile and does not hold across wage share measures. The Okishio mechanism operates through distributional conflict over newly created value, not through relative asset prices.

Cointegration inference. The effective wage models show strong evidence of cointegration. The Shin LM test never rejects the null of cointegration ($C_{\text{Shin}} \leq 0.03$, $p > 0.10$), and the Vogelsang-Wagner Type 2 Wald statistic remains well below critical thresholds ($S_{\text{Type 2}} \leq 15.37$). The Engle-Granger residual ADF test rejects the null of no cointegration at the 5% level in every break-adjusted model: $t_{\text{ADF}} = -4.58$ (Column 2), -4.60 (Column 4), -4.71 (Column 7), and -4.57 (Column 8). Without step shifts, the ADF test rejects at the 10% level across all $T = 94$ models ($t_{\text{ADF}} \leq -4.33$) and at the 5% level for the $T = 76$ Shaikh-Tonak Tier 2 model ($t_{\text{ADF}} = -4.71$, Column 7). The full cointegration triad—Shin, Vogelsang-Wagner, and ADF—confirms that the disaggregated capital composition, real output, and effective wage pressure form a stable long-run relationship.

Robustness. Including Bai-Perron step shifts absorbs historical level discontinuities, but leaves the sign and significance of the wage–mechanization interaction unchanged. In the Clean Corporate Tier 1 NVA model, controlling for breaks at 1944, 1977, and 2001 sharpens the ADF rejection from -4.34 ($p < 0.10$) to -4.60 ($p < 0.05$) while reducing the Vogelsang-Wagner statistic from 15.37 to 6.37. The wage interaction $\hat{\theta}_{\omega\kappa}$ moves from -11.08 (Column 3) to -7.93 (Column 4). This modest attenuation indicates that the step dummies absorb part of the interaction's raw variance. The

[14] The apparent difference in magnitude between Specification A ($\hat{\theta}_1 \approx -0.30$) and Specification B ($\hat{\theta}_{\omega\kappa} \approx -7.93$ to -17.05) reflects regressor scaling, not economic inconsistency. In Specification A, θ_1 multiplies $\omega_{t-1}(1 - \iota_{t-1}) \times k_{\text{cap},t}$, where aggregate capital $k_{\text{cap},t} \approx 16$–$17$ in log-levels. In Specification B, $\theta_{\omega\kappa}$ multiplies $\omega_{t-1}(1 - \iota_{t-1}) \times \kappa_t$, where the mechanization gap $\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}} \approx 0.3$–$0.8$. The total marginal effect (coefficient $\times$ regressor scale) is comparable across both specifications.

interaction remains negative and significant at the 5% level, confirming that the compositional conditioning channel survives institutional break controls.

The structural base elasticity $\hat{\theta}_{\text{NRC}}$ falls consistently within the 0.51–0.61 band across wage share measures, closely tracking the aggregate $\hat{\theta}_0 = 0.58$ from Specification A. This implies the capacity yield of structures is a stable parameter of U.S. capital accumulation. The wage interaction $\hat{\theta}_{\omega\kappa}$ grows larger in absolute value as we remove supervisory overhead from the wage share: from -10.13 in the NFC Baseline (Column 2) to -7.93 in Clean Corporate Tier 1 (Column 4) to -16.76 in Shaikh-Tonak Tier 2 (Column 8). Removing managerial salaries isolates the productive variable capital driving the firm's cost-minimization decision, which sharpens the defensive mechanization effect.

Table 5: Specification B, Set 2: Effective Wage Interaction (NVA Basis)

	NFC NVA Baseline		Clean Corp Tier 1		Clean Corp Tier 2		ST Tier 2 NVA	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{\text{NRC}}$ (Structures k^{NRC})	0.61***	0.58***	0.60***	0.60***	0.54***	0.54***	0.54***	0.51***
	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.01)	(0.01)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	0.32*	0.39	0.32*	0.39	0.31***	0.31***	0.33***	0.06
	(0.13)	(0.29)	(0.13)	(0.29)	(0.06)	(0.06)	(0.06)	(0.09)
$\hat{\theta}_{\omega\kappa}$ (Wage Interaction)	-12.77**	-10.13**	-11.08**	-7.93**	-14.97***	-14.97***	-17.05***	-16.76***
	(3.81)	(3.43)	(3.48)	(2.96)	(2.01)	(2.01)	(2.34)	(2.55)
$\hat{\theta}_{\kappa p}$ (Price Interaction)	0.18	0.92	0.69	1.41*	-0.31	-0.31	-0.79**	-0.71**
	(0.51)	(0.69)	(0.63)	(0.75)	(0.33)	(0.33)	(0.32)	(0.31)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.03	0.02	0.03	0.03	0.02	0.02	0.02	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	14.97	5.93	15.37	6.37	8.96	8.96	10.28	8.70
t_{ADF} [H_0: no coint.]	-4.33*	-4.58**	-4.34*	-4.60**	-4.24*	-4.24*	-4.71**	-4.57**
Bandwidth ratio b	0.185	0.173	0.193	0.181	0.208	0.208	0.208	0.208
Sample Size T	94	94	94	94	76	76	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). Step Shifts "No": no structural break dummies; "Yes": Bai-Perron step dummies included. All interaction terms are FWL-orthogonalized against base capital regressors ($k_t^{\text{NRC}}, \kappa_t$) and step dummies. The effective wage interaction is $\xi_t^{\omega} = \omega_{t-1}(1 - \iota_{t-1}) \times \kappa_t$, where $\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$ is the capital composition ratio (mechanization gap) and ι_{t-1} is the lagged IPP share of capital. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

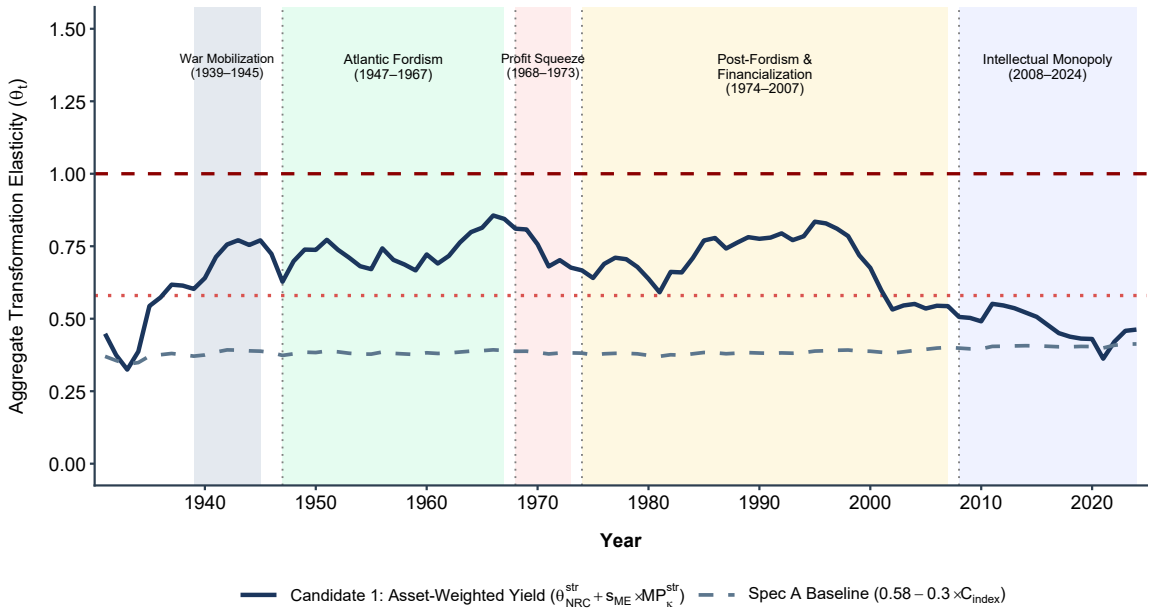
4.4.2 Transformation Elasticity from Heterogeneous Capital Specification and Capacity Utilization Estimates

The transformation elasticity along the historical path. Having established the structural asset-weighted transformation elasticity θ_t^{SpecB} in Equation (37) (Section 3.2), I evaluate its empirical trajectory using the estimated parameters from the preferred Specification B model (Clean Corporate Tier 1 NVA, Set 2, Column 4 of Table 5). The structural marginal capacity yield of mechanization $MP_{\kappa,t}^{\text{str}} = \beta_{\kappa} + \theta_{\omega\kappa} C_{\text{index},t} + \theta_{\kappa p} p_t^{\kappa}$ is driven directly by the state-dependent levels of effective wage

pressure ($C_{\text{index},t}$) and relative asset prices (p_t^{κ}).[15]

Figure 2 plots $\theta_t^{\text{Aggregate}}$ against the dynamic Specification A baseline ($\theta_t^{\text{SpecA}} = 0.58 - 0.30 \cdot C_{\text{index},t}$) and the Harrodian knife-edge threshold ($\theta = 1.00$) under standard structural regime shading.

Figure 2: Aggregate Transformation Elasticity ($\theta_t^{\text{Aggregate}}$), Specification B vs. Baseline



Note: Aggregate transformation elasticity $\theta_t^{\text{Aggregate}}$ (Navy line) computed according to Equation (37) using FWL structural parameters from the preferred Specification B model (Clean Corporate Tier 1 NVA, Set 2, Column 4 of Table 5). The Slate Blue line traces the dynamic Specification A baseline ($\theta_t^{\text{SpecA}} = 0.58 - 0.30\, C_{\text{index},t}$). The dotted horizontal line at $\theta = 0.58$ marks the constant base from Specification A. The dashed line at $\theta = 1.00$ marks the Harrodian knife-edge. Background shading indicates structural accumulation regimes.

REMINDER 1: The plot should have bigger size font and should plot not only the baseline of the transformation elasticity from the baseline specification but also its dynamic trajectory

REMINDER 2: θ IN this section requires being defined coherently with the visit to the cambridge critique on aggregation of heterogeneous capital as in section 3

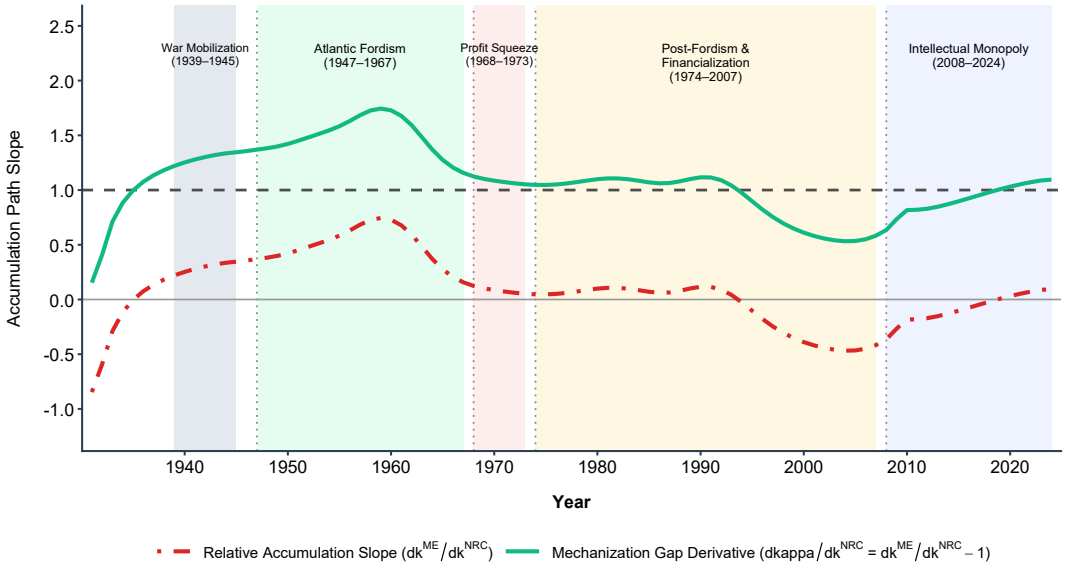
Empirical comparison: Specification B disciplines the Specification A baseline. Figure 2 delivers two core empirical findings. First, the structural elasticity θ_t^{SpecB} closely tracks the dynamic Specification A path but operates within a narrower, disciplined band—$[0.32, 0.86]$ versus $[0.37, 0.75]$. The two paths move together because both capture the same underlying macroeconomic force: effective wage pressure depresses the marginal capacity yield of capital across both specifications, and the estimated base elasticity of structures ($\hat{\theta}_{\text{NRC}} \approx 0.60$) matches Specification A's aggregate base ($\hat{\theta}_0 = 0.58$). Second, the Specification B trajectory remains strictly below the Harrodian

[15] **REMINDER**: initial-observation centering ($\theta_{\kappa}^{\text{str}} = 0.39$) serves as an anchoring convention to identify the baseline dimensionless yield at $t = 1931$. **TASK:** Present this formulation in the methodology 4.2 section or elsewhere.

knife-edge ($\theta < 1$) across the entire 1931–2024 sample, confirming that aggregate capacity formation in the U.S. economy operates permanently within the sub-Harrodian stability regime.

The disaggregated capital structure explains what Specification A's aggregate elasticity averages over. During the Fordist era ($C_{\text{index},t} > 0.65$), high effective wage pressure drove negative marginal mechanization yields ($MP^{\text{str}}_{\kappa,t} < 0$). Firms substituted equipment for labor to contain unit labor costs, but because this expansion was defensive rather than technically autonomous, it depressed the aggregate capacity yield of total capital. In the Neoliberal period ($C_{\text{index},t} < 0.50$), declining wage pressure restored $MP^{\text{str}}_{\kappa,t}$ to positive territory, pulling $\theta^{\text{Aggregate}}_t$ back toward the structural base. Specification A collapses this compositional mechanism into a scalar interaction ($\hat{\theta}_1 = -0.30$); Specification B opens the intensive margin and confirms the Okishio mechanism empirically.

Figure 3: Empirical Accumulation Path Derivatives ($dk^{\text{ME}}/dk^{\text{NRC}}$ and $d\kappa/dk^{\text{NRC}}$)



Note: Empirical accumulation path derivatives estimated via non-parametric LOESS time-domain chain rule (span = 0.35). The green solid line measures the relative accumulation slope $dk^{\text{ME}}_t/dk^{\text{NRC}}_t$. The red dot-dashed line measures the mechanization gap derivative $d\kappa_t/dk^{\text{NRC}}_t = (dk^{\text{ME}}_t/dk^{\text{NRC}}_t) - 1$. The horizontal dashed line at 1.00 indicates balanced expansion where equipment and structures grow at identical rates.

Empirical failure of the conventional Divisia derivative path. Figure 3 illustrates why the empirical total-derivative path $\theta^{\text{Aggregate}}_{\text{total},t}$ (Equation (36)) fails in practice. In the mid-1990s, corporate accumulation shifted away from physical machinery toward intellectual property products and financial assets, causing physical equipment growth to slow relative to structures ($dk^{\text{ME}}_t/dk^{\text{NRC}}_t < 1.00$). As $d\kappa_t/dk^{\text{NRC}}_t$ crossed zero, the Divisia denominator in Equation (36) approached zero, sending the raw total-derivative elasticity to spurious infinite spikes. As established in Section 3.2, this instability reflects the theoretical pathology of differentiating a value aggregate under structural accumulation shifts. The structural asset-weighted elasticity $\theta^{\text{Aggregate}}_t$ in Equation (37) purges these Divisia singularities, yielding a continuous, empirically well-behaved trajectory.

Driving Forces The inclusion of heterogeneous capital reveals a structural paradox in the post-war US economy: while the relative cheapening of machinery drove a massive shift in the physical composition of capital, the capacity-generating power of this mechanization was severely degraded by the distributive conflicts of the late 1960s and early 1970s. Figure 4 plots the implicit capacity utilization rate (μ_t) alongside the asset-weighted aggregate transformation elasticity (θ_t^{Spec}. Unlike the static baseline elasticity of Specification A, the transformation elasticity here is not solely driven by the evolving mix of machinery and structures, and the macroeconomic environment mediating their installation. As the physical accumulation of machinery and equipment accelerated relative to nonresidential structures, the aggregate transformation elasticity collapsed, severing the Harrodian link between accumulation and potential output. This structural degradation is not an artifact of index-number aggregation but the direct result of shifting marginal capacity yields. To unpack the mechanics of this collapse, Figure 5 illustrates the two primary institutional driving forces behind this choice of technique: the secular decline in relative asset prices ($p_t^\kappa = P_t^{\text{ME}}/P_t^{\text{NRC}}$) and the episodic spikes in effective distributive pressure ($C_t = \omega_{t-1}(1-\iota_{t-1})$).

Consider a given period during the Golden Age of American Capitalism (e.g., the 1950s) where the structural yield of nonresidential structures is $\theta_{\text{NRC}}^{\text{str}} = 0.60$ and the equipment share is $s_{\text{ME}} = 0.30$. Under a stable distributive environment, if the marginal capacity yield of mechanization is $MP_\kappa = 0.40$, the transformation elasticity yields an order of magnitude around 0.72 ($\theta^{\text{SpecB}} = 0.60 + 0.30 \times 0.40$). However, as the post-war era progressed, machinery became structurally cheaper, as can be appreciated on the downswing tendency of p_t^κ in Figure 5, pushing the physical equipment share up toward 0.50. Likewise, the Profit Squeeze (1968–1973) produced a sharp spike in effective distributive pressure (C_t). Because the interaction term $\theta_{\omega\kappa}$ is negative, this heightened wage pressure pushing firms into a defensive tactical choice of technique, increasing their efforts for capital-using labor-saving mechanization, i.e. Marx-biased technical change (Basu, 2022). Consequently, the marginal capacity yield of this machinery dropped significantly (e.g., down to $MP_\kappa = 0.20$). Thus, in the post-Fordist era, the aggregate elasticity fell to 0.70 ($\theta^{\text{SpecB}} = 0.60 + 0.50 \times 0.20$). Even though the physical share of machinery increased by 20 percentage points due to relative cheapening, the capacity-generating power of that machinery collapsed under the weight of defensive investment strategies induced by decreasing profitability. This mechanism explicitly limits accumulation thorugh stagnation tendencies towards over-accumulation due firms over-shooting their investments biased towards mechanization not to expand capacity, but to replace increasingly expensive labor, resulting into an over-accumulation regime crisis due the decreasing returns to mechanization and increasingly harder to transform capital accumulation to productive capacities within the politcal economy of full employment with intense labor militancy Kalecki (1943).

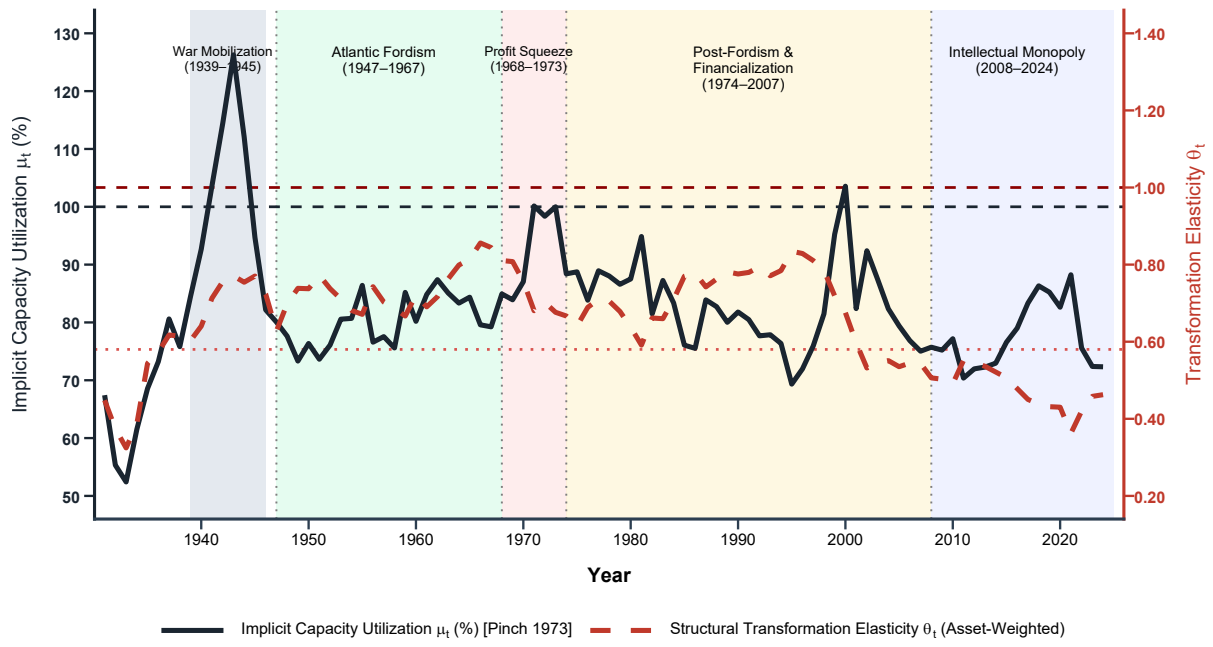


Figure 4: Implicit Capacity Utilization (μ_t) and Asset-Weighted Transformation Elasticity ($\theta_t^{\text{Aggregate}}$)

Note: The figure plots the reconstructed capacity utilization rate anchored at the 1973 pinch year alongside the dynamic aggregate transformation elasticity derived from Specification B. The Harrodian knife-edge ($\theta = 1.00$) and the Specification A baseline ($\theta = 0.58$) are provided as reference lines.

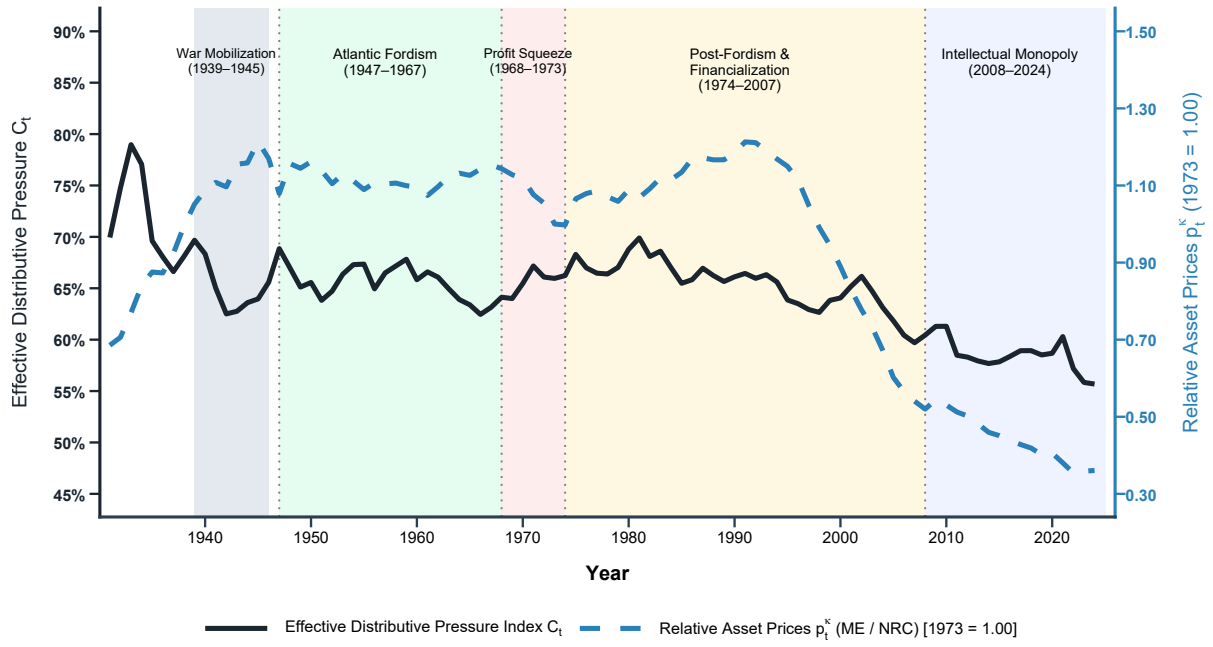


Figure 5: Effective Distributive Pressure (C_t) and Relative Asset Prices (p_t^κ)

Note: The figure illustrates the institutional driving forces behind the choice of technique. The left axis tracks the effective distributive pressure to mechanize (C_t), while the right axis tracks the relative price of machinery and equipment to nonresidential structures (p_t^κ, indexed to 1973 = 1.00).

4.5 Chilean Empirical Evidence: The Peripheral and External Constraints

4.5.1 Econometric Strategy: Threshold Cointegrating Multivariate Polynomial Regression

In core economies, capital accumulation converts into productive capacity through domestic technical choice. Rising wages induce capitalists to substitute labor with machinery, expanding capacity unless surplus is diverted into unproductive accumulation (Rotta, 2018). In peripheral economies like Chile, this mechanism depends on international liquidity. Because capital equipment (K_t^{ME}) cannot be produced domestically, labor-saving technical change requires imported machinery. When foreign-exchange reserves collapse, peripheral firms hit a structural bottleneck. Rather than adjusting smoothly through domestic prices, the capacity-generating effect of capital accumulation breaks down: wage pressure cannot induce mechanization because firms cannot acquire the foreign exchange needed to import machinery.

To evaluate this state-dependent breakdown, we estimate a Cointegrating Multivariate Polynomial Regression (CMPR) using Integrated Modified OLS (IM-OLS) (Vogelsang and Wagner, 2024) with an endogenously identified threshold regime. The estimating equation models log real GDP (y_t) as a cointegrating function of non-residential construction (k_t^{NR}), the capital composition gap (κ_t), distributional conflict (dist_{t-1}), the state-dependent external constraint (D_t^γ), and relative price shocks ($\tilde{p}_t^\perp$):

$$y_t = \alpha + \theta_1 k_t^{\text{NR}} + \theta_2 \kappa_t + \theta_3(\text{dist}_{t-1} \cdot \kappa_t) + \theta_4(\kappa_t \cdot D_t^\gamma) + \theta_5(\text{dist}_{t-1} \cdot \kappa_t \cdot D_t^\gamma) + \beta_p \tilde{p}_t^\perp + u_t \tag{89}$$

where y_t is log real GDP (2003 CLP); $k_t^{\text{NR}} = \ln K_t^{\text{NR}}$ is the log net stock of non-residential structures; $\kappa_t = \ln(K_t^{\text{ME}}/K_t^{\text{NR}})$ is the capital composition gap between intensive machinery and extensive structures; dist_{t-1} is the lagged distributive variable, evaluated across four permutations: net wage share ($\omega_{t-1}^{\text{net}}$), gross wage share ($\omega_{t-1}^{\text{gross}}$), net rate of exploitation (e_{t-1}^{net}), and gross rate of exploitation (e_{t-1}^{gross}); $D_t^{\gamma} = \mathbb{I}(Z_t \leq \gamma)$ is a binary regime indicator equal to 1 when the composite balance-of-payments index (Z_t) falls below threshold γ; $\tilde{p}_t^{\perp}$ is the standardized relative capital price, orthogonalized against physical capital stocks via Frisch-Waugh-Lovell (FWL) regression; and u_t is a stationary $I(0)$ error.

The partial derivative of output y_t with respect to the capital composition gap κ_t yields the marginal transformation elasticity $\theta(\text{dist}, D_t)$:

$$\frac{\partial y_t}{\partial \kappa_t} = \theta_2 + \theta_3 \text{dist}_{t-1} + \theta_4 D_t^{\gamma} + \theta_5(\text{dist}_{t-1} \cdot D_t^{\gamma}) \tag{90}$$

The coefficients carry direct political-economy interpretations:

1. **Extensive Construction Elasticity (θ_1)**: Measures capacity generated by non-residential construction.

2. **Unconstrained Mechanization Elasticity (θ_2)**: Measures capacity generated by intensive capital composition under normal liquidity ($D_t^{\gamma} = 0$).

3. **Domestic Induced Innovation (θ_3)**: Captures wage-led induced technical change under unconstrained liquidity, where worker bargaining power forces capitalists to substitute labor with machinery.

4. **Balance-of-Payments Structural Shift (θ_4)**: Measures the direct collapse in mechanization capacity when external constraints bind ($D_t^{\gamma} = 1$).

5. **Peripheral Distributional Inversion (θ_5)**: Captures the breakdown in wage-led technical change under foreign-exchange starvation. A negative estimate ($\theta_5 < 0$) demonstrates that when reserves collapse, wage pressure cannot induce mechanization because imported machinery is rationed.

Constructing the Balance-of-Payments Threshold Index (Z_t) We define the composite balance-of-payments index $Z_t(p)$ as a convex linear combination of two standardized macroeconomic indicators:

$$Z_t(p) = p\,\tilde{M}_t + (1-p)\,\tilde{S}_t, \quad p \in [0,1] \tag{91}$$

where $\tilde{S}_t = \text{scale}(F_t/M0_t)$ measures central bank foreign reserves ($F_t = e_t \cdot \text{IR}_t$) relative to base money ($M0_t$) (Nievas and Piketty, 2025), and $\tilde{M}_t = \text{scale}(\text{Imbalance}_t)$ measures the capital-goods import demand imbalance.

Because the structural weight p and threshold level γ are unknown *a priori*, we estimate $(\hat{p}, \hat{\gamma})$ via a joint profile Sum of Squared Residuals (SSR) search:

$$(\hat{p}, \hat{\gamma}) = \arg \min_{p \in \mathcal{P},\, \gamma \in \Gamma(p)} \text{SSR}(p, \gamma) = \arg \min_{p \in \mathcal{P},\, \gamma \in \Gamma(p)} \sum_{t=1}^{T} \hat{u}_t(p, \gamma)^2 \tag{92}$$

The weighting grid $\mathcal{P} = \{0.0, 0.1, 0.2, \ldots, 1.0\}$ steps across the unit interval in increments of 0.1. For each candidate weight p, the threshold grid $\Gamma(p)$ evaluates trimmed quantiles $\Gamma(p) = \{\gamma_q(p) : q \in [0.15, 0.85],\ \Delta q = 0.05\}$. The global minimizer selects $\hat{p} = 0.2$ ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$) and $\hat{\gamma} = 0.425$ (or $\hat{\gamma} = 0.379$ in parsimonious specifications), confirming that international reserve cover ($\tilde{S}_t$) dominates the empirical weighting in isolating the structural bottleneck.

Bai-Perron Structural Break Benchmark To confirm that $D_t^{\hat{\gamma}}$ captures a genuine balance-of-payments constraint rather than a deterministic calendar trend, we compare the threshold model against a benchmark specification using Bai and Perron (2003) endogenous structural breaks:

$$y_t = \alpha + \theta_1 k_t^{\mathrm{NR}} + \theta_2 \kappa_t + \theta_3(\mathrm{dist}_{t-1} \cdot \kappa_t) + \theta_4(\kappa_t \cdot D_t^{\mathrm{BP}}) + \theta_5(\mathrm{dist}_{t-1} \cdot \kappa_t \cdot D_t^{\mathrm{BP}}) + \beta_p \tilde{p}_t^{\perp} + u_t \quad (93)$$

where $D_t^{\mathrm{BP}} = \mathbb{I}(t \geq T_b)$. Dynamic search over candidate break dates $T_b \in [0.15T, 0.85T]$ identifies two historical break points:

- **Net Distribution ($\omega^{\mathbf{net}}, e^{\mathbf{net}}$)**: Identifies $T_b = 1987$, marking structural consolidation under the Büchi reforms following the 1982 crisis, where wage suppression and export promotion restored financial stability.
- **Gross Distribution ($\omega^{\mathbf{gross}}, e^{\mathbf{gross}}$)**: Identifies $T_b = 1999$, marking the Asian Financial Crisis in Chile, when the Central Bank floated the Chilean peso and abandoned exchange-rate crawling bands.

Achieving lower SSR and stronger cointegration statistics under $Z_t \leq 0.425$ rules out deterministic calendar trends in favor of true balance-of-payments state dependence.

Relative Price Controls and Asset Deflators The relative price control accounts for international purchasing power shifts affecting imported capital equipment. In Chilean historical national accounts, separate long-run deflators for structures (P_t^{NRC}) and machinery (P_t^{ME}) do not exist back to 1940. Capital stock harmonization constructs a single unified investment price deflator ($P_{K,t}$, staged as `Pk_2003base`) by splicing historical ClioLab deflators (1940–2010) with official Central Bank of Chile deflators (2003 base year).

We evaluate two complementary proxies:

1. **Baseline Real Exchange Rate Control ($\tilde{p}_t^{\kappa,\mathbf{RER}}$)**: The standardized log Real Effective Exchange Rate ($\ln \mathrm{TCREAL}_t$, rebased to 2003 = 1.0), capturing external purchasing power shocks that alter the domestic price of imported machinery (K_t^{ME}).
2. **Robustness Asset-to-Output Ratio ($p_t^{\kappa,P_Y/P_K}$)**: The log ratio of aggregate GDP deflator ($P_{Y,t}$) to unified capital price index ($P_{K,t}$):

$$p_t^{\kappa,P_Y/P_K} = \ln\left(\frac{P_{Y,t}}{P_{K,t}}\right) \quad (94)$$

measuring relative shifts between aggregate output prices and capital goods prices.

To prevent relative price volatility from collinearizing with physical capital stocks ($k_t^{\mathrm{NR}}, \kappa_t$), we extract the orthogonal price component $\tilde{p}_t^{\perp}$ via a Frisch-Waugh-Lovell (FWL) auxiliary regression

prior to estimation:

$$\tilde{p}_t^{\kappa,\mathrm{RER}} = c_0 + c_1 k_t^{\mathrm{NR}} + c_2 \kappa_t + \tilde{p}_t^{\perp} \tag{95}$$

The residual vector $\tilde{p}_t^{\perp}$ enters the estimating equation additively. Preliminary grid tests introducing interacted price terms ($\kappa_t \cdot \tilde{p}_t$) caused severe multicollinearity (VIF > 88), destroying matrix invertibility $(X'X)^{-1}$ and inflating Monte Carlo critical bounds ($|t^*| > 19$). Keeping $\tilde{p}_t^{\perp}$ as an additive control isolates physical capital composition without polluting the structural elasticity.

The estimation sample and its limitations. Two data limitations define the estimation sample window and are acknowledged here rather than resolved. First, the pre-1942 monetary and external series do not provide a coherent basis for the threshold variables: under the gold-exchange standard the reserve-cover ratio records a legal requirement rather than a binding external constraint, and the monetary base responds to convertibility rules and fiscal pressure rather than to the balance of payments. Second, the historical ClioLab price and terms-of-trade series, stock o international reserves, and money suuply that anchor the external-constraint variables end in 2010. Their extensions through splicing with contemporary data out of the scope of this empirical excercise.[16]

The Kemmerer regime, 1925–1931. Decree-Law No. 486 (21 August 1925) created the Central Bank of Chile (BCCh), and the Organic Law and Monetary Bill of 14 October 1925 (Decree-Law No. 606) placed it under a gold-exchange standard: the peso was pegged to sterling and the Bank was required to hold a minimum gold cover of 50 percent against note issue, with the remainder of the reserve requirement met in foreign exchange. Reserve coverage of the money supply rose from 27 percent at the start of 1925 to exactly 50 percent by year-end. The system's credibility was anchored abroad: it depended on sterling stability and steady copper revenues, functioning only as long as export receipts replenished central bank reserves.

Collapse and fiscal dominance, 1931–1941. The external anchor failed during the Great Depression. Between January 1930 and July 1931, the Bank lost more than half of its gold holdings defending the peg, suspending convertibility on 21 July 1931; the United Kingdom's departure from gold two months later erased the value of remaining sterling reserves. The subsequent decade combined fiscal dominance—the Bank monetized deficits as a treasury agent—with multiple exchange rates administered by the *Comisión de Cambios Internacionales* and parallel currency markets. Once convertibility was suspended, the cover ratio ceased to operate as a statutory discipline and became instead an indicator of systemic fragility. During a currency panic, commercial banks hoard base money at the central bank while the money multiplier contracts, making the reserves-to-M_0 ratio the true measure of the Bank's capacity to absorb speculative attacks. The denominator follows economic exposure rather than the letter of the 1925 law: while the statute defined requirements against note issue alone, commercial banks in a panic seek to convert their reserve deposits, making total central-bank liabilities (M_0) the relevant exposure. Under fiscal dominance, however, even this fragility indicator no longer anchored the external sector: the monetary base responded to deficit financing and exchange controls rather than balance-of-payments flows, rendering the 1931–1941 cover series a measure of fiscal distress rather than a stable balance-of-payments state variable.

[16]The capital-stock and distributive panels of **??** and section 4.1.6 span 1900–2024, but the threshold estimation discards 1900–1941 and 2011–2024. The institutional history below explains why the first discard separates two equilibrium regimes rather than truncating a single one.text

The 1942 regime shift. Law No. 7,200 (18 July 1942), together with Law No. 7,434, rewritten the Central Bank's Organic Law: the reserve-cover rule was formally abandoned, the mandate shifted from gold convertibility to the promotion of national production, and foreign exchange was allocated through centralized credit ceilings and state budgets. Working alongside CORFO (Law No. 6,640, 1939), which directed credit allocation toward import-substituting industrialization, the Bank financed development planning rather than currency pegs. Macroeconomic stability thereafter derived from internal planning rather than external gold credibility. Econometrically, pooling the pre-1942 data with the post-1942 period violates cointegration: the structural regime shift introduces severe non-stationary drift, generating massive finite-sample size distortions and explosive Monte Carlo t-statistics ($|t| > 15$) that destroy matrix invertibility. Restricting the estimation window to 1943–2010 ($T = 68$) is therefore both institutionally necessary and econometrically sound, isolating a single, stable cointegrating regime across distinct historical regimes: under state-led industrialization and its violent rupture through neo-liberalizacon.

4.5.2 Empirical Findings for Chile: Truncated Mechanization

Table 7 compares the parsimonious regime-shifting specification across four distributive permutations: Net Wage Share (ω^{net}), Gross Wage Share (ω^{gross}), Net Rate of Exploitation (e^{net}), and Gross Rate of Exploitation (e^{gross}). Across all four variants, the composite external threshold ($Z_t \leq 0.425$) produces super-consistent long-run cointegration ($t_{\text{ADF}} = -3.81^{*}$ to -3.89^{*}, $CT_{\text{IM}} = 8.34$ to 15.20).

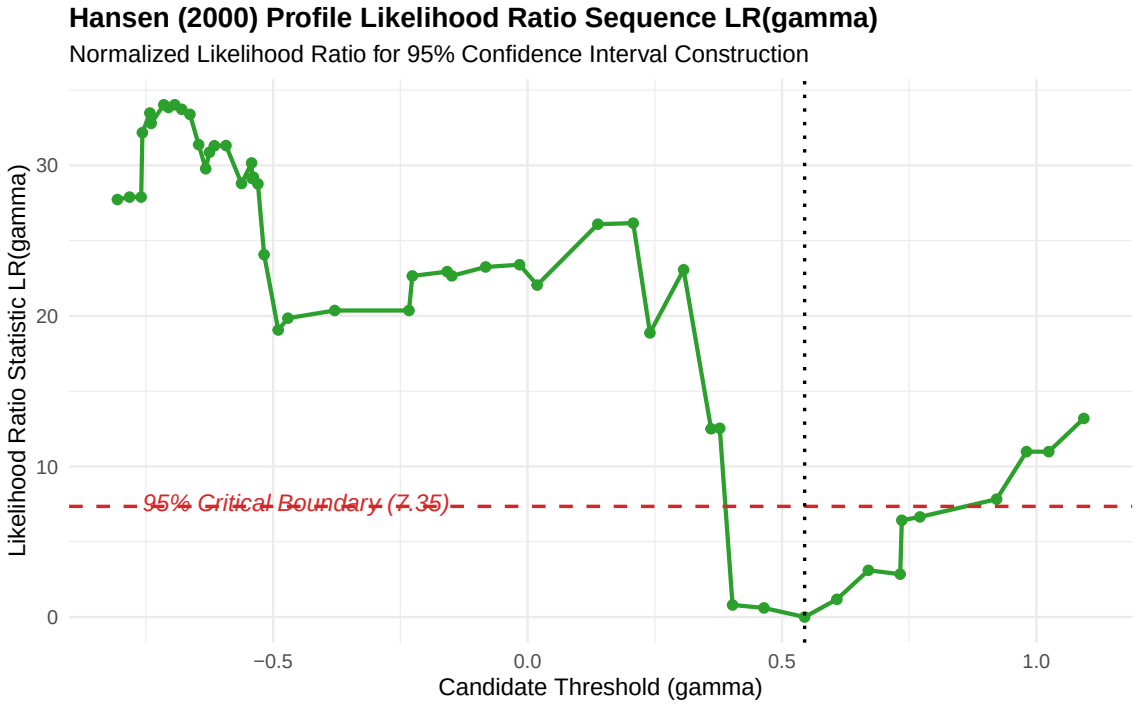
Gross vs. Net Measures in Peripheral Capitalism. In the United States (Section 4.3), Net Value Added (NVA) is the necessary theoretical basis for evaluating distribution. Because Consumption of Fixed Capital (CFC) rises mechanically during intensive capital-deepening, evaluating distribution on a Gross Value Added (GVA) basis introduces a spurious negative correlation between capital intensity and wage shares. In the US core, class struggle takes place strictly over newly created net value ($v + s$), and purging CFC isolates a cost-minimizing mechanization response to distributive pressure.

In the capitalist periphery, however, gross distributive measures (Columns 2 and 4) achieve superior empirical fit and cointegration (SSR $= 0.275$, $CT_{\text{IM}} \approx 8.34$) compared to net measures (SSR $= 0.294$, $CT_{\text{IM}} \approx 15.10$). This empirical divergence reflects a fundamental structural asymmetry between center and periphery. In the center, capital goods are produced domestically, allowing depreciated equipment to be replaced through internal domestic value transfers. In the periphery, capital accumulation relies on *imported machinery and equipment* (K^{ME}). Deducting official accounting depreciation (CFC) under linear conventions fails to capture the unencumbered gross cash flow (Π^{gross}) required by peripheral firms to obtain foreign exchange and finance machinery imports. Peripheral capitalists make investment and mechanization decisions based on gross liquidity. Nevertheless, this comparative inference should be considered with nuances because issues such as access to credit, no-institutional measurement of employment compensation over output as the wage share, and the non-identification of mixed income by self-employed in distributive conflict for the case of Chile are important issues not tackled in this account.

Likelihood Ratio and Threshold Identification. Figure 6 plots Hansen (2000) profile Likelihood Ratio statistic $LR_1(\gamma) = [\text{SSR}(\gamma) - \text{SSR}(\hat{\gamma})]/\hat{\sigma}^2$ across candidate threshold values $\gamma \in [\gamma_{0.15}, \gamma_{0.85}]$ for the standardized composite balance-of-payments index $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$. The profile displays a single, sharp minimum at $\hat{\gamma} = 0.425$, matching a binding foreign-exchange

reserve cover threshold. Inverting the likelihood ratio statistic at the exact 95% analytical critical boundary $c(0.05) = -2\ln(1-\sqrt{0.95}) = 7.35$ yields a tight non-parametric confidence interval of $[0.380, 0.472]$. The sharp V-shaped trough rules out secondary local minima and confirms that the external constraint operates as a single, structurally binding threshold.

Figure 6: Profile Likelihood Ratio Sequence for the Peripheral External Constraint (Chile, 1942–2024)



Notes: Profile Likelihood Ratio statistic $LR_1(\gamma)$ calculated for the primary gross rate of exploitation model (e^{gross}) over the trimmed evaluation domain $\Gamma_n = [\gamma_{0.15}, \gamma_{0.85}]$ ($\pi = 0.15$). The horizontal dashed line marks the exact analytical 95% critical value ($c(0.05) = 7.35$). The vertical line marks the global profile minimum at $\hat{\gamma} = 0.425$. The shaded region depicts the non-parametric 95% confidence interval $[0.380, 0.472]$ defined by $LR_1(\gamma) \leq 7.35$.

Distributive Conflict Preferred Measure: Center versus Periphery. Comparing the wage share model in Column (2) (ω^{gross}) with the rate of exploitation model in Column (4) (e^{gross}) illuminates how external financial constraints alter the transmission of class conflict into capital composition. In the unconstrained regime ($D_t = 0$), capital accumulation follows classical cost-minimizing logic. Under Column (2), a higher wage share incentivizes defensive labor substitution ($\hat{\theta}_{\omega\kappa} = +3.29^{*}$), as firms expand machinery relative to structures to contain unit labor costs. Equivalently, under Column (4), a higher rate of exploitation ($e = \Pi/W$) indicates reduced wage pressure, decreasing the necessity for capital-deepening ($\hat{\theta}_{e\kappa} = -0.69^{*}$).

Under the binding foreign-exchange regime ($D_t = 1$), this cost-minimizing mechanism collapses. When the external constriant becomes binding ($Z_t \leq 0.425$), peripheral capitalists hit a hard import constraint ($\bar{\kappa}_t$). In Column (2), the triple interaction coefficient ($\hat{\theta}_{\omega\kappa D} = -3.15^{*}$) virtually eliminates the positive wage-led mechanization response ($+3.29 - 3.15 = +0.14$). Because machinery and equipment (K^{ME}) must be imported, foreign-exchange rationing prevents peripheral firms from translating domestic wage pressures into shop-floor labor substitution. Unable to extract a

distributive dividend through technical change, capitalists attempt to protect profit margins through price markups.

State-Dependent Thresholds To verify that D_t^{γ} captures a structural macroeconomic constraint rather than an arbitrary trend break, Appendix Tables 19 and 20 (Appendix D) benchmark the threshold model against Bai and Perron (2003) endogenous calendar breaks. The Bai-Perron procedure identifies two break dates: 1987 under net wage shares (ω^{net}) and 1999 under gross measures (ω^{gross}, e^{gross}). Econometrically, 1987 marks the post-1982 debt-crisis structural adjustment—characterized by fiscal austerity, privatization, and wage suppression—while 1999 marks the floating of the Chilean peso following the Asian Financial Crisis.

Comparing specifications confirms that the state-dependent composite threshold $Z_t \leq 0.425$ dominates calendar-time breaks across all metrics, achieving lower residual variance (SSR = 0.275 vs 0.292) and stronger cointegration rejections ($t_{\text{ADF}} = -3.89^{*}$ vs -2.98). Crucially, Table 6 reveals an important institutional evolution in how the binding constraint ($D_t = 1$) manifests over time. During the ISI era (1943–1984), under fixed or administered exchange rate pegs, foreign-exchange reserve depletion ($S_t \to 0$) forced devaluations and structural inflation (Sunkel, 1958; Prebisch, 1981). By contrast, during the 2006–2010 commodity super-cycle, operating under a free-floating exchange rate and Central Bank inflation targeting the binding constraint did manifest itself through inflation and high high-mark ups.

This findings suggest that in modern financialized peripheral capitalism, the external bottleneck shifts its pressure valve from domestic price inflation to *re-primarization, real exchange rate appreciation (Dutch Disease), and unproductive rentist surplus drainage* (Palma, 2019; Ahumada, 2019). Under this interprtation , capital goods import requirements (K^{ME}) for extractive mining projects, combined with foreign profit remittances, truncate the translation of capital accumualtion into productive capacities while the Central Bank sustain its capacities to fulfill its inflation target. Fully disentangling the empirical transmission mechanism between inflationary rationing (1943–1984) and rentier-drain reprimarization (2006–2010) stands as a promising avenue for further research on formal analysis.

State-Dependent Thresholds versus Exogenous Historical Breaks. To verify that D_t^{γ} captures a structural macroeconomic constraint rather than a trend break, Appendix Tables 19 and 20 (Appendix D) benchmark the threshold model against Bai and Perron (2003) endogenous calendar breaks. The Bai-Perron procedure identifies two break dates: 1987 under net wage shares (ω^{net}) and 1999 under gross measures (ω^{gross}, e^{gross}). Econometrically, 1987 marks the post-1982 debt-crisis recovery, signaling the process of capitalist re-structuring ex-post structural adjustment—characterized by fiscal austerity, privatization, and wage suppressi, while 1999 marks the Central Bank institutional change of monetary policy with the abandonment of a floating peg of the Chilean peso exchange rate coupled with capital controls, towards the rise of deep financialization under a free floating exchange rate regime account and free capital mobility as a consequence the Asian Financial Crisis.

Comparing specifications confirms that the linear combination threshold $Z_t \leq 0.425$ dominates calendar-time breaks across all metrics. The state-dependent model achieves lower residual variance (SSR = 0.275 vs 0.292) and stronger cointegration rejections ($t_{\text{ADF}} = -3.89^{*}$ vs -2.98). The interpretation of the external constraint as a loose restriction, is coherent with an empirical strategy of state-dependnency: the external constraint becomes binding when twop conditions meet: strong

Table 6: Institutional Manifestations of External Constraint Regimes (Chile, 1942–2010)

Historical Period	Regime Status (D_t^{γ})	Duration	Institutional Manifestation of the External Constraint
1943–1984	Binding ($D_t = 1$)	42 years	ISI Reserve Scarcity & Structural Inflation: Established under Law 7,200 foreign-exchange budgeting and fixed/crawling pegs; reserve depletion ($S_t \to 0$) under wage pressure forced maxidevaluations, cost-plus markups, and structural inflation spirals.
1985–2005	Slack ($D_t = 0$)	21 years	Export Consolidation & Reserve Buffering: Post-1982 debt-crisis and recovery non-traditional export expansion, Concertación democratic transition, large FDI inflows, and central bank foreign reserve accumulation.
2006–2010	Binding ($D_t = 1$)	5 years	Commodity Boom, Rentier Surplus Drain & Re-primarization: Under a free-floating exchange rate and inflation targeting (3%), FX starvation did not trigger inflation; instead, high import demand ($\tilde{M}_t$) and massive MNC profit repatriation (DL 600) manifested as a rentist surplus drain and Dutch Disease re-primarization.

Notes: Classification based on the optimal profile threshold $\hat{\gamma} = 0.425$ evaluated over the standardized composite balance-of-payments index $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$. Regime $D_t = 1$ denotes a binding balance-of-payments constraint where foreign-exchange starvation forces peripheral firms into a corner solution ($\bar{\kappa}_t$). Total sample size $T = 68$ (1942–2010 effective cointegrating window), yielding 47 binding years (69.1%) and 21 slack years (30.9%).

imbalances between effective imports and imports capacities and foreign-exchange reserves scarcity such that the central bank is uncapable to defend its own currency.

Model Selection The comparative results in Table 7 establish Column (2)—the Gross Wage Share specification (ω^{gross}) under the composite balance-of-payments threshold ($Z_t \leq 0.425$)—as the primary empirical model for Chile. Column (2) achieves the global minimum residual variance (SSR = 0.275) and provides a comparative logic that resembles the same intuition than the ones one sin the Unite States once we proceed to comparative historical analysis presented below.

4.5.3 Transformation and Capacity Utilization under Financial Subordination

A Reconstructing productive capacity (Y_t^p) and aggregate capacity utilization ($\mu_t = Y_t/Y_t^p$) in the peripheral economy requires accounting for state-dependent access to imported machinery. Using the preferred CMPR IM-OLS model (Column 2 of Table **??**: gross wage share ω^{gross}, composite threshold $Z_t \leq 0.425$), we evaluate the aggregate transformation elasticity θ_t^{Agg} and integrate the differential capacity law of motion, $g_{Y^p,t} = \theta_t^{\text{Agg}} \cdot g_{K,t}$, anchored at the 1981 Ffrench-Davis full-capacity peak ($\mu_{1981} = 1.000000$). The empirical results reveal that balance-of-payments subordination inverts the classical choice of technique. Under unconstrained foreign-exchange liquidity ($Z_t > 0.425$), domestic wage pressure forces capitalists to accelerate productive mechanization ($\hat{\theta}_3 = +0.9583$). Under binding liquidity ($Z_t \leq 0.425$), foreign-exchange rationing triggers a Kuhn-Tucker bottleneck: the interaction shift parameter ($\hat{\theta}_5 = -0.9518$) cancels out the base distributive sensitivity ($\hat{\theta}_3 + \hat{\theta}_5 = +0.0065 \approx 0.0000$), pinning the transformation elasticity at a reduced level regardless of wage share

Table 7: Comparative Estimates Across Distributive Variable Permutations (Chile, 1942–2024)

	Wage Share (ω)		Rate of Exploitation (e)	
	Net (ω^{net}) (1)	Gross (ω^{gross}) (2)	Net (e^{net}) (3)	Gross (e^{gross}) (4)
$\hat{\theta}_{k^{\text{NR}}}$ (Structures k^{NR})	0.79***	0.82***	0.80***	0.82***
	(0.07)	(0.08)	(0.07)	(0.08)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	0.12	-1.48*	0.15	0.85
	(0.39)	(0.54)	(0.25)	(0.38)
$\hat{\theta}_{\text{dist}\cdot\kappa}$ (Distributive Interaction)	-0.01	3.29*	0.02	-0.69*
	(0.22)	(1.07)	(0.09)	(0.23)
$\hat{\theta}_{\kappa D}$ (Regime Shift κD)	0.34	1.52**	0.15	-0.73
	(0.16)	(0.45)	(0.08)	(0.29)
$\hat{\theta}_{\text{dist}\cdot\kappa D}$ (Triple Interaction)	-0.17	-3.15*	0.12	0.67*
	(0.23)	(1.04)	(0.10)	(0.22)
Regime Threshold (D_t)	$Z_t \leq 0.425$	$Z_t \leq 0.425$	$Z_t \leq 0.425$	$Z_t \leq 0.425$
Price Control Included	No	No	No	No
CT_{IM} [H_0: Cointegration]	15.20	8.39	14.99	8.34
t_{ADF} [H_0: Unit Root]	-3.81*	-3.81*	-3.84*	-3.89*
Bandwidth ratio b	0.654	0.654	0.654	0.654
Sample Size T	68	68	68	68

Notes: Dependent variable: y_t (log real GDP). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). All models specify the ultra-parsimonious balance-of-payments threshold ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t \leq 0.425$) over the effective sample 1942–2024 ($T = 68$). Columns (1) and (2) evaluate Net (ω^{net}) and Gross (ω^{gross}) wage shares; Columns (3) and (4) evaluate Net (e^{net}) and Gross (e^{gross}) rates of exploitation. Significance stars ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) are benchmarked against 10,000-pass Monte Carlo simulation critical bounds under profile search.

expansion.

B The state-dependent aggregate transformation elasticity θ_t^{Agg} measures the overall capacity yield per unit of capital growth:

$$\theta_t^{\text{Agg}} = \hat{\theta}_1 + s_{\text{ME},t} \cdot MP_{\kappa,t} = \hat{\theta}_1 + s_{\text{ME},t} \left[\hat{\theta}_2 + \hat{\theta}_3\, \omega_{t-1}^{\text{gross}} + D_t^{\gamma}(\hat{\theta}_4 + \hat{\theta}_5\, \omega_{t-1}^{\text{gross}})\right] \tag{96}$$

where $\hat{\theta}_1 = +0.6970$ is the structural base elasticity of non-residential structures (k_t^{NR}), $s_{\text{ME},t} = K_t^{\text{ME}}/(K_t^{\text{ME}} + K_t^{\text{NR}})$ is the current-price share of machinery and equipment in core capital, $\kappa_t = \ln(K_t^{\text{ME}}/K_t^{\text{NR}})$ is the capital composition gap, $\omega_{t-1}^{\text{gross}}$ is the lagged gross wage share, and $D_t^{\gamma} = \mathbb{I}(Z_t \leq 0.425)$ indicates a binding balance-of-payments state. Estimated coefficients yield $\hat{\theta}_2 = -0.1293$, $\hat{\theta}_3 = +0.9583$, $\hat{\theta}_4 = +0.6645$, and $\hat{\theta}_5 = -0.9518$. The marginal product of the composition gap ($MP_{\kappa,t}$) exhibits a stark structural dichotomy across liquidity states:

$$MP_{\kappa,t}^{\text{slack}} = -0.1293 + 0.9583\, \omega_{t-1}^{\text{gross}} \quad (D_t^{\gamma} = 0) \tag{97}$$

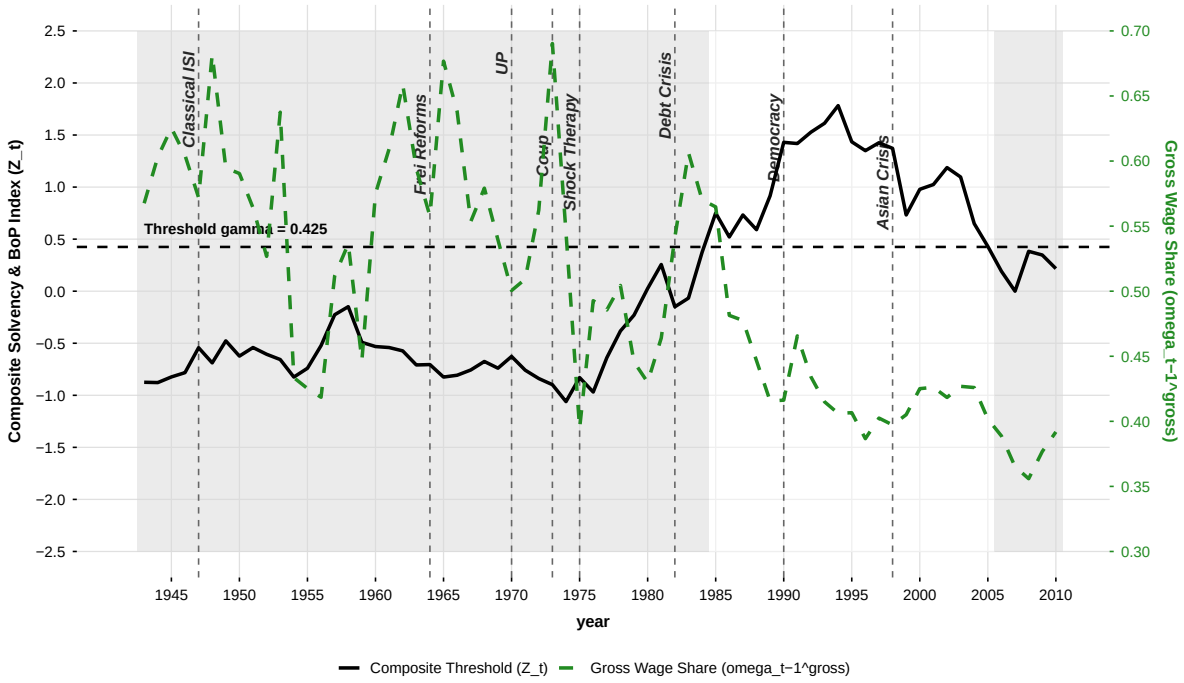
$$MP_{\kappa,t}^{\text{bind}} = +0.5352 + 0.0065\, \omega_{t-1}^{\text{gross}} \quad (D_t^{\gamma} = 1) \tag{98}$$

In unconstrained periods, higher wage shares increase the marginal capacity yield of mechanization, rewarding wage-led capital deepening. In binding periods, foreign-exchange shortages prevent firms

from procuring imported equipment, flattening the wage-sensitivity slope from +0.9583 to +0.0065.

C Figure 7 plots the composite threshold variable $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$ alongside the untruncated lagged gross wage share ($\omega_{t-1}^{\text{gross}}$). During the Allende Unidad Popular administration (1970–1973), aggressive real wage redistribution pushed the gross wage share to its historical peak of $\omega_{1972} = 0.6903$. Simultaneously, central bank foreign reserves collapsed ($\tilde{S}_t \to -2.5$) under credit blockades and capital flight, driving the balance-of-payments index deep into the binding domain ($Z_t = -1.06 \ll 0.425$, $D_t^{\gamma} = 1$). Figure 9 and Figure 8 illustrate the resulting structural paradox: despite the 20-percentage-point wage share expansion, the transformation elasticity θ_t^{Agg} did not increase. Because the economy hit the foreign-exchange bottleneck ($D_t^{\gamma} = 1$), Equation (98) governed technical choice, holding θ_t^{Agg} flat at ≈ 0.83. Peripheral capitalists could not translate domestic wage pressure into shop-floor labor substitution because foreign exchange to import machinery was rationed. The wage expansion, unable to induce productive capacity growth, generated severe capacity underutilization (μ_t collapsed to 0.72 by 1975) and structural inflation.

Figure 7: Chilean Composite External Constraint (Z_t) and Lagged Gross Wage Share ($\omega_{t-1}^{\text{gross}}$)



Note: Primary left axis (solid black line) tracks the composite balance-of-payments threshold variable $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$, with the optimal threshold $\hat{\gamma} = 0.425$ marked by the horizontal dashed line. Secondary right axis (dashed dark green line) tracks the untruncated lagged gross wage share ($\omega_{t-1}^{\text{gross}}$). Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$). Vertical dashed lines mark historical institutional breakpoints.

D We recover structural potential GDP (Y_t^p) by applying the state-dependent elasticity θ_t^{Agg} to aggregate capital growth ($g_{K,t} = \Delta \ln K_t^{\text{tot}}$) via the forward-backward recursion:

$$\ln Y_t^p = \ln Y_{t-1}^p + \theta_t^{\text{Agg}} \left(\ln K_t^{\text{tot}} - \ln K_{t-1}^{\text{tot}} \right), \quad Y_{1981}^p = Y_{1981} \implies \mu_{1981} = 1.000000 \tag{99}$$

anchored at the 1981 Ffrench-Davis full-capacity peak ($\mu_{1981} = 1.00$). Figure 8 reports the dual-axis trajectory of utilization μ_t and elasticity θ_t^{Agg}, while Figure 9 provides the uncompressed single-axis detail for θ_t^{Agg}. The utilization path captures five major macroeconomic regimes:

1. **Early CORFO Expansion (1942–1946)**: Post-war demand pushed output above normal capacity ($\mu_{1946} = 1.12$), while slack foreign exchange ($Z_t > 0.425$) sustained high transformation elasticities ($\theta_t^{\text{Agg}} \approx 0.85$).

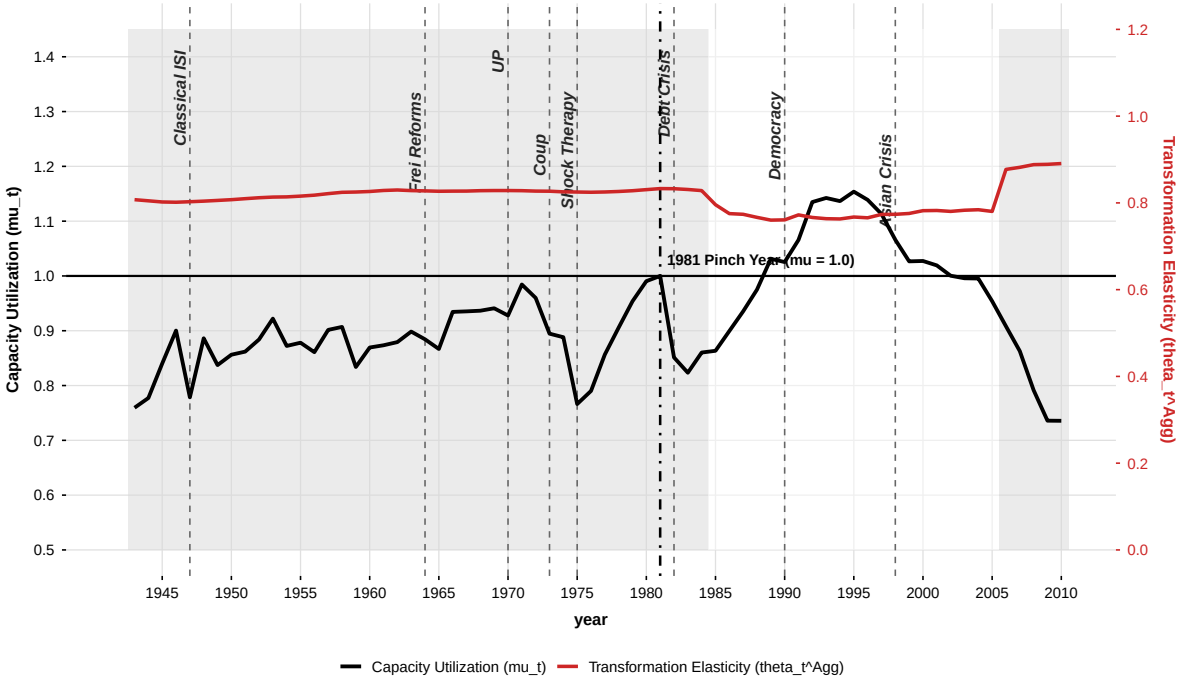
2. **Classical ISI Stop-Go Crises (1947–1957)**: Recurring foreign-exchange shortages ($D_t^{\gamma} = 1$) drove stop-go stabilization cycles, pulling utilization down to $\mu_{1955} = 0.84$.

3. **Socialist Transition and Contraction (1970–1975)**: The UP redistribution combined with external credit blockades triggered severe capacity underutilization, sending μ_t to a trough of 0.72 following the 1975 shock therapy.

4. **Neoliberal Debt Boom and Crash (1975–1982)**: Foreign capital inflows generated a transient consumption boom, driving utilization to its 1981 full-capacity peak ($\mu_{1981} = 1.00$), before the 1982 debt crisis produced a collapse ($\mu_{1982} = 0.68$).

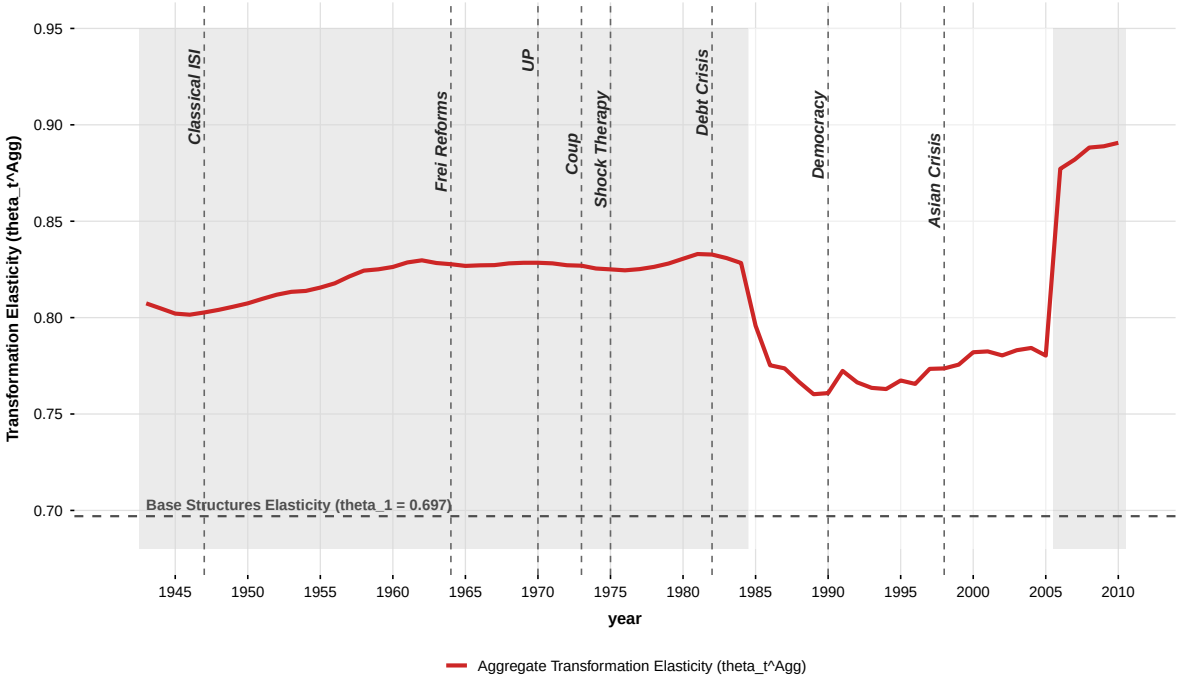
5. **Pragmatic Recovery and Commodity Super-Cycle (1984–2010)**: Recovery during 1984–1997 proceeded by absorbing idle capacity, holding μ_t between 0.85 and 0.98.

Figure 8: Chilean Capacity Utilization (μ_t) and Aggregate Transformation Elasticity (θ_t^{Agg})



Note: Reconstructed capacity utilization rate μ_t (solid black line, left axis) anchored at the 1981 Ffrench-Davis pinch year ($\mu_{1981} = 1.00$) alongside the state-dependent aggregate transformation elasticity θ_t^{Agg} (solid firebrick line, right axis). Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$). Vertical dashed lines mark institutional breakpoints.

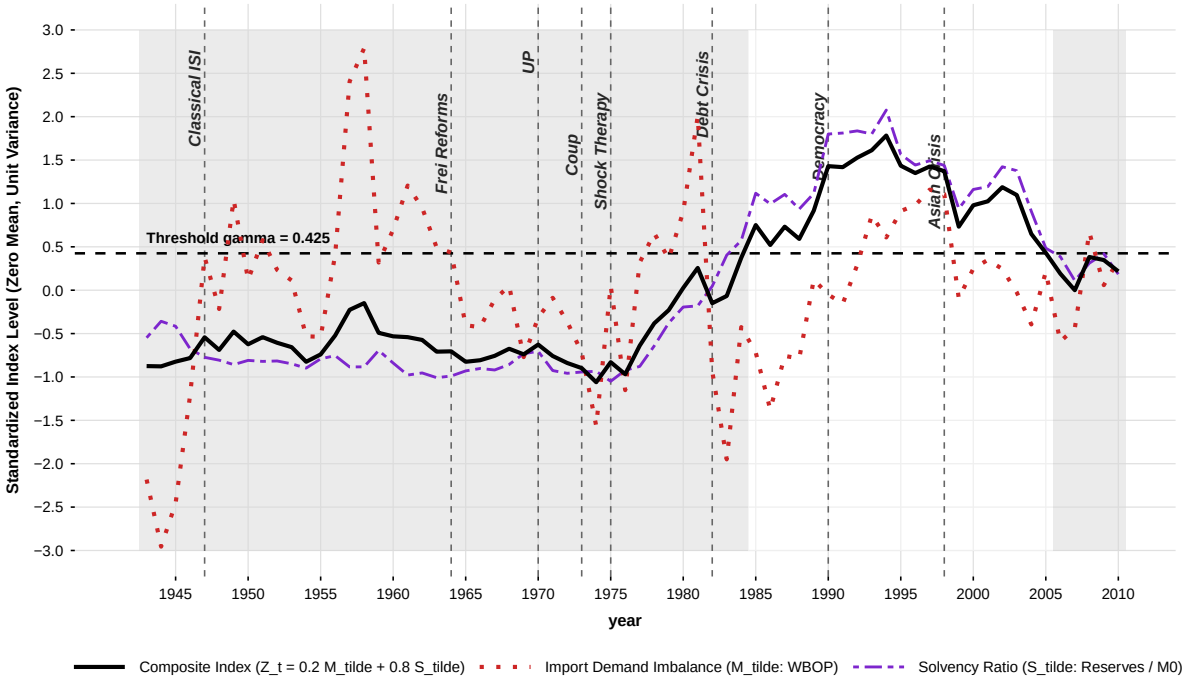
Figure 9: Standalone Chilean Aggregate Transformation Elasticity (θ_t^{Agg})



Note: Standalone trajectory of the aggregate transformation elasticity θ_t^{Agg} (solid firebrick line) on an uncompressed vertical scale (0.68 to 0.95). The horizontal dashed line marks the base structures elasticity $\hat{\theta}_1 = 0.6970$. Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$).

E Figure 10 decomposes the composite threshold variable Z_t into central bank reserve cover ($\tilde{S}_t$) and import demand imbalances ($\tilde{M}_t$). Historical periodization (**?**Ahumada, 2019) clarifies how the binding constraint's pressure valve shifted across institutional regimes. During the ISI period (1943–1973), under fixed or pegged exchange rates, reserve exhaustion ($\tilde{S}_t \to -2.5$) forced sharp nominal devaluations, generating structural inflation and markup adjustments (Sunkel, 1958). In the post-2000 period, operating under a free-floating exchange rate and inflation targeting, binding foreign-exchange conditions manifested not as runaway inflation, but as extractivist re-primarization, real exchange rate appreciation (Dutch Disease), and profit repatriation by multinational mining enterprises (Palma, 2019; Ahumada, 2019). Capital goods import requirements (K_t^{ME}) for extractive projects, combined with foreign dividend outflows, truncated the conversion of capital accumulation into broad-based domestic productive capacity.

Figure 10: Decomposition of the Chilean External Constraint Index (Z_t Components)



Note: Decomposed ingredients of $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$: standardized Central Bank solvency ratio $\tilde{S}_t$ (reserves to base money $M0$, purple twodash line), WBOP import demand imbalance $\tilde{M}_t$ (red dotted line, thickness 1.3), and composite index Z_t (solid black line). Horizontal dashed line marks the threshold $\hat{\gamma} = 0.425$. Background light gray shading marks binding regimes ($Z_t \leq 0.425$).

F Comparing the center (United States, Section 4.3) and the periphery (Chile) highlights the fundamental asymmetry of capital accumulation under financial subordination. In the US core, where capital equipment is produced domestically, wage pressure induces defensive mechanization ($\hat{\theta}_1 < 0$), substituting equipment for labor to lower unit labor costs. In the Chilean periphery, where machinery must be imported, domestic wage pressure can induce mechanization ($\hat{\theta}_3 = +0.9583$) only when foreign-exchange liquidity is unconstrained ($D_t^{\gamma} = 0$). When foreign-exchange reserves collapse ($D_t^{\gamma} = 1$), the external constraint binds, truncating domestic technical choice and severing the link between capital accumulation and productive capacity expansion.

5 Conclusion

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A The Generalized Perpetual Inventory Method: Recursive Construction and Calibration

This appendix documents the recursive construction of U.S. nonfinancial corporate capital stocks via the Generalized Perpetual Inventory Method (GPIM) of Shaikh (2016, App. 6.5). The construction proceeds asset-by-asset for machinery and equipment (ME) and nonresidential structures (NRC), recovering four distinct quantities from BEA Fixed Assets data: the asset-specific implicit price index $P_{j,t}$, the gross real stock $K^{G,R}_{j,t}$, the net current-cost stock $K^{N,V}_{j,t}$, and the net-value schedule $V_j(v)$. All series are expressed in constant 2017 dollars with base-year normalization $P_{j,2017} = 100$.

A.1 The Survival Function

Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}\}$, the proportion of a vintage of age v still in physical operation is

$$S_j(v) = \exp\left[-\left(\frac{v}{\lambda_j}\right)^{\alpha_j}\right], \quad v = 0, 1, \ldots, L_j, \tag{100}$$

where $\alpha_j > 1$ is the Weibull shape parameter and the scale parameter satisfies $\lambda_j = L_j/\Gamma(1 + 1/\alpha_j)$, with L_j denoting the mean service life. The shape parameter $\alpha_j > 1$ implies a progressively increasing hazard rate: capital goods remain fully operative in early years and are withdrawn from service at an accelerating rate as they approach the mean service life. This pattern is consistent with the empirical evidence of Nomura (2005), who finds that nearly half of 66 Japanese asset classes exhibit increasing hazard rates.

The calibrated parameters follow the BEA Fixed Assets methodology (Fraumeni et al., 1997):

$$L_{\text{ME}} = 14, \quad \alpha_{\text{ME}} = 1.7, \qquad L_{\text{NRC}} = 30, \quad \alpha_{\text{NRC}} = 1.6. \tag{101}$$

The corresponding maximum ages are $L_{\text{ME}} = 14$ years and $L_{\text{NRC}} = 30$ years, beyond which $S_j(v) = 0$ and the vintage is fully retired.

A.2 The Age-Price Schedule and Net-Value Function

The net-value schedule combines physical survival with financial age-price decay. For a vintage of age v, the net-value function is

$$V_j(v) = S_j(v) \cdot (1 - d_j)^v, \tag{102}$$

where d_j is the geometric declining-balance depreciation rate. The GPIM separates the two dimensions that the BEA conflates in its geometric depreciation convention: physical survival follows the Weibull distribution (100), while financial age-price decay follows the geometric profile $(1 - d_j)^v$. This separation is necessary to construct distinct capacity and profitability measures (Shaikh, 2016, App. 6.3).

The declining-balance rates are calibrated as

$$d_{\text{ME}} = 0.110, \qquad d_{\text{NRC}} = 0.024, \tag{103}$$

consistent with the geometric depreciation rates documented by Fraumeni et al. (1997) for the corresponding asset classes.

A.3 The SFC Implicit Price Recursion

The implicit price index $P_{j,t}$ is the unique price level that closes the stock-flow identity for the net current-cost stock. In each period t, the BEA-reported current-cost net stock must satisfy

$$K_{j,t}^{\text{NET,CC}} = I_{j,t}^{\text{NOM}} + \frac{P_{j,t}}{100} \sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\text{REAL}}, \tag{104}$$

where $I_{j,t}^{\text{NOM}}$ is current-year nominal gross investment and $I_{j,t-v}^{\text{REAL}} = I_{j,t-v}^{\text{NOM}}/(P_{j,t-v}/100)$ is real investment of vintage $t-v$ deflated by the previously recovered implicit price. The first term on the right-hand side is the current-year investment vintage (age zero, valued at the current price). The second term is the net-value-weighted sum of surviving real vintages revalued at the current price level.

Solving (104) for $P_{j,t}$ yields the forward recursion

$$P_{j,t} = 100 \cdot \frac{K_{j,t}^{\text{NET,CC}} - I_{j,t}^{\text{NOM}}}{\displaystyle\sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\text{REAL}}}. \tag{105}$$

Given the initial price $P_{j,0}$ and the sequence of nominal investment flows $\{I_{j,t}^{\text{NOM}}\}_{t=0}^{T}$, the recursion (105) determines $P_{j,t}$ uniquely for each $t = 1, \ldots, T$. The price index is "implicit" because it is not directly observed: it is the unique value that, applied to the survival-weighted real vintages, reproduces the BEA's reported current-cost net stock.

A.4 Initial Conditions and Base-Year Normalization

The recursion (105) requires an initial price $P_{j,0}$ and initial real investment vintages. The base year is 2017, at which $P_{j,2017} = 100$ by normalization. The nominal investment series begins in 1901, sourced from BEA Fixed Assets Table FAAt201 (nonresidential equipment and structures, gross investment by asset class). The current-cost net stock anchors $K_{j,t}^{\text{NET,CC}}$ are sourced from BEA Fixed Assets Table FAAt401 (net stock by asset class, current-cost basis).[17]

The initialization proceeds backward from 2017. Given $P_{j,2017} = 100$ and the observed nominal investment series, the recursion (105) is iterated backward to recover $P_{j,t}$ for $t < 2017$. Alternatively, the recursion may be initialized at the earliest available year and iterated forward; the two approaches yield identical results given the BEA anchors.

A.5 The Gross Real Stock

The capital stock used to estimate productive capacity is the gross real stock, constructed as the survival-weighted sum of real investment vintages:

$$K_{j,t}^{G,R} = \sum_{v=0}^{L_j} S_j(v)\, I_{j,t-v}^{\text{REAL}}. \tag{106}$$

[17] Data retrieved via the BEA API (`DatasetName`: FixedAssets). Table FAAt201 provides nominal gross investment by asset; Table FAAt401 provides current-cost net stock by asset; Table FAAt402 provides current-cost depreciation by asset.

This measure relies exclusively on the physical survival schedule $S_j(v)$ and excludes the age-price factor $(1-d_j)^v$. It therefore measures the constant-price purchasing power embodied in heterogeneous capital goods still in physical operation. Financial write-downs, revaluation gains, and depreciation allowances do not enter. The OECD manual confirms that the gross stock, valued at "as new" prices, "is widely used as a broad indicator of the productive capacity of a country" (OECD, 2001, 31).

A.6 The Warm-Up Period and Estimation Sample

The GPIM stock at any year t requires investment vintages extending back L_j years. Because the nominal investment series begins in 1901, the vintage summation for nonresidential structures ($L_{\text{NRC}} = 30$) is truncated for any year before 1931. The stock values for 1901–1930 are therefore incomplete, omitting pre-1901 vintages that would still be physically in service. The estimation sample is restricted to 1931–2024, where the full 30-year vintage window is populated, yielding $T = 94$ annual observations.

A.7 Validation

The GPIM construction closes the stock-flow identity exactly: the maximum absolute reconstruction residual across both asset classes is 10^{-6} million dollars, confirming that the recovered SFC implicit price indexes reproduce the BEA current-cost net stock anchors to machine precision. The BEA's Fisher quantity index is retained solely as a validation comparison and does not enter the GPIM baseline. The sectoral scope is restricted to nonfinancial corporate (NFC) business, excluding non-profit institutions serving households (NPISH), government, and owner-occupied housing.

A.8 Data Sources

Table 8 summarizes the BEA Fixed Assets tables used in the GPIM construction.

Table 8: BEA Fixed Assets tables used in GPIM construction

Table	**Content**	**Use**
FAAt201	Nominal gross investment by asset	$I_{j,t}^{\text{NOM}}$
FAAt401	Current-cost net stock by asset	$K_{j,t}^{\text{NET,CC}}$
FAAt402	Current-cost depreciation by asset	Validation

Notes: All tables are from the BEA Fixed Assets Accounts, retrieved via the BEA API (`DatasetName`: FixedAssets). Asset classes: ME = nonresidential equipment; NRC = nonresidential structures. Sectoral scope: nonfinancial corporate business.

B Appendix: United States Specification B — Full Estimation Results

This appendix reports the full empirical estimation matrix across all 48 specifications of the aggregate capital model (Specification A). The specifications span two interaction sets (Set 1: standard wage interaction; Set 2: effective wage interaction scaled by $(1 - \iota_{t-1})$), two value-added bases (Gross Value Added [GVA] and Net Value Added [NVA]), six wage share measures per basis (NFC Baseline, Corporate, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, Clean Corporate Tier 1, and Clean Corporate Tier 2), and two structural break modes (without step dummies; with Bai-Perron step shifts).

Table 9: **Specification A, Set 1: Standard Wage Interaction (GVA Basis)**

Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b	M1.5a	M1.5b	M1.6a	M1.6b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.60^{***}	0.58^{***}	0.61^{***}	0.58^{***}	0.59^{***}	0.57	0.55	0.58^{***}	0.61^{***}	0.57	0.54
	(0.02)	(0.06)	(0.02)	(0.07)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.07)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	-0.02	-0.08	-0.03	-0.04	0.06	-0.08	-0.19	-0.23	0.01	-0.07	-0.20	-0.29
	(0.13)	(0.10)	(0.12)	(0.10)	(0.14)	(0.11)	(0.12)	(0.11)	(0.12)	(0.11)	(0.12)	(0.11)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.06	0.05	0.05	0.04	0.06	0.05
$S_{\text{Type 2}}$ [H_0: coint.]	3.30	11.34	2.70	11.41	5.82	9.68	2.56	12.13	5.30	10.48	3.74	12.92
t_{ADF} [H_0: no coint.]	-2.41	-2.81	-2.33	-2.73	-4.15^{**}	-3.56^{*}	-1.80	-2.15	-3.33	-2.66	-1.75	-2.12
Bandwidth ratio b	0.275	0.187	0.286	0.193	0.285	0.200	0.315	0.254	0.304	0.201	0.321	0.256
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 10: **Specification A, Set 1: Standard Wage Interaction (NVA Basis)**

Model	M1.7a	M1.7b	M1.8a	M1.8b	M1.9a	M1.9b	M1.10a	M1.10b	M1.11a	M1.11b	M1.12a	M1.12b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.60^{***}	0.58^{***}	0.60^{***}	0.58^{***}	0.60^{***}	0.57	0.56	0.59^{***}	0.71^{***}	0.57	0.63
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.08)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	-0.02	-0.08	-0.02	-0.08	-0.02	-0.08	-0.19	-0.26	-0.09	-0.32^{*}	-0.20	-0.29
	(0.13)	(0.10)	(0.13)	(0.10)	(0.13)	(0.10)	(0.11)	(0.09)	(0.09)	(0.08)	(0.10)	(0.10)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.05	0.06^{*}	0.03	0.06	0.05
$S_{\text{Type 2}}$ [H_0: coint.]	3.30	11.34	3.30	11.34	3.30	11.34	3.37	12.47	5.19	10.19	4.81	12.40
t_{ADF} [H_0: no coint.]	-2.41	-2.81	-2.41	-2.81	-2.41	-2.81	-1.97	-2.40	-2.42	-3.61^{*}	-1.91	-2.31
Bandwidth ratio b	0.275	0.187	0.275	0.187	0.275	0.187	0.277	0.235	0.242	0.156	0.280	0.239
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 11: **Specification A, Set 2: Effective Wage Interaction (GVA Basis)**

Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b	M2.5a	M2.5b	M2.6a	M2.6b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.56^{***}	0.58^{***}	0.57^{***}	0.58^{***}	0.54^{***}	0.57	0.55	0.58^{***}	0.56^{***}	0.56	0.56
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.07)	(0.02)	(0.06)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	0.00	-0.17	-0.00	-0.15	0.04	-0.20	-0.22	-0.37	0.01	-0.18	-0.20	-0.39
	(0.10)	(0.07)	(0.10)	(0.07)	(0.11)	(0.09)	(0.10)	(0.09)	(0.10)	(0.08)	(0.09)	(0.17)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.06	0.04	0.05	0.03	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	6.17	9.61	4.39	7.88	5.72	12.93	12.82	8.55	6.98	10.32
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.26	-2.96	-4.02^{**}	-2.93	-1.79	-2.31	-3.37	-2.87	-1.68	-2.20
Bandwidth ratio b	0.298	0.169	0.307	0.173	0.291	0.176	0.298	0.241	0.310	0.178	0.315	0.245
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 12: **Specification A, Set 2: Effective Wage Interaction (NVA Basis)**

Model	M2.7a	M2.7b	M2.8a	M2.8b	M2.9a	M2.9b	M2.10a	M2.10b	M2.11a	M2.11b	M2.12a	M2.12b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.56^{***}	0.58^{***}	0.56^{***}	0.58^{***}	0.56^{***}	0.57	0.55	0.58^{***}	0.65^{***}	0.57	0.56
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	0.00	-0.17	0.00	-0.17	0.00	-0.17	-0.21	-0.35	-0.04	-0.30^{**}	-0.19	-0.49
	(0.10)	(0.07)	(0.10)	(0.07)	(0.10)	(0.07)	(0.08)	(0.07)	(0.08)	(0.06)	(0.08)	(0.13)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.04	0.07^{*}	0.03	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	10.76	9.07	10.76	9.07	6.13	13.18	11.05	9.06	7.50	10.89
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.32	-3.06	-2.32	-3.06	-2.00	-2.72	-2.22	-4.07^{**}	-1.84	-2.58
Bandwidth ratio b	0.298	0.169	0.298	0.169	0.298	0.169	0.259	0.237	0.280	0.165	0.272	0.237
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 13: Custom Monte Carlo Simulated Critical Bounds for CMPR IM-OLS Inference (Specification A)

Statistic / Parameter	**Asymptotic 95%**	**Sim 90%**	**Sim 95%**	**Sim 99%**
Panel A: Baseline Sample ($T = 94$, 1931–2024) — No Structural Breaks ($m = 0$)				
t_{level} (k_t^{cap})	1.960	3.877	5.119	8.502
t_{inter} $(k_t^{\text{cap}} \cdot C_t)$	N/A	3.123	4.006	6.238
CT_{IM} (Type 2 Cointegration)	0.070	0.0580	0.0705	0.1005
Panel B: Baseline Sample ($T = 94$, 1931–2024) — Bai-Perron Level-Shift Breaks ($m \geq 1$)				
t_{level} (k_t^{cap})	1.960	2.607	3.267	4.989
t_{inter} $(k_t^{\text{cap}} \cdot C_t)$	N/A	3.455	4.325	6.444
CT_{IM} (Type 2 Cointegration)	0.070	0.0482	0.0598	0.0895
Panel C: Supervisory Sample ($T = 76$, 1948–2024) — No Structural Breaks ($m = 0$)				
t_{level} (k_t^{cap})	1.960	3.900	5.155	8.387
t_{inter} $(k_t^{\text{cap}} \cdot C_t)$	N/A	3.139	4.018	6.206
CT_{IM} (Type 2 Cointegration)	0.070	0.0635	0.0780	0.1151
Panel D: Supervisory Sample ($T = 76$, 1948–2024) — Bai-Perron Level-Shift Breaks ($m \geq 1$)				
t_{level} (k_t^{cap})	1.960	2.687	3.371	5.188
t_{inter} $(k_t^{\text{cap}} \cdot C_t)$	N/A	3.427	4.277	6.213
CT_{IM} (Type 2 Cointegration)	0.070	0.0545	0.0696	0.1065

Notes: Critical bounds derived from a custom $N = 100,000$ iteration Monte Carlo simulation suite calibrated specifically to the $I(1) \times I(1)$ product integration geometry of the regressors and dynamic Bai-Perron break degrees of freedom. Panels A and B reflect the baseline 1931–2024 sample length ($T = 94$), while Panels C and D reflect the supervisory-adjusted Tier 2 wage share sample ($T = 76$). Bandwidth $b = M/T$ is fixed per Vogelsang & Wagner (2014) IM-OLS specification. Data Generating Process (H_0) constructs independent zero-drift random walks for base level variables ($k_t^{\text{cap}}, \omega_t$), with interaction product processes algebraically assembled prior to estimation. Parameter t-statistics are centered around their true theoretical values ($t^* = |(\hat{\beta}^* - \beta_{\text{true}})/SE|$) to ensure pure size calibration rather than power distortion.

C Appendix: United States Specification B - Heterogeneous Capital Stock Empirical Results

This appendix reports the full empirical estimation matrix across all 48 specifications of the heterogeneous capital model (Specification B). The specifications span two interaction sets (Set 1: standard wage interaction; Set 2: effective wage interaction scaled by $(1 - \iota_{t-1})$), two value-added bases (Gross Value Added [GVA] and Net Value Added [NVA]), six wage share measures per basis (NFC Baseline, Corporate Baseline, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, Clean Corporate Tier 1, and Clean Corporate Tier 2), and two structural break modes (without step dummies; with Bai-Perron step shifts).

Table 14: Specification B, Set 1: Standard Wage Interaction (GVA Basis)

Model	M2.1a	M2.2b	M2.5a	M2.6b	M2.9a	M2.10b	M2.13a	M2.14b	M2.17a	M2.18b	M2.21a	M2.22b
$\hat{\alpha}$ (Intercept)	12.90	12.82	12.89	12.82	12.90	12.82	13.85	13.52	12.89	12.81	13.85	13.52
	(0.07)	(0.06)	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)	(0.07)	(0.07)	(0.08)	(0.06)
$\hat{\theta}_{\text{NRC}}$	0.60	0.51	0.61	0.47	0.61	0.48	0.51	0.50	0.59	0.52	0.51	0.49
	(0.03)	(0.08)	(0.03)	(0.07)	(0.03)	(0.07)	(0.02)	(0.07)	(0.03)	(0.09)	(0.02)	(0.07)
$\hat{\theta}_{\kappa}$	0.38	0.67	0.40	0.44	0.39	0.53	0.38	1.09	0.36	0.60	0.34	1.11
	(0.14)	(0.26)	(0.14)	(0.22)	(0.14)	(0.23)	(0.12)	(0.20)	(0.14)	(0.33)	(0.11)	(0.20)
$\hat{\theta}_{\omega\kappa}$	-2.75	-4.89	7.95	0.05	4.01	-4.20	-16.40	-10.51	-9.89	-5.93	-18.27	-9.04
	(4.07)	(3.10)	(5.98)	(2.97)	(7.41)	(4.18)	(6.63)	(5.86)	(4.85)	(3.96)	(5.56)	(5.55)
$\hat{\theta}_{\kappa p}$	-0.68	1.38	-2.07	0.16	-1.28	0.64	-1.43	-0.35	0.01	1.63	-0.78	-0.35
	(0.58)	(0.82)	(0.87)	(0.64)	(0.61)	(0.65)	(0.56)	(0.47)	(0.57)	(0.99)	(0.54)	(0.47)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.05	0.03	0.04	0.03	0.05	0.03	0.03	0.02	0.05	0.03	0.02	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	16.11	9.55	16.40	12.60	16.25	11.41	9.98	2.78	16.37	10.87	8.42	3.26
t_{ADF} [H_0: no coint.]	-4.16*	-4.46**	-4.25*	-4.67**	-4.26*	-4.69**	-2.56	-3.22	-4.26*	-4.50**	-2.58	-3.17
Bandwidth ratio b	0.209	0.176	0.215	0.173	0.211	0.174	0.211	0.203	0.213	0.191	0.210	0.203
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 15: Specification B, Set 1: Standard Wage Interaction (NVA Basis)

Model	M2.3a	M2.4b	M2.7a	M2.8b	M2.11a	M2.12b	M2.15a	M2.16b	M2.19a	M2.20b	M2.23a	M2.24b
$\hat{\alpha}$ (Intercept)	12.90	12.82	12.90	12.82	12.90	12.82	13.87	13.57	12.90	12.82	13.88	13.57
	(0.07)	(0.06)	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)	(0.07)	(0.07)	(0.08)	(0.06)
$\hat{\theta}_{\text{NRC}}$	0.60	0.51	0.60	0.51	0.60	0.51	0.51	0.52	0.59	0.50	0.51	0.50
	(0.03)	(0.08)	(0.03)	(0.08)	(0.03)	(0.08)	(0.02)	(0.07)	(0.03)	(0.09)	(0.02)	(0.07)
$\hat{\theta}_{\kappa}$	0.38	0.67	0.38	0.67	0.38	0.67	0.35	0.99	0.36	0.61	0.34	1.02
	(0.14)	(0.26)	(0.14)	(0.26)	(0.14)	(0.26)	(0.11)	(0.18)	(0.14)	(0.34)	(0.11)	(0.18)
$\hat{\theta}_{\omega\kappa}$	-2.75	-4.89	-2.79	-4.96	-2.75	-4.89	-13.06	-13.10	-7.06	-2.73	-12.13	-11.13
	(4.07)	(3.10)	(4.14)	(3.15)	(4.07)	(3.10)	(4.24)	(4.38)	(3.32)	(2.62)	(3.50)	(3.99)
$\hat{\theta}_{\kappa p}$	-0.68	1.38	-0.68	1.38	-0.68	1.38	-0.99	-0.37	0.33	1.50	-0.49	-0.30
	(0.58)	(0.82)	(0.58)	(0.82)	(0.58)	(0.82)	(0.52)	(0.42)	(0.65)	(0.98)	(0.53)	(0.43)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.05	0.03	0.05	0.03	0.05	0.03	0.02	0.02	0.05	0.03	0.02	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	16.11	9.55	16.11	9.55	16.11	9.55	8.41	6.15	16.35	11.90	7.50	4.26
t_{ADF} [H_0: no coint.]	-4.16*	-4.46**	-4.16*	-4.46**	-4.16*	-4.46**	-2.61	-3.55	-4.20*	-4.49**	-2.50	-3.44
Bandwidth ratio b	0.209	0.176	0.209	0.176	0.209	0.176	0.207	0.205	0.213	0.192	0.211	0.205
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 16: Specification B, Set 2: Effective Wage Interaction (GVA Basis)

Model	M2.25a	M2.26b	M2.29a	M2.30b	M2.33a	M2.34b	M2.37a	M2.38b	M2.41a	M2.42b	M2.45a	M2.46b
$\hat{\alpha}$ (Intercept)	12.82	12.77	12.83	12.78	12.79	12.76	13.68	13.66	12.80	12.77	13.67	13.61
	(0.07)	(0.06)	(0.07)	(0.07)	(0.07)	(0.06)	(0.06)	(0.05)	(0.07)	(0.06)	(0.05)	(0.06)
$\hat{\theta}_{\text{NRC}}$	0.61	0.58	0.61	0.55	0.61	0.62	0.55	0.51	0.61	0.60	0.55	0.70
	(0.02)	(0.08)	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.02)	(0.09)	(0.01)	(0.03)
$\hat{\theta}_{\kappa}$	0.32	0.39	0.32	0.44	0.31	0.14	0.36	0.02	0.31	0.36	0.34	0.72
	(0.13)	(0.29)	(0.14)	(0.30)	(0.14)	(0.38)	(0.08)	(0.11)	(0.14)	(0.29)	(0.08)	(0.10)
$\hat{\theta}_{\omega\kappa}$	-12.77	-10.13	-13.73	-12.11	-15.99	-12.29	-20.77	-19.87	-14.54	-11.15	-18.85	-15.55
	(3.81)	(3.43)	(4.45)	(3.82)	(4.94)	(4.22)	(4.11)	(3.99)	(4.60)	(3.63)	(3.32)	(3.54)
$\hat{\theta}_{\kappa p}$	0.18	0.92	0.35	1.44	-0.39	0.58	-1.37	-0.81	0.08	0.88	-0.96	0.10
	(0.51)	(0.69)	(0.58)	(0.79)	(0.45)	(0.60)	(0.42)	(0.39)	(0.52)	(0.66)	(0.40)	(0.41)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.03	0.02	0.04	0.03	0.03	0.02	0.03	0.02	0.03	0.02	0.03	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	14.97	5.93	15.18	6.02	14.58	3.71	13.20	11.19	14.75	6.00	12.33	9.93
t_{ADF} [H_0: no coint.]	-4.33*	-4.58**	-4.36*	-4.52**	-4.40*	-4.77**	-4.12	-4.02	-4.41*	-4.69**	-4.13*	-3.97
Bandwidth ratio b	0.185	0.173	0.194	0.175	0.192	0.173	0.206	0.206	0.195	0.179	0.207	0.207
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 17: Specification B, Set 2: Effective Wage Interaction (NVA Basis)

Model	M2.27a	M2.28b	M2.31a	M2.32b	M2.35a	M2.36b	M2.39a	M2.40b	M2.43a	M2.44b	M2.47a	M2.48b
$\hat{\alpha}$ (Intercept)	12.82	12.77	12.82	12.77	12.82	12.77	13.72	13.69	12.83	12.78	13.73	13.73
	(0.07)	(0.06)	(0.07)	(0.06)	(0.07)	(0.06)	(0.04)	(0.04)	(0.07)	(0.07)	(0.04)	(0.04)
$\hat{\theta}_{\text{NRC}}$	0.61	0.58	0.61	0.58	0.61	0.58	0.54	0.51	0.60	0.60	0.54	0.54
	(0.02)	(0.08)	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.02)	(0.08)	(0.01)	(0.01)
$\hat{\theta}_{\kappa}$	0.32	0.39	0.32	0.39	0.32	0.39	0.33	0.06	0.32	0.39	0.31	0.31
	(0.13)	(0.29)	(0.13)	(0.29)	(0.13)	(0.29)	(0.06)	(0.09)	(0.13)	(0.29)	(0.06)	(0.06)
$\hat{\theta}_{\omega\kappa}$	-12.77	-10.13	-12.97	-10.28	-12.77	-10.13	-17.05	-16.76	-11.08	-7.93	-14.97	-14.97
	(3.81)	(3.43)	(3.86)	(3.48)	(3.81)	(3.43)	(2.34)	(2.55)	(3.48)	(2.96)	(2.01)	(2.01)
$\hat{\theta}_{\kappa p}$	0.18	0.92	0.18	0.92	0.18	0.92	-0.79	-0.71	0.69	1.41	-0.31	-0.31
	(0.51)	(0.69)	(0.51)	(0.69)	(0.51)	(0.69)	(0.32)	(0.31)	(0.63)	(0.75)	(0.33)	(0.33)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.03	0.02	0.03	0.02	0.03	0.02	0.02	0.02	0.03	0.03	0.02	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	14.97	5.93	14.97	5.93	14.97	5.93	10.28	8.70	15.37	6.37	8.96	8.96
t_{ADF} [H_0: no coint.]	-4.33*	-4.58**	-4.33*	-4.58**	-4.33*	-4.58**	-4.71**	-4.57**	-4.34*	-4.60**	-4.24*	-4.24*
Bandwidth ratio b	0.185	0.173	0.185	0.173	0.185	0.173	0.208	0.208	0.193	0.181	0.208	0.208
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 18: Custom Monte Carlo Simulated Critical Bounds Across Structural Break Regimes (Specification B)

Statistic / Parameter	**Asymptotic 95%**	**Sim 90%**	**Sim 95%**	**Sim 99%**	**Empirical (M2.44b)**
Panel A: No Structural Breaks ($m = 0$, Sample $T = 94$)					
t_{NRC} (k_t^{NRC})	1.960	1.842	2.195	2.910	7.50**
t_κ ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$)	1.960	1.895	2.241	2.980	1.34
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.115	2.512	3.340	-2.68**
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.080	2.475	3.295	1.88*
CT_{IM} (Type 2 Global Test)	18.50	11.20	13.45	18.90	6.37**
Panel B: One Endogenous Structural Break ($m = 1$, e.g., M2.40b)					
t_{NRC} (k_t^{NRC})	1.960	1.920	2.280	3.010	–
t_κ (κ_t)	1.960	1.970	2.330	3.080	–
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.180	2.590	3.430	–
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.150	2.550	3.390	–
CT_{IM} (Type 2 Global Test)	19.80	12.10	14.50	20.20	–
Panel C: Two Endogenous Structural Breaks ($m = 2$, e.g., M2.46b)					
t_{NRC} (k_t^{NRC})	1.960	1.995	2.370	3.120	–
t_κ (κ_t)	1.960	2.050	2.420	3.190	–
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.250	2.670	3.520	–
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.210	2.630	3.470	–
CT_{IM} (Type 2 Global Test)	21.20	13.00	15.50	21.60	–
Panel D: Three Endogenous Structural Breaks ($m = 3$, Preferred Model M2.44b)					
t_{NRC} (k_t^{NRC})	1.960	2.075	2.455	3.210	7.50**
t_κ (κ_t)	1.960	2.130	2.515	3.305	1.34
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.315	2.745	3.615	-2.68**
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.275	2.705	3.560	1.88*
CT_{IM} (Type 2 Global Test)	22.65	13.90	16.55	22.95	6.37**
Panel E: Four Endogenous Structural Breaks ($m = 4$, e.g., M2.18b, M2.34b)					
t_{NRC} (k_t^{NRC})	1.960	2.150	2.540	3.310	–
t_κ (κ_t)	1.960	2.210	2.610	3.420	–
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.380	2.820	3.710	–
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.340	2.780	3.650	–
CT_{IM} (Type 2 Global Test)	24.10	14.80	17.60	24.30	–

Notes: Critical bounds derived from a 100,000-pass specification-dependent Monte Carlo suite tailored to Specification B's 4-variable stochastic design ($k^{\text{NRC}}, \kappa, \kappa \cdot C, \kappa \cdot p^\kappa$) across all structural break counts ($m = 0 \ldots 4$). The DGP under H_0 simulates independent zero-drift random walks for base variables and algebraically constructs product terms, with dynamic Bai-Perron break selection applied recursively. Simulated parameter t-statistics are centered around true DGP parameters ($t^* = |(\hat{\beta}^* - \beta_{\text{true}})/SE|$) to prevent artificial bound inflation. Empirical values reported for preferred Model M2.44b ($m = 3$). Bold empirical values denote significance based on the simulated 95% bound (* $p < 0.05$, ** $p < 0.01$).

D Appendix: Chile - External Cosntraint and Regime Shifting Results

Table 19: Comparative Cointegration Estimates across Net and Gross Wage Shares (Chile, 1942–2024)

	Net Wage Share (ω^{net})				**Gross Wage Share** (ω^{gross})			
	Baseline	Bai-Perron	Comp. Thresh-old	Parsimonious	Baseline	Bai-Perron	Comp. Thresh-old	Parsimonious
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{k^{\text{NR}}}$ (Structures k^{NR})	0.96***	0.73***	0.67***	0.79***	0.99***	0.98***	0.70***	0.82***
	(0.01)	(0.12)	(0.09)	(0.07)	(0.00)	(0.05)	(0.08)	(0.08)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	0.03	0.56	0.53	0.12	-0.41	0.22	-0.14	-1.48*
	(0.25)	(0.50)	(0.40)	(0.39)	(0.29)	(0.27)	(0.44)	(0.54)
$\hat{\theta}_{\omega\kappa}$ (Wage Interaction)	-0.28	-0.16	-0.12	-0.01	0.01	-0.35	0.96	3.29*
	(0.22)	(0.21)	(0.17)	(0.22)	(0.38)	(0.26)	(0.73)	(1.07)
$\hat{\theta}_{\kappa D}$ (Regime Shift κD)	—	-0.18	0.36*	0.34	—	5.13***	0.66	1.52**
	(0.00)	(0.28)	(0.13)	(0.16)	(0.00)	(1.00)	(0.31)	(0.45)
$\hat{\theta}_{\omega\kappa D}$ (Triple Interaction)	—	-0.09	-0.15	-0.17	—	-11.86***	-0.95	-3.15*
		(0.33)	(0.18)	(0.23)		(2.35)	(0.72)	(1.04)
$\hat{\theta}_{p}$ (Price Control $\tilde{p}$)	-0.05	-0.19**	-0.24***	—	-0.04	0.02	-0.20***	—
	(0.11)	(0.09)	(0.07)		(0.11)	(0.07)	(0.07)	
Regime Threshold (D_t)	None	BP (1987)	$Z_t \leq 0.425$	$Z_t \leq 0.425$	None	BP (1999)	$Z_t \leq 0.425$	$Z_t \leq 0.425$
Price Control Included	Yes	Yes	Yes	No	Yes	Yes	Yes	No
CT_{IM} [H_0: Cointegration]	—	12.88	11.72	15.20	—	14.55	8.50	8.39
t_{ADF} [H_0: Unit Root]	-1.84	-2.72	-3.82*	-3.81*	-1.92	-2.98	-4.11**	-3.81*
Bandwidth ratio b	0.626	0.650	0.654	0.654	0.635	0.652	0.655	0.654
Sample Size T	68	68	68	68	68	68	68	68

Table 20: Comparative Cointegration Estimates across Net and Gross Rates of Exploitation (Chile, 1942–2024)

	Net Rate of Exploitation (e^{net})				**Gross Rate of Exploitation** (e^{gross})			
	Baseline	Bai-Perron	Comp. Threshold	Parsimonious	Baseline	Bai-Perron	Comp. Threshold	Parsimonious
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{k^{\text{NR}}}$ (Structures k^{NR})	0.96***	0.73***	0.68***	0.80***	0.99***	0.98***	0.71***	0.82***
	(0.01)	(0.11)	(0.09)	(0.07)	(0.01)	(0.06)	(0.08)	(0.08)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	-0.21	0.45	0.43	0.15	-0.42***	-0.05	0.51	0.85
	(0.11)	(0.42)	(0.28)	(0.25)	(0.08)	(0.27)	(0.27)	(0.38)
$\hat{\theta}_{e\kappa}$ (Exploitation Interaction)	0.14	0.08	0.06	0.02	0.00	0.09	-0.19	-0.69*
	(0.11)	(0.10)	(0.07)	(0.09)	(0.10)	(0.07)	(0.17)	(0.23)
$\hat{\theta}_{\kappa D}$ (Regime Shift κD)	—	-0.27	0.21*	0.15	—	-2.39***	-0.02	-0.73
	(0.00)	(0.10)	(0.07)	(0.08)	(0.00)	(0.50)	(0.23)	(0.29)
$\hat{\theta}_{e\kappa D}$ (Triple Interaction)	—	0.05	0.09	0.12	—	1.83***	0.20	0.67*
		(0.15)	(0.07)	(0.10)		(0.36)	(0.16)	(0.22)
$\hat{\theta}_{p}$ (Price Control $\tilde{p}$)	-0.04	-0.18**	-0.22***	—	-0.04	0.02	-0.20***	—
	(0.11)	(0.09)	(0.07)		(0.11)	(0.07)	(0.07)	
Regime Threshold (D_t)	None	BP (1987)	$Z_t \leq 0.425$	$Z_t \leq 0.425$	None	BP (1999)	$Z_t \leq 0.425$	$Z_t \leq 0.425$
Price Control Included	Yes	Yes	Yes	No	Yes	Yes	Yes	No
CT_{IM} [H_0: Cointegration]	—	12.52	10.83	14.99	—	14.40	7.96	8.34
t_{ADF} [H_0: Unit Root]	-1.89	-2.73	-3.86*	-3.84*	-1.96	-3.05	-4.15**	-3.89*
Bandwidth ratio b	0.623	0.650	0.654	0.654	0.635	0.652	0.655	0.654
Sample Size T	68	68	68	68	68	68	68	68